

AWS CCP questions

339) For its steady-state database, a corporation runs a self-managed Oracle database directly on Amazon EC2. The corporation wishes to cut its computing expenses.

Which option should the business choose in order to optimize savings over a three-year period?

- A. EC2 Dedicated Instances
- B. EC2 Spot Instances
-  C. EC2 Reserved Instances
- D. EC2 On-Demand Instances

Correct Answer: C

Amazon EC2 Reserved Instances (RI) provide a significant discount (up to 72%) compared to On-Demand pricing and provide a capacity reservation when used in a specific Availability Zone.

1) Which AWS service offers threat detection and continuously monitors for malicious activity and unauthorized behavior in AWS accounts?

- A. Amazon Macie
- B. AWS Config
- C. Amazon GuardDuty
- D. Amazon Inspector

Correct Answer: C

Amazon GuardDuty is a threat detection service that continuously monitors your AWS accounts and workloads for malicious activity and delivers detailed security findings for visibility and remediation. Continuously monitor your AWS accounts, instances, serverless and container workloads, users, databases, and storage for potential threats.

2) Which AWS service or feature enables users to block the incoming or outgoing traffic associated with specific IP addresses flowing through a VPC?

- A. Network ACLs
- B. Security groups
- C. AWS Identity and Access Management (IAM)
- D. AWS WAF

Correct Answer: A

Security groups are tied to an instance whereas Network ACLs are tied to the subnet. Network ACLs are applicable at the subnet level, so any instance in the subnet with an associated NACL will follow the rules of NACL. That's not the case with security groups, security groups have to be assigned explicitly to the instance

3) What is the customer ALWAYS responsible for managing, according to the AWS shared responsibility model?

- A. Software licenses
- B. Networking
- C. Customer data
- D. Encryption keys

Correct Answer: C

Customers are responsible for managing their data (including encryption options), classifying their assets, and using IAM tools to apply the appropriate permissions.

4) Which component of the AWS global infrastructure is made up of one or more discrete data centers that have redundant power, networking, and connectivity?

- A. AWS Region
- B. Availability Zone
- C. Edge location
- D. AWS Outposts

Correct Answer: B

An Availability Zone (AZ) is a component of the AWS global infrastructure that is made up of one or more discrete data centers that have redundant power, networking, and connectivity. Each AZ is physically separated from other AZs within a region, and is designed to be fault-tolerant and provide low-latency networking.

5) Which pillar of the AWS Well-Architected Framework refers to the ability of a system to recover from infrastructure or service disruptions and dynamically acquire computing resources to meet demand?

- A. Security
- B. Reliability
- C. Performance efficiency
- D. Cost optimization

Correct Answer: B

The Reliability pillar includes the ability of a workload to perform its intended function correctly and consistently when it's expected to. This includes the ability to operate and test the workload through its total lifecycle.

6) Who is responsible for configuration management under the AWS shared responsibility model?

- A. It is solely the responsibility of the customer.
- B. It is solely the responsibility of AWS.
- C. It is shared between AWS and the customer.
- D. It is not part of the AWS shared responsibility model.

Correct Answer: C

AWS maintains the configuration of its infrastructure devices, but a customer is responsible for configuring their own guest operating systems, databases, and applications.

7) Which duties are the responsibility of a company that is using AWS Lambda? (Choose two.)

- A. Security inside of code
- B. Selection of CPU resources
- C. Patching of operating system
- D. Writing and updating of code
- E. Security of underlying infrastructure

Correct Answer: AD

Lambda runs your code on high availability compute infrastructure and performs all the administration of your compute resources. This includes server and operating system maintenance, capacity provisioning and automatic scaling, code and security patch deployment, and code monitoring and logging. All you need to do is supply the code.

8) Which compute hosting model should be accounted for in the Total Cost of Ownership (TCO) when undertaking a cost analysis that allows physical isolation of a customer workload?

- A. Dedicated Hosts
- B. Reserved Instances
- C. On-Demand Instances
- D. No Upfront Reserved Instances

Correct Answer: A

Use Dedicated Hosts to launch Amazon EC2 instances on physical servers that are dedicated for your use. Dedicated Hosts give you additional visibility and control over how instances are placed on a physical server, and you can reliably use the same physical server over time. As a result, Dedicated Hosts enable you to use your existing server-bound software licenses like Windows Server and address corporate compliance and regulatory requirements.

9) Who is accountable for security and compliance under the AWS shared responsibility model?

- A. The customer is responsible.
- B. AWS is responsible.
- C. AWS and the customer share responsibility.
- D. AWS shares responsibility with the relevant governing body.

Correct Answer: C

Security and Compliance is a shared responsibility between AWS and the customer. This shared model can help relieve the customer's operational burden as AWS operates, manages and controls the components from the host operating system and virtualization layer down to the physical security of the facilities in which the service operates.

10) Which of the following is a critical design concept for architecting cloud applications?

- A. Use the largest instance possible
- B. Provision capacity for peak load
- C. Use the Scrum development process
- D. Implement elasticity

Correct Answer: D

Cloud services main proposition is to provide elasticity through horizontal scaling. It's already there. As for using largest instance possible, it is not a design principle that helps cloud applications in anyway. Scrum development process is not related to architecting. Therefore, a key principle is to provision your application for on-demand capacity. Peak loads is something that cloud applications experience everyday. Peak load management should be a necessary part of cloud application design principle.

11) Which AWS services or features provide disaster recovery solutions for Amazon EC2 instances? (Choose two.)

- A. Reserved Instances
- B. EC2 Amazon Machine Images (AMIs)
- C. Amazon Elastic Block Store (Amazon EBS) snapshots
- D. AWS Shield
- E. Amazon GuardDuty

Correct Answer: BC

The AWS services or features that provide disaster recovery solutions for Amazon EC2 instances are:

C. Amazon Elastic Block Store (Amazon EBS) snapshots: Amazon EBS is a block storage service for use with Amazon EC2 instances. EBS snapshots are point-in-time copies of an EBS volume that can be used to create a new volume or restore an existing volume. EBS snapshots can be used to recover data in the event of an instance failure, or to create a new instance in a different region or Availability Zone.

B. Amazon Machine Images (AMIs): An AMI is a pre-configured virtual machine image that contains the operating system, application software, and any other required components needed to launch an instance. AMIs can be used to create new instances in the same or a different region, which can be useful for disaster recovery purposes.

... ❌ ...

12) A company wants to review its monthly costs of using Amazon EC2 and Amazon RDS for the past year. Which AWS service or tool provides this information?

- A. AWS Trusted Advisor
- B. Cost Explorer
- C. Amazon Forecast
- D. Amazon CloudWatch

Correct Answer: B

After you enable Cost Explorer, AWS prepares the data about your costs for the current month and the last 12 months, and then calculates the forecast for the next 12 months. The current month's data is available for viewing in about 24 hours. The rest of your data takes a few days longer. Cost Explorer updates your cost data at least once every 24 hours.

13) Who is the main point of contact for billing or account questions if a user has an AWS account with an Enterprise-level AWS Support plan?

- A. Solutions architect
- B. AWS Concierge Support team
- C. An AWS Marketplace seller
- D. AWS Partner Network (APN) partner

Correct Answer: B

AWS Support Concierge Included as part of the Enterprise Support plan, the Support Concierge Team are **AWS billing and account experts** that specialize in working with enterprise accounts.

14) What is one technique to provide unified billing if each department within a firm has its own AWS account?

- A. Use AWS Budgets on each account to pay only to budget.
- B. Contact AWS Support for a monthly bill.
- C. Create an AWS Organization from the payer account and invite the other accounts to join.
- D. Put all invoices into one Amazon Simple Storage Service (Amazon S3) bucket, load data into Amazon Redshift, and then run a billing report.

Correct Answer: C

You can invite an account to join an organization that has only the consolidated billing features enabled. If you later want to enable all features for the organization, invited accounts must approve the change.

15) Which cloud architecture design concept is supported by distributing workloads across various Availability Zones?

- A. Implement automation.
- B. Design for agility.
- C. Design for failure.
- D. Implement elasticity.

Correct Answer: C

Design of failure : Distribute workload data and resources across multiple Availability Zones (AZs) or, where necessary, across AWS Regions. These locations can be as diverse as required.

16) Which security-related duty is AWS accountable for under the AWS shared responsibility model?

- A. Lifecycle management of IAM credentials
- B. Physical security of global infrastructure
- C. Encryption of Amazon EBS volumes
- D. Firewall configuration

Correct Answer: B

Security and Compliance is a shared responsibility between AWS and the customer. This shared model can help relieve the customer's operational burden as AWS operates, manages and controls the components from the host operating system and virtualization layer down to the physical security of the facilities in which the service operates.

17) A company wants to migrate a critical application to AWS. The application has a short runtime. The application is invoked by changes in data or by shifts in system state. The company needs a compute solution that maximizes operational efficiency and minimizes the cost of running the application. Which AWS solution should the company use to meet these requirements?

- A. Amazon EC2 On-Demand Instances
- B. AWS Lambda
- C. Amazon EC2 Reserved Instances
- D. Amazon EC2 Spot Instances

Correct Answer: B

AWS Lambda supports function packaging and deployment as container images, making it easy for customers to build Lambda-based applications using familiar container image tooling, workflows, and dependencies. Customers also benefit from Lambda's operational simplicity, automatic scaling with sub-second startup times, high availability, pay-for-use billing model, and native integrations with over 200 AWS services and software-as-a service (SaaS) applications.

18) Which AWS service or feature allows users to connect with and deploy AWS services programmatically?

- A. AWS Management Console
- B. AWS Cloud9
- C. AWS CodePipeline
- D. AWS software development kits (SDKs)

Correct Answer: D

AWS software development kits (SDKs) allow users to connect with and deploy AWS services programmatically. AWS provides SDKs for several popular programming languages, including Java, Python, JavaScript, and C++.

19) According to the AWS shared responsibility model, who is responsible for managing IAM user access and secret keys?

- A. IAM access and secret keys are static, so there is no need to rotate them.
- B. The customer is responsible for rotating keys.
- C. AWS will rotate the keys whenever required.
- D. The AWS Support team will rotate keys when requested by the customer.

Correct Answer: B

20) To install a PCI-compliant workload on AWS, which of the following tasks is required?

- A. Use any AWS service and implement PCI controls at the application layer
- B. Use an AWS service that is in-scope for PCI compliance and raise an AWS support ticket to enable PCI compliance at the application layer
- C. Use any AWS service and raise an AWS support ticket to enable PCI compliance on that service
- D. Use an AWS service that is in scope for PCI compliance and apply PCI controls at the application layer

Correct Answer: D

21) Which of the following is a suggestion made by an AWS Trusted Advisor? (Select two.)

- A. Cost optimization
- B. Auditing
- C. Serverless architecture
- D. Performance
- E. Scalability

Correct Answer: AD



You can use the Trusted Advisor Recommendations page of the Trusted Advisor console to review check results for your AWS account and then follow the recommended steps to fix any issues. For example, Trusted Advisor might recommend that you delete unused resources to reduce your monthly bill, such as an Amazon Elastic Compute Cloud (Amazon EC2) instance.

22) A company plans to create a data lake that uses Amazon S3. Which factor will have the MOST effect on cost?

- A. The selection of S3 storage tiers
- B. Charges to transfer existing data into Amazon S3
- C. The addition of S3 bucket policies
- D. S3 ingest fees for each request

Correct Answer: A

Which storage tier they pick out of all the options can largely effect the final cost.

23) A company is launching an ecommerce application that must always be available. The application will run on Amazon EC2 instances continuously for the next 12 months. What is the MOST cost-effective instance purchasing option that meets these requirements?

- A. Spot Instances
- B. Savings Plans
- C. Dedicated Hosts
- D. On-Demand Instances

Correct Answer: B

Amazon EC2 Savings Plans enable you to reduce your compute costs by committing to a consistent amount of compute usage for a 1-year or 3-year term. This results in savings of up to 72% over On-Demand Instance costs. Any usage up to the commitment is charged at the discounted Savings Plan rate (for example, \$10 an hour). Any usage beyond the commitment is charged at regular On-Demand Instance rates.

24) Which pillar of the AWS Well-Architected Framework is supported by the design philosophy of performing operations as code?

- A. Performance efficiency
- B. Operational excellence
- C. Reliability
- D. Security

Correct Answer: B

1. Operational Excellence

The Operational Excellence pillar includes the ability to support development and run workloads effectively, gain insight into their operation, and continuously improve supporting processes and procedures to deliver business value. You can find prescriptive guidance on implementation in the [Operational Excellence Pillar whitepaper](#).

Design Principles

There are five design principles for operational excellence in the cloud:

- Perform operations as code »
- Make frequent, small, reversible changes
- Refine operations procedures frequently
- Anticipate failure »
- Learn from all operational failures

25) What is the customer's responsibility while using Amazon RDS?

- A. Patching and maintenance of the underlying operating system.
- B. Managing automatic backups of the database.
- C. Controlling network access through security groups.
- D. Replacing failed instances in the event of a hardware failure.

Correct Answer: C

Amazon RDS is responsible for hosting the software components and infrastructure of DB instances and DB cluster. You are responsible for query tuning, which is the process of adjusting SQL queries to improve performance. Query performance is highly dependent on database design, data size, data distribution, application workload, and query patterns, which can vary greatly. Monitoring and tuning are highly individualized processes that you own for your RDS databases. You can use Amazon RDS Performance Insights and other tools to identify problematic queries.

26) What is the customer's obligation while using an AWS managed service under the AWS shared responsibility model?

- A. Physical security of the data centers
- B. Server-side encryption
- C. Customer data
- D. Operating system patching

Correct Answer: C

Customers are responsible for managing their data (including encryption options), classifying their assets, and using IAM tools to apply the appropriate permissions.

27) Which AWS service or feature can a company use to determine which business unit is using specific AWS resources?

- A. Cost allocation tags
- B. Key pairs
- C. Amazon Inspector
- D. AWS Trusted Advisor

Correct Answer: A

You can use tags to organize your resources, and cost allocation tags to track your AWS costs on a detailed level. After you activate cost allocation tags, AWS uses the cost allocation tags to organize your resource costs on your cost allocation report, to make it easier for you to categorize and track your AWS costs.

28) Which AWS service or tool should a company use to centrally request and track service limit increases?

- A. AWS Config
- B. Service Quotas
- C. AWS Service Catalog
- D. AWS Budgets

Correct Answer: B

You can now use Service Quotas in AWS GovCloud (US) Regions to view and manage your service quotas at scale as your AWS workloads grow.

29) Which pillar of the AWS Well-Architected Framework is designed on the idea of frequent, minor, reversible changes?

- A. Reliability
- B. Operational excellence
- C. Performance efficiency
- D. Cost optimization

Correct Answer: B

There are five design principles for operational excellence in the cloud:

- 1) **Perform operations as code**
- 2) **Make frequent, small, reversible changes**
- 3) **Refine operations procedures frequently**
- 4) **Anticipate failure**
- 5) **Learn from all operational failures**

30) Which AWS service can identify the person who made the API request when an Amazon EC2 instance is terminated?

- A. Amazon CloudWatch
- B. AWS CloudTrail
- C. AWS X-Ray
- D. AWS Identity and Access Management (IAM)

Correct Answer: B

AWS CloudTrail allows AWS customers to **record API calls, sending log files to Amazon S3 buckets for storage.**

31) All AWS users have access to which AWS Trusted Advisor check?

- A. Core checks
- B. All checks
- C. Cost optimization checks
- D. Fault tolerance checks

Correct Answer: A

AWS Basic Support and AWS Developer Support customers can access core security checks

32) Which of the following is an example of security in the AWS Cloud under the AWS shared responsibility model?

- A. Managing edge locations
- B. Physical security
- C. Firewall configuration
- D. Global infrastructure

Correct Answer: C

The customer assumes responsibility and management of the guest operating system (including updates and security patches), other associated application software as well as the configuration of the AWS provided security group firewall.

33) Which documentation does AWS Artifact provide?

- A. Amazon EC2 terms and conditions
- B. AWS ISO certifications
- C. A history of a company's AWS spending
- D. A list of previous-generation Amazon EC2 instance types

Correct Answer: B

AWS Artifact provides on-demand downloads of AWS security and compliance documents, such as AWS ISO certifications, Payment Card Industry (PCI), and Service Organization Control (SOC) reports.

34) Which task requires using AWS account root user credentials?

- A. Viewing billing information
- B. Changing the AWS Support plan
- C. Starting and stopping Amazon EC2 instances
- D. Opening an AWS Support case

Correct Answer: B

There are only a few tasks that require you to use the root user:

- Change your account settings. This includes the account name, email address, root user password, and root user access keys. ...
- View certain tax invoices. ...
- Close your AWS account.
- Restore IAM user permissions. ...
- Change your AWS Support plan or Cancel your AWS Support plan. ...

36) Which of the following AWS services should a client utilize to audit the change management of AWS resources?

- A. AWS Config
- B. AWS Trusted Advisor
- C. Amazon CloudWatch
- D. Amazon Inspector

Correct Answer: A

AWS Config is a service that enables you to assess, audit, and evaluate the configurations of your AWS resources. Config continuously monitors and records your AWS resource configurations and allows you to automate the evaluation of recorded configurations against desired configurations. With Config, you can review changes in configurations and relationships between AWS resources, dive into detailed resource configuration histories, and determine your overall compliance against the configurations specified in your internal guidelines. This enables you to simplify compliance auditing, security analysis, change management, and operational troubleshooting.

37) What is raised when a business deploys web servers across several AWS Regions?

- A. Coupling
- B. Availability
- C. Security
- D. Durability

Correct Answer: B

Availability (also known as *service availability*) is both a commonly used metric to quantitatively measure resiliency, as well as a target resiliency objective.

•**Availability** is the percentage of time that a workload is available for use.

Available for use means that it performs its agreed function successfully when required.

38) Which of the following is a shared control between a client and AWS under the shared responsibility model?

- A. Physical controls
- B. Patch management
- C. Zone security
- D. Data center auditing

Correct Answer: B

Patch Management: AWS is responsible for patching and fixing flaws within the infrastructure, but customers are responsible for patching their guest OS and applications

39) What is the scope of a VPC within the AWS network?

- A. A VPC can span all Availability Zones globally.
- B. A VPC must span at least two subnets in each AWS Region.
- C. A VPC must span at least two edge locations in each AWS Region.
- D. A VPC can span all Availability Zones within an AWS Region.

Correct Answer: D

Amazon VPC lets you provision a logically isolated section of the Amazon Web Services (AWS) cloud where you can launch AWS resources in a virtual network that you define. You have complete control over your virtual networking environment, including selection of your own IP address ranges, creation of subnets, and configuration of route tables and network gateways.

40) Which of the following are components of an AWS Site-to-Site VPN connection? (Choose two.)

- A. AWS Storage Gateway
- B. Virtual private gateway
- C. NAT gateway
- D. Customer gateway
- E. Internet gateway

Correct Answer: BD

B. Virtual private gateway: A virtual private gateway is the Amazon VPC side of a VPN connection. It acts as the termination point for VPN connections.

D. Customer gateway: A customer gateway is the on-premises side of a VPN connection. It is a physical or software appliance that is connected to your on-premises network and is responsible for establishing the VPN connection to your VPC.

41) According to the AWS shared responsibility model, which job is shared between AWS and the customer?

- A. Physical and environmental controls
- B. Server hardware management and encryption
- C. Application security
- D. Patch management and configuration management

Correct Answer: D

42) How should an application be created to function on the AWS Cloud in accordance with best practices?

- A. Use tightly coupled components.
- B. Use loosely coupled components.
- C. Use infrequently coupled components.
- D. Use frequently coupled components.

Correct Answer: B

The more **loosely coupled** the application components are, the better they will scale.

43) Which cloud architectural concept is supported by a system that can scale in terms of users, traffic, or data quantity without sacrificing performance?

- A. Think parallel
- B. Implement elasticity
- C. Decouple your components
- D. Design for failure

Correct Answer: B

Elasticity - AWS Well-Architected Framework Elasticity The ability to acquire resources as you need them and release resources when you no longer need them. In the cloud, you want to do this automatically.

44) The following describes an application that spans various Availability Zones:

- A. being highly available
- B. having global reach
- C. using an economy of scale
- D. having elasticity

Correct Answer: A

An **Availability Zone (AZ)** is one or more discrete data centers with redundant power, networking, and connectivity in an **AWS Region**. AZs give customers the ability to operate production applications and databases that are more highly available, fault tolerant, and scalable than would be possible from a single data center.

45) A company needs to establish a connection between two VPCs. The VPCs are located in two different AWS Regions. The company wants to use the existing infrastructure of the VPCs for this connection. Which AWS service or feature can be used to establish this connection?

- A. AWS Client VPN
- B. VPC peering
- C. AWS Direct Connect
- D. VPC endpoints

Correct Answer: B

A VPC peering connection is a networking connection between two VPCs that enables you to route traffic between them using private IPv4 addresses or IPv6 addresses. Instances in either VPC can communicate with each other as if they are within the same network. You can create a VPC peering connection between your own VPCs, or with a VPC in another AWS account. The VPCs can be in different Regions (also known as an inter-Region VPC peering connection).

46) According to the AWS shared responsibility model, what responsibility does a customer have when using Amazon RDS to host a database?

- A. Manage connections to the database
- B. Install Microsoft SQL Server
- C. Design encryption-at-rest strategies
- D. Apply minor database patches

Correct Answer: A

Amazon RDS encryption is currently available for all database engines and storage types, except for SQL Server Express Edition. Amazon RDS can encrypt your Amazon RDS DB instances. Data that is encrypted at rest includes the underlying storage for DB instances, its automated backups, read replicas, and snapshots.

47) Which duty is the customer's responsibility while administering AWS Lambda functions under the AWS shared responsibility model?

- A. Creating versions of Lambda functions
- B. Maintaining server and operating systems
- C. Scaling Lambda resources according to demand
- D. Updating the Lambda runtime environment

Correct Answer: A

You can use versions to manage the deployment of your functions. For example, you can publish a new version of a function for beta testing without affecting users of the stable production version. Lambda creates a new version of your function each time that you publish the function. The new version is a copy of the unpublished version of the function.

48) Which of the following is not a duty of the client under the AWS shared responsibility model? (Select two.)

- A. Decommissioning of physical storage devices
- B. Security group and ACL configuration
- C. Patch management of an Amazon RDS instance operating system
- D. Controlling physical access to data centers
- E. Patch management of an Amazon EC2 instance operating system

Correct Answer: AD

AWS assets are centrally managed through an inventory management system that stores and tracks owner, location, status, maintenance, and descriptive information for AWS-owned assets. Following procurement, assets are scanned and tracked, and assets undergoing maintenance are checked and monitored for ownership, status, and resolution.

49) Which service or functionality does AWS Enterprise Support provide that other AWS Support subscriptions do not?

- A. AWS Trusted Advisor
- B. AWS Support case
- C. Concierge team
- D. Amazon Connect

Correct Answer: C

•AWS Support Concierge Included as part of the Enterprise Support plan, the Support Concierge Team are **AWS billing and account experts** that specialize in working with enterprise accounts.

50) What attributes of an AWS account can AWS Trusted Advisor monitor and advise on? (Select two.)

- A. Compliance with security best practices
- B. Application performance
- C. Network utilization
- D. Cost optimization
- E. Compliance status

Correct Answer: AD



51) Which feature enables Amazon EC2 instances to be more elastic in response to changing workload demand?

- A. Amazon GuardDuty
- B. AWS WAF
- C. AWS Shield
- D. Amazon Inspector

Correct Answer: C

AWS Shield is a managed Distributed Denial of Service (DDoS) protection service that safeguards applications running on AWS.

52) Which AWS service or feature acts as a firewall for Amazon EC2 instances?

- A. Network ACL
- B. Elastic network interface
- C. Amazon VPC
- D. Security group

Correct Answer: D

A security group is a virtual firewall that controls inbound and outbound traffic for an Amazon EC2 instance.

53) Which feature enables Amazon EC2 instances to be more elastic in response to changing workload demand?

- A. Virtualization Management
- B. Hardware management
- C. Encryption management
- D. Facilities management
- E. Firewall management

Correct Answer: A

•Virtual computing environments, known as *instances*

54) What is the unique advantage that Enterprise Support customers receive?

- A. Access to a Technical Project Manager
- B. Access to a Technical Account Manager
- C. Access to a Cloud Support Engineer
- D. Access to a Solutions Architect

Correct Answer: B

With Enterprise Support, you get 24x7 technical support from high-quality engineers, tools and technology to automatically manage health of your environment, consultative architectural guidance delivered in the context of your applications and use-cases, and a designated Technical Account Manager (TAM) to coordinate access to proactive / preventative programs and AWS subject matter experts. AWS Enterprise Support is recommended if you have business and/or mission critical workloads in AWS.

55) What are the advantages of AWS Cloud elasticity?

- A. Ensure web traffic is automatically spread across multiple AWS Regions.
- B. Minimize storage costs by automatically archiving log data.
- C. Enable AWS to automatically select the most cost-effective services.
- D. Automatically adjust the required compute capacity to maintain consistent performance.

Correct Answer: D

Elasticity

The ability to acquire resources as you need them and release resources when you no longer need them. In the cloud, you want to do this automatically.

56) What is Amazon CloudFront?

- A. A global content delivery network
- B. A web service to schedule regular data movement
- C. A development front-end to Amazon Web Services
- D. An encrypted endpoint to upload files to the cloud

Correct Answer: A

Amazon CloudFront is a global content delivery network (CDN) service that accelerates delivery of your websites, APIs, video content or other web assets through CDN caching. It integrates with other Amazon Web Services products to give developers and businesses an easy way to accelerate content to end users with no minimum usage commitments.

<https://aws.amazon.com/cloudfront/>

57) Which of the following are capabilities of the AWS Cost Management tools? (Select two.)

- A. Terminate all AWS resources automatically if budget thresholds are exceeded.
- B. Break down AWS costs by day, service, and linked AWS account.
- C. Create budgets and receive notifications if current or forecasted usage exceeds the budgets.
- D. Switch automatically to Reserved Instances or Spot Instances, whichever is most cost-effective.
- E. Move data stored in Amazon S3 to a more cost-effective storage class.

Correct Answer: BC

AWS offers the following tools for Cost Management:

- Billing Dashboard
- Cost Explorer
- Cost Usage and Report

58) How should an Amazon EC2 instance be granted access to an Amazon S3 bucket in accordance with security best practices?

- A. Hard code an IAM user's secret key and access key directly in the application, and upload the file.
- B. Store the IAM user's secret key and access key in a text file on the EC2 instance, read the keys, then upload the file.
- C. Have the EC2 instance assume a role to obtain the privileges to upload the file.
- D. Modify the S3 bucket policy so that any service can upload to it at any time.

Correct Answer: C

To connect to your S3 buckets from your EC2 instances, you must do the following:

- 1. Create an AWS Identity and Access Management (IAM) profile role that grants access to Amazon S3.
- 2. Attach the IAM instance profile to the instance.
- 3. Validate permissions on your S3 bucket.
- 4. Validate network connectivity from the EC2 instance to Amazon S3.
- 5. Validate access to S3 buckets.

59) A company has a single Amazon EC2 instance. The company wants to adopt a highly available architecture. What can the company do to meet this requirement?

- A. Scale vertically to a larger EC2 instance size.
- B. Scale horizontally across multiple Availability Zones.
- C. Purchase an EC2 Dedicated Instance.
- D. Change the EC2 instance family to a compute optimized instance.

Correct Answer: B

AWS Auto Scaling monitors your applications and automatically adjusts capacity to maintain steady, predictable performance at the lowest possible cost. Using AWS Auto Scaling, it's easy to setup application scaling for multiple resources across multiple services in minutes.

60) A company's on-premises application deployment cycle was 3-4 weeks. After migrating to the AWS Cloud, the company can deploy the application in 2-3 days. Which benefit has this company experienced by moving to the AWS Cloud?

- A. Elasticity
- B. Flexibility
- C. Agility
- D. Resilience

Correct Answer: C

In a cloud computing environment, new IT resources are only a click away, which means that you reduce the time to make those resources available to your developers from weeks to just minutes. This results in a dramatic increase in agility for the organization, since the cost and time it takes to experiment and develop is significantly lower.

61) Which of the following price options would result in the highest reduction after making an Amazon EC2 Dedicated Host reservation?

- A. No upfront payment
- B. Hourly on-demand payment
- C. Partial upfront payment
- D. All upfront payment

Correct Answer: D

With the **All Upfront option**, you pay for the entire Dedicated Host Reservation with one upfront payment. This option provides you with the largest discount compared to On-Demand pricing.

62) Which AWS services and/or features increase availability and mitigate the effect of failures while constructing a typical three-tier web application? (Select two.)

- A. AWS Auto Scaling for Amazon EC2 instances
- B. Amazon VPC subnet ACLs to check the health of a service
- C. Distributed resources across multiple Availability Zones
- D. AWS Server Migration Service (AWS SMS) to move Amazon EC2 instances into a different Region
- E. Distributed resources across multiple AWS points of presence

Correct Answer: AC

You can use Auto Scaling to help make sure that you are running the desired number of healthy EC2 instances across multiple Availability Zones. Auto Scaling can also automatically increase the number of EC2 instances during demand spikes to maintain performance and decrease capacity during less busy periods to optimize costs

63) Which of the following IT responsibilities may AWS relieve a company's IT team of? (Select two.)

- A. Patching database software
- B. Storage capacity planning
- C. Creating database schemas
- D. Setting up access controls for data
- E. Writing application code

Correct Answer: AB

AWS maintains the configuration of its infrastructure devices, but a customer is responsible for configuring their own guest operating systems, databases, and applications.

64) Which of the following is included in the AWS Trusted Advisor checks? (Select two.)

- A. Information on Amazon S3 bucket permissions
- B. AWS service outages
- C. Multi-factor authentication enabled on the AWS account root user
- D. Available software patches
- E. Number of users in the account

Correct Answer: AC

Trusted Advisor can help improve the security of your AWS environment by suggesting foundational security best practices curated by security experts. Examples include identifying RDS security group access risk, exposed access keys, and unnecessary S3 bucket permissions.

65) What is AWS accountable for under the AWS shared responsibility model?

- A. Configuring Amazon VPC
- B. Managing application code
- C. Maintaining application traffic
- D. Managing the network infrastructure

Correct Answer: D

AWS is responsible for protecting the infrastructure that runs all of the services offered in the AWS Cloud. This infrastructure is composed of the hardware, software, networking, and facilities that run AWS Cloud services.

66) Which of the following tasks may be used to improve Amazon EC2 costs? (Select two.)

- A. Implementing Auto Scaling groups to add and remove instances based on demand.
- B. Creating a policy to restrict IAM users from creating new instances.
- C. Setting a budget to limit spending on EC2 instances using AWS Budgets.
- D. Purchasing Reserved Instances.
- E. Adding EC2 instances to a second AWS Region that is geographically close to the end users.

Correct Answer: AD

Take advantage of unused EC2 capacity for up to 90% off On-Demand pricing. Spot Instances are perfect for fault-tolerant, stateless applications like containers, big data, AI/ML and more.

Improve application resiliency and availability by automatically adding or removing EC2 instances and by mixing multiple purchase options and instance types. EC2 Auto Scaling ensures you are not over or under-provisioning compute to match application demand.

67) Which of the following are included in AWS Enterprise Support? (Choose two.)

- A. AWS technical account manager (TAM)
- B. AWS partner-led support
- C. AWS Professional Services
- D. Support of third-party software integration to AWS
- E. 5-minute response time for critical issues

Correct Answer: AD

Designated point of contact

A Technical Account Manager (TAM) is your designated technical point of contact who helps you onboard, provides advocacy and guidance to help plan and build solutions using best practices, coordinates access to subject matter experts, assists with case management, presents insights and recommendations on your AWS spend, workload optimization, and event management, and proactively keeps your AWS environment healthy.

Third-party software support

Guidance, configuration, and troubleshooting of AWS interoperability with many common operating systems, platforms, and application stack components.

68) A company wants to limit its employees' AWS access to a portfolio of predefined AWS resources. Which AWS solution should the company use to meet this requirement?

- A. AWS Config
- B. AWS software development kits (SDKs)
- C. AWS Service Catalog
- D. AWS AppSync

Correct Answer: C

AWS Service Catalog allows organizations to create and manage catalogs of IT services that are approved for use on AWS. These IT services can include everything from virtual machine images, servers, software, and databases to complete multi-tier application architectures.

69) Which of the following is a customer duty under the AWS shared responsibility model?

- A. Installing security patches for the Xen and KVM hypervisors
- B. Installing operating system patches for Amazon DynamoDB
- C. Installing operating system security patches for Amazon EC2 database instances
- D. Installing operating system security patches for Amazon RDS database instances

Correct Answer: C

Customers that deploy an Amazon EC2 instance are responsible for management of the guest operating system (including updates and security patches), any application software or utilities installed by the customer on the instances, and the configuration of the AWS-provided firewall (called a security group) on each instance.

70) A firm wishes to develop a new line of business application.

Which design concepts should be applied in accordance with the AWS Well-Architected Framework? (Select two.)

- A. Consolidate multiple AWS accounts into a single account.
- B. Buy and host hardware in the AWS Cloud.
- C. Decouple the AWS Cloud architecture to break up monolithic deployments.
- D. Move on-premises network hardware to VPCs.
- E. Design elasticity into the AWS Cloud design.

Correct Answer: CE

AWS Well-Architected helps cloud architects build secure, high-performing, resilient, and efficient infrastructure for a variety of applications and workloads.

71) Which scenarios should a user report to the AWS Abuse team? (Select two.)

- A. A DDoS attack is being made on an AWS resource.
- B. A SQL injection attack is being made from an IP address that is not an AWS address.
- C. AWS resources are being used to host objectionable or illegal content.
- D. A company's resources are being used in a way that is inconsistent with corporate policy.
- E. A company is receiving HTTPS requests on a web server that is serving HTTP.

Correct Answer: AC

The AWS Trust & Safety team can assist you when AWS resources are used to engage in the following types of abusive behavior:

- Spam
- Port scanning
- Denial-of-service (DoS) attacks
- Intrusion attempts
- Hosting prohibited content
- Distributing malware

72) Which of the following is the customer's obligation under the AWS shared responsibility model?

- A. Patching guest OS and applications
- B. Patching and fixing flaws in the infrastructure
- C. Physical and environmental controls
- D. Configuration of AWS infrastructure devices

Correct Answer: A

The customer assumes responsibility and management of the guest operating system (including updates and security patches), other associated application software as well as the configuration of the AWS provided security group firewall. Customers should carefully consider the services they choose as their responsibilities vary depending on the services used, the integration of those services into their IT environment, and applicable laws and regulations. The nature of this shared responsibility also provides the flexibility and customer control that permits the deployment.

73) An online company was running a workload on premises and was struggling to launch new products and features. After migrating the workload to AWS, the company can quickly launch products and features and can scale its infrastructure as required.

Which AWS Cloud value proposition does this scenario describe?

- A. Business agility
- B. High availability
- C. Security
- D. Centralized auditing

Correct Answer: A

Increase speed and agility – In a cloud computing environment, new IT resources are only a click away, which means that you reduce the time to make those resources available to your developers from weeks to just minutes. This results in a dramatic increase in agility for the organization, since the cost and time it takes to experiment and develop is significantly lower.

74) Which of the following are advantages of the AWS Cloud? (Choose two.)

- A. AWS management of user-owned infrastructure
- B. Ability to quickly change required capacity
- C. High economies of scale
- D. Increased deployment time to market
- E. Increased fixed expenses

Correct Answer: BC

More customers lead to higher **AWS** usage which leads to more infrastructure.

The **economies of scale** effect occurs because costs can now be spread over a larger number of customers. The infrastructure becomes cheaper, prices can be reduced and new customers benefit from the lower prices.

75) AWS has the ability to achieve lower pay-as-you-go pricing by aggregating usage across hundreds of thousands of users. This describes which advantage of the AWS Cloud?

- A. Launch globally in minutes
- B. Increase speed and agility
- C. High economies of scale
- D. No guessing about compute capacity

Correct Answer: C

•**Benefit from massive economies of scale** – By using cloud computing, you can achieve a lower variable cost than you can get on your own. Because usage from hundreds of thousands of customers is aggregated in the cloud, providers such as AWS can achieve higher economies of scale, which translates into lower pay as-you-go prices.

76) Which AWS serverless platform services are included?

- A. Amazon EC2, Amazon S3, Amazon Athena
- B. Amazon Kinesis, Amazon SQS, Amazon EMR
- C. AWS Step Functions, Amazon DynamoDB, Amazon SNS
- D. Amazon Athena, Amazon Cognito, Amazon EC2

Correct Answer: C

AWS provides a set of fully managed services that you can use to build and run serverless applications. Serverless applications don't require provisioning, maintaining, and administering servers for backend components such as compute, databases, storage, stream processing, message queueing, and more. You also no longer need to worry about ensuring application fault tolerance and availability. Instead, AWS handles all of these capabilities for you. Serverless platform includes: AWS lambda, Amazon S3, DynamoDB, API gateway, Amazon SNS, AWS step functions, Amazon kinesis and developing tools and services.

77) According to the AWS shared responsibility model, which of the following is the customer's obligation?

- A. Patching underlying infrastructure
- B. Physical security
- C. Patching Amazon EC2 instances
- D. Patching network infrastructure

Correct Answer: C

To ensure the best results from running Windows on Amazon EC2, AWS recommend that you perform the following best practices.

- Update drivers
- Use the latest Windows AMIs
- Security
- Storage
- Resource management
- Backup and recovery
- Networking

78) Which of the following is the customer's obligation under the shared responsibility model?

- A. Ensuring that disk drives are wiped after use.
- B. Ensuring that firmware is updated on hardware devices.
- C. Ensuring that data is encrypted at rest.
- D. Ensuring that network cables are category six or higher.

Correct Answer: C

In short AWS responsibility **"Security of the Cloud"** and Customer responsibility **"Security in the Cloud"**. In the cases where you store data in RDS and S3, the customer takes full responsibility for encryption. However both RDS and S3 provides you necessary tools to encrypt the data at rest.

79) What is the AWS Cloud's value proposition?

- A. AWS is responsible for security in the AWS Cloud
- B. No long-term contract is required
- C. Provision new servers in days
- D. AWS manages user applications in the AWS Cloud

Correct Answer: B

All paid **AWS Support** plans are billed monthly, with no long-term contracts. AWS Support fees are calculated on a per-account basis for Business and Developer Support plans. For Enterprise Support, you are billed based on the aggregate monthly AWS charges for all your account IDs subscribed to Enterprise Support.

80) A company has a database server that is always running. The company hosts the server on Amazon EC2 instances. The instance sizes are suitable for the workload. The workload will run for 1 year. Which EC2 instance purchasing option will meet these requirements MOST cost-effectively?

- A. Standard Reserved Instances
- B. On-Demand Instances
- C. Spot Instances
- D. Convertible Reserved Instances

Correct Answer: A

Standard Reserved Instance (RI) AWS offers the standard RI through the **one-year or the three-year term purchase plans**. AWS Standard RIs allow users to modify the scope, type of networking, Availability Zone and the type of instance (as long as the new instance is within the same instance type).

81) Which tasks are the responsibility of AWS, according to the AWS shared responsibility model? (Choose two.)

- A. Patch the Amazon EC2 guest operating system.
- B. Upgrade the firmware of the network infrastructure.
- C. Apply password rotation for IAM users.
- D. Maintain the physical security of edge locations.
- E. Maintain least privilege access to the root user account.

Correct Answer: BD

AWS responsibility "Security of the Cloud" - AWS is responsible for protecting the infrastructure that runs all of the services offered in the AWS Cloud. This infrastructure is composed of the hardware, software, networking, and facilities that run AWS Cloud services.

82) Which of the following are features of network ACLs as they are used in the AWS Cloud? (Choose two.)

- A. They are stateless.
- B. They are stateful.
- C. They evaluate all rules before allowing traffic.
- D. They process rules in order, starting with the lowest numbered rule, when deciding whether to allow traffic.
- E. They operate at the instance level.

Correct Answer: AD

- Network ACLs are stateless, which means that responses to allowed inbound traffic are subject to the rules for outbound traffic (and vice versa).
- Evaluate the network ACL rules when traffic enters and leaves the subnet, not as it is routed within a subnet.

83) A company has designed its AWS Cloud infrastructure to run its workloads effectively. The company also has protocols in place to continuously improve supporting processes. Which pillar of the AWS Well-Architected Framework does this scenario represent?

- A. Security
- B. Performance efficiency
- C. Cost optimization
- D. Operational excellence

Correct Answer: D

The Operational Excellence pillar includes the ability to support development and run workloads effectively, gain insight into their operations, and to continuously improve supporting processes and procedures to deliver business value.

84) Which of the following describes elasticity in the AWS Cloud? (Select two.)

- A. How quickly an Amazon EC2 instance can be restarted
- B. The ability to rightsize resources as demand shifts
- C. The maximum amount of RAM an Amazon EC2 instance can use
- D. The pay-as-you-go billing model
- E. How easily resources can be produced when they are needed

Correct Answer: BE

In **AWS**, the process of getting the resources dynamically when you actually require them and then release the resources when you are done and do not need them is known as **elasticity**. In another way, growing or shrinking the resources dynamically when needed is known as **Elasticity**.

85) After studying the AWS Cost and Usage report in the AWS Management Console, a Cloud Practitioner detects a billing problem.

Which course of action should be done to rectify this?

- A. Open a detailed case related to billing and submit it to AWS Support for help.
- B. Upload data describing the issue to a new object in a private Amazon S3 bucket.
- C. Create a pricing application and deploy it to a right-sized Amazon EC2 instance for more information.
- D. Proceed with creating a new dashboard in Amazon QuickSight.

Correct Answer: A

If you have an account or billing related question, do the following:

1. Sign in to your AWS account.
2. Open the AWS Support Center.
3. Choose Create case.
4. Select Account and billing support.
5. Enter the required information in the Case details section.

86) Which action, in terms of AWS Cloud architecture, is compatible with the concept of least privilege?

- A. Allow users the minimum access that is needed to do a task.
- B. Prevent managers from accessing important source code.
- C. Provide only the permissions that are needed for users to do their jobs in the current month.
- D. Assign permissions that are based on job titles.

Correct Answer: A

Least privilege is a principle of granting only the permissions required to complete a task. Least privilege is also one of many [Amazon Web Services \(AWS\) Well-Architected](#) best practices that can help you build securely in the cloud. For example, if you have an [Amazon Elastic Compute Cloud \(Amazon EC2\)](#) instance that needs to access an [Amazon Simple Storage Service \(Amazon S3\)](#) bucket to get configuration data, you should only allow read access to the specific S3 bucket that contains the relevant data.

87) Which of the following are customer duties under the AWS shared responsibility model? (Select two.)

- A. Setting up server-side encryption on an Amazon S3 bucket
- B. Amazon RDS instance patching
- C. Network and firewall configurations
- D. Physical security of data center facilities
- E. Compute capacity availability

Correct Answer: AC

Customers are responsible for **managing their data** (including encryption options), classifying their assets, and using IAM tools to apply the appropriate permissions. This customer/AWS shared responsibility model also extends to IT controls.

88) A company needs to graphically visualize AWS billing and usage over time. The company also needs information about its AWS monthly costs.

Which AWS Billing and Cost Management tool provides this data in a graphical format?

- A. AWS Bills
- B. Cost Explorer
- C. AWS Cost and Usage Report
- D. AWS Budgets

Correct Answer: B

AWS Cost Explorer is a graphical cost analysis tool available from the Billing and Cost Management console's menu. It uses the data from each month's spending for up to the last thirteen months and can forecast a spend pattern for the next three months. The spend graphs can be tailored for certain time periods as well.

89) A company wants to run production workloads on AWS. The company needs concierge service, a designated AWS technical account manager (TAM), and technical support that is available 24 hours a day, 7 days a week.

Which AWS Support plan will meet these requirements?

- A. AWS Basic Support
- B. AWS Enterprise Support
- C. AWS Business Support
- D. AWS Developer Support

Correct Answer: B

AWS Enterprise Support provides you with concierge-like service where the main focus is helping you achieve your outcomes and find success in the cloud.

With Enterprise Support, you get 24x7 technical support from high-quality engineers, tools and technology to automatically manage health of your environment, consultative architectural guidance delivered in the context of your applications and use-cases, and a designated Technical Account Manager (TAM) to coordinate access to proactive / preventative programs and AWS subject matter experts. AWS Enterprise Support is recommended if you have business and/or mission critical workloads in AWS.

90) Which of the following tasks is the AWS customer's duty under the shared responsibility model? (Select two.)

- A. Ensuring that application data is encrypted at rest
- B. Ensuring that AWS NTP servers are set to the correct time
- C. Ensuring that users have received security training in the use of AWS services
- D. Ensuring that access to data centers is restricted
- E. Ensuring that hardware is disposed of properly

Correct Answer: AC

Cloud security operates on a **shared responsibility model** that defines the balance of security responsibilities of the cloud service provider (CSP) and of the cloud **customer**. Generally, it stipulates that the CSP is **responsible** for the security of the cloud and the **customer** is **responsible** for the security in the cloud.

91) Before transferring an environment to the AWS Cloud, a cloud practitioner must receive AWS compliance reports.

How are these reports produced?

- A. Contact the AWS Compliance team
- B. Download the reports from AWS Artifact
- C. Open a case with AWS Support
- D. Generate the reports with Amazon Macie

Correct Answer: B

AWS Artifact provides a central resource for AWS security and compliance reports. The artifacts available in AWS Artifact include Service Organization Control (SOC) reports, Payment Card Industry (PCI) reports, and certifications from accreditation bodies that validate the implementation and operating effectiveness of AWS security controls.

92) Which change management actions, according to the AWS Well-Architected Framework, should be followed to ensure AWS Cloud reliability? (Select two.)

- A. Use AWS Config to generate an inventory of AWS resources
- B. Use service limits to prevent users from creating or making changes to AWS resources
- C. Use AWS CloudTrail to record AWS API calls into an auditable log file
- D. Use AWS Certificate Manager to whitelist approved AWS resources and services
- E. Use Amazon GuardDuty to validate configuration changes made to AWS resources

Correct Answer: AC

AWS Config continually assesses, audits, and evaluates the configurations and relationships of your resources on AWS, on premises, and on other clouds.

AWS CloudTrail is an AWS service that helps you enable operational and risk auditing, governance, and compliance of your AWS account. Actions taken by a user, role, or an AWS service are recorded as events in CloudTrail. Events include actions taken in the AWS Management Console, AWS Command Line Interface, and AWS SDKs and APIs.

93) What may be associated to an Amazon EC2 instance through AWS Identity and Access Management (IAM) to initiate service requests?

- A. Group
- B. Role
- C. Policy
- D. Access key

Correct Answer: B

IAM role management

Q: What is an IAM role?

An IAM role is an IAM entity that defines a set of [permissions](#) for making AWS service requests. IAM roles are not associated with a specific user or group. Instead, trusted entities assume roles, such as IAM users, applications, or AWS services such as EC2.

Q: What problems do IAM roles solve?

IAM roles allow you to delegate access with defined permissions to trusted entities without having to share long-term access keys. You can use IAM roles to delegate access to IAM users managed within your account, to IAM users under a different AWS account, or to an AWS service such as EC2.

Q: How do I get started with IAM roles?

You create a role in a way similar to how you create a user—name the role and attach a policy to it. For details, see [Creating IAM Roles](#).

94) Which of the following actions is the customer's responsibility under the AWS shared responsibility model? (Select two.)

- A. Patching operating system components for Amazon Relational Database Server (Amazon RDS)
- B. Encrypting data on the client-side
- C. Training the data center staff
- D. Configuring Network Access Control Lists (ACL)
- E. Maintaining environmental controls within a data center

Correct Answer: BD

Customer responsibility will be determined by the AWS Cloud services that a customer selects. This determines the amount of configuration work the customer must perform as part of their security responsibilities. For example, a service such as Amazon Elastic Compute Cloud (Amazon EC2) is categorized as Infrastructure as a Service (IaaS) and, as such, requires the customer to perform all of the necessary security configuration and management tasks.

95) Which architecture design principle describes the need to isolate failures between dependent components in the AWS Cloud?

- A. Use a monolithic design.
- B. Design for automation.
- C. Design for single points of failure.
- D. Loosely couple components.

Correct Answer: D

In the cloud practitioner course states that in a microservices approach, application components are loosely coupled. In this case, if a single component fails, the other components continue to work because they are communicating with each other. The loose coupling prevents the entire application from failing.

1) A company plans to use an Amazon Snowball Edge device to transfer files to the AWS Cloud. Which activities related to a Snowball Edge device are available to the company at no cost?

- A. Use of the Snowball Edge appliance for a 10-day period
- B. The transfer of data out of Amazon S3 and to the Snowball Edge appliance
- C. The transfer of data from the Snowball Edge appliance into Amazon S3
- D. Daily use of the Snowball Edge appliance after 10 days

Correct Answer: C

Data transfer IN to Amazon S3 is \$0.00 per GB (except for small files as explained below). Data transfer OUT of Amazon S3 is priced by region.

<https://aws.amazon.com/snowball/pricing/>

2) A company has deployed applications on Amazon EC2 instances. The company needs to assess application vulnerabilities and must identify infrastructure deployments that do not meet best practices. Which AWS service can the company use to meet these requirements?

- A. AWS Trusted Advisor
- B. Amazon Inspector
- C. AWS Config
- D. Amazon GuardDuty

Correct Answer: B

Amazon Inspector is an automated vulnerability management service that continually scans AWS workloads for software vulnerabilities and unintended network exposure. The key word here is "vulnerabilities".

3) A company has a centralized group of users with large file storage requirements that have exceeded the space available on premises. The company wants to extend its file storage capabilities for this group while retaining the performance benefit of sharing content locally.

What is the MOST operationally efficient AWS solution for this scenario?

- A. Create an Amazon S3 bucket for each user. Mount each bucket by using an S3 file system mounting utility.
- B. Configure and deploy an AWS Storage Gateway file gateway. Connect each user's workstation to the file gateway.
- C. Move each user's working environment to Amazon WorkSpaces. Set up an Amazon WorkDocs account for each user.
- D. Deploy an Amazon EC2 instance and attach an Amazon Elastic Block Store (Amazon EBS) Provisioned IOPS volume. Share the EBS volume directly with the users.

Correct Answer: B

4) According to security best practices, how should an Amazon EC2 instance be given access to an Amazon S3 bucket?

- A. Hard code an IAM user's secret key and access key directly in the application, and upload the file.
- B. Store the IAM user's secret key and access key in a text file on the EC2 instance, read the keys, then upload the file.
- C. Have the EC2 instance assume a role to obtain the privileges to upload the file.
- D. Modify the S3 bucket policy so that any service can upload to it at any time.

Correct Answer: C

<https://repost.aws/knowledge-center/ec2-instance-access-s3-bucket#>

5) Which option is a customer responsibility when using Amazon DynamoDB under the AWS Shared Responsibility Model?

- A. Physical security of DynamoDB
- B. Patching of DynamoDB
- C. Access to DynamoDB tables
- D. Encryption of data at rest in DynamoDB

Correct Answer: C

Under the AWS Shared Responsibility Model, controlling and managing access to AWS services, including Amazon DynamoDB tables, is a customer responsibility. While AWS takes care of the physical infrastructure, patching, and encryption of data at rest in DynamoDB, customers are responsible for setting up proper access controls, authentication, and authorization to protect their data and resources

6) Which option is a perspective that includes foundational capabilities of the AWS Cloud Adoption Framework (AWS CAF)?

- A. Sustainability
- B. Performance efficiency
- C. Governance
- D. Reliability

Correct Answer: C

The 6 AWS CAF perspectives are: Business, People, Governance, Platform, Security and Operations.

7) A company is running and managing its own Docker environment on Amazon EC2 instances. The company wants an alternative to help manage cluster size, scheduling, and environment maintenance. Which AWS service meets these requirements?

- A. AWS Lambda
- B. Amazon RDS
- C. AWS Fargate
- D. Amazon Athena

Correct Answer: C

AWS Fargate is a serverless, pay-as-you-go compute engine that lets you focus on building applications without managing servers. AWS Fargate is compatible with both Amazon Elastic Container Service (Amazon ECS) and Amazon Elastic Kubernetes Service (Amazon EKS).

<https://aws.amazon.com/fargate/>

8) A company wants to run a NoSQL database on Amazon EC2 instances. Which task is the responsibility of AWS in this scenario?

- A. Update the guest operating system of the EC2 instances.
- B. Maintain high availability at the database layer.
- C. Patch the physical infrastructure that hosts the EC2 instances.
- D. Configure the security group firewall.

Correct Answer: C

Patch the physical infrastructure that hosts the EC2 instances. Guest operating system is always responsibility of customer and host of AWS.

<https://aws.amazon.com/compliance/shared-responsibility-model/>

9) Which AWS services or tools can identify rightsizing opportunities for Amazon EC2 instances? (Choose two.)

- A. AWS Cost Explorer
- B. AWS Billing Conductor
- C. Amazon CodeGuru
- D. Amazon SageMaker
- E. AWS Compute Optimizer

Correct Answer: AE

<https://docs.aws.amazon.com/cost-management/latest/userguide/ce-rightsizing.html>
<https://aws.amazon.com/compute-optimizer/>

10) Which of the following are benefits of using AWS Trusted Advisor? (Choose two.)

- A. Providing high-performance container orchestration
- B. Creating and rotating encryption keys
- C. Detecting underutilized resources to save costs
- D. Improving security by proactively monitoring the AWS environment
- E. Implementing enforced tagging across AWS resources

Correct Answer: CD

Benefits of Trusted Advisor:

- Cost optimization - Trusted Advisor can help you save cost with actionable recommendations by analyzing usage, configuration and spend.
- Performance - Trusted Advisor can help improve the performance of your services with actionable recommendations by analyzing usage and configuration.
- Security - Trusted Advisor can help improve the security of your AWS environment by suggesting foundational security best practices curated by security experts.
- Fault tolerance - Trusted Advisor can help improve the reliability of your services.
- Service quotas - Service quotas are the maximum number of resources that you can create in an AWS account.

11) Which of the following is an advantage that users experience when they move on-premises workloads to the AWS Cloud?

- A. Elimination of expenses for running and maintaining data centers
- B. Price discounts that are identical to discounts from hardware providers
- C. Distribution of all operational controls to AWS
- D. Elimination of operational expenses

Correct Answer: A

<https://docs.aws.amazon.com/whitepapers/latest/aws-overview/six-advantages-of-cloud-computing.html>

12) A company wants to manage deployed IT services and govern its infrastructure as code (IaC) templates. Which AWS service will meet this requirement?

- A. AWS Resource Explorer
- B. AWS Service Catalog
- C. AWS Organizations
- D. AWS Systems Manager

Correct Answer: B

AWS Service Catalog lets you centrally manage your cloud resources to achieve governance at scale of your infrastructure as code (IaC) templates, written in CloudFormation or Terraform configurations. With AWS Service Catalog, you can meet your compliance requirements while making sure your customers can quickly deploy the cloud resources they need.

<https://aws.amazon.com/servicecatalog/>

13) Which AWS service or tool helps users visualize, understand, and manage spending and usage over time?

- A. AWS Organizations
- B. AWS Pricing Calculator
- C. AWS Cost Explorer
- D. AWS Service Catalog

Correct Answer: C

AWS Cost Explorer has an easy-to-use interface that lets you visualize, understand, and manage your AWS costs and usage over time. Get started quickly by creating custom reports that analyze cost and usage data. Analyze your data at a high level (for example, total costs and usage across all accounts), or dive deeper into your cost and usage data to identify trends, pinpoint cost drivers, and detect anomalies.

<https://aws.amazon.com/aws-cost-management/aws-cost-explorer/>

14) A company is using a central data platform to manage multiple types of data for its customers. The company wants to use AWS services to discover, transform, and visualize the data.

Which combination of AWS services should the company use to meet these requirements? (Choose two.)

- A. AWS Glue
- B. Amazon Elastic File System (Amazon EFS)
- C. Amazon Redshift
- D. Amazon QuickSight
- E. Amazon Quantum Ledger Database (Amazon QLDB)

Correct Answer: AD

AWS Glue is a serverless data integration service that makes it easier to discover, prepare, move, and integrate data from multiple sources for analytics, machine learning (ML), and application development. <https://aws.amazon.com/glue/>

Amazon QuickSight powers data-driven organizations with unified business intelligence (BI) at hyperscale. With QuickSight, all users can meet varying analytic needs from the same source of truth through modern interactive dashboards, paginated reports, embedded analytics, and natural language queries. <https://aws.amazon.com/quicksight/>

15) A global company wants to migrate its third-party applications to the AWS Cloud. The company wants help from a global team of experts to complete the migration faster and more reliably in accordance with AWS internal best practices.

Which AWS service or resource will meet these requirements?

- A. AWS Support
- B. AWS Professional Services
- C. AWS Launch Wizard
- D. AWS Managed Services (AMS)

Correct Answer: B

AWS Partner Network (APN) Consulting Partners help customers design, architect, build, migrate, and manage workloads and applications on Amazon Web Services.

https://d1.awsstatic.com/partner-network/APN_Consulting-Benefits_Brochure-Digital.pdf

16) An e-learning platform needs to run an application for 2 months each year. The application will be deployed on Amazon EC2 instances. Any application downtime during those 2 months must be avoided. Which EC2 purchasing option will meet these requirements MOST cost-effectively?

- A. Reserved Instances
- B. Dedicated Hosts
- C. Spot Instances
- D. On-Demand Instances

Correct Answer: D

"On-Demand Instances are recommended for: • Users that prefer the low cost and flexibility of EC2 without any upfront payment or long-term commitment • Applications with short-term, spiky, or unpredictable workloads that cannot be interrupted • Applications being developed or tested on EC2 for the first time"

Reference: <https://aws.amazon.com/ec2/pricing/>

17) A developer wants to deploy an application quickly on AWS without manually creating the required resources. Which AWS service will meet these requirements?

- A. Amazon EC2
- B. AWS Elastic Beanstalk
- C. AWS CodeBuild
- D. Amazon Personalize

Correct Answer: B

With Elastic Beanstalk, you can quickly deploy and manage applications in the AWS Cloud without having to learn about the infrastructure that runs those applications. Elastic Beanstalk reduces management complexity without restricting choice or control. You simply upload your application, and Elastic Beanstalk automatically handles the details of capacity provisioning, load balancing, scaling, and application health monitoring.

18) A company is storing sensitive customer data in an Amazon S3 bucket. The company wants to protect the data from accidental deletion or overwriting.

Which S3 feature should the company use to meet these requirements?

- A. S3 Lifecycle rules
- B. S3 Versioning
- C. S3 bucket policies
- D. S3 server-side encryption

Correct Answer: B

Versioning in Amazon S3 is a means of keeping multiple variants of an object in the same bucket. You can use the S3 Versioning feature to preserve, retrieve, and restore every version of every object stored in your buckets. Versioning-enabled buckets can help you recover objects from accidental deletion or overwrite. For example, if you delete an object, Amazon S3 inserts a delete marker instead of removing the object permanently.

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/Versioning.html>

19) Which AWS service provides the ability to manage infrastructure as code?

- A. AWS CodePipeline
- B. AWS CodeDeploy
- C. AWS Direct Connect
- D. AWS CloudFormation

Correct Answer: D

AWS CloudFormation lets you model, provision, and manage AWS and third-party resources by treating infrastructure as code.

<https://aws.amazon.com/cloudformation/>

20) An online gaming company needs to choose a purchasing option to run its Amazon EC2 instances for 1 year. The web traffic is consistent, and any increases in traffic are predictable. The EC2 instances must be online and available without any disruption.

Which EC2 instance purchasing option will meet these requirements MOST cost-effectively?

- A. On-Demand Instances
- B. Reserved Instances
- C. Spot Instances
- D. Spot Fleet

Correct Answer: B

Amazon EC2 Reserved Instances (RI) provide a significant discount (up to 72%) compared to On-Demand pricing and provide a capacity reservation when used in a specific Availability Zone. Terms (avg. discount off On-Demand) - 1yr (40%), 3yr (60%)
<https://aws.amazon.com/ec2/pricing/reserved-instances/>

21) Which AWS service or feature allows a user to establish a dedicated network connection between a company's on-premises data center and the AWS Cloud?

- A. AWS Direct Connect
- B. VPC peering
- C. AWS VPN
- D. Amazon Route 53

Correct Answer: A

Create a dedicated network connection to AWS. The AWS Direct Connect cloud service is the shortest path to your AWS resources. While in transit, your network traffic remains on the AWS global network and never touches the public internet.

<https://aws.amazon.com/directconnect/>

22) Which option is a physical location of the AWS global infrastructure?

- A. AWS DataSync
- B. AWS Region
- C. Amazon Connect
- D. AWS Organizations

Correct Answer: B

"AWS has the concept of a Region, which is a physical location around the world where we cluster data centers"

https://aws.amazon.com/about-aws/global-infrastructure/regions_az/

23) A company wants to protect its AWS Cloud information, systems, and assets while performing risk assessment and mitigation tasks.

Which pillar of the AWS Well-Architected Framework is supported by these goals?

- A. Reliability
- B. Security
- C. Operational excellence
- D. Performance efficiency

Correct Answer: B

This is conclusion of security pillar "help you build and operate architectures that protect information, systems, and assets while delivering business value"

<https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/conclusion.html>

24) What is the purpose of having an internet gateway within a VPC?

- A. To create a VPN connection to the VPC
- B. To allow communication between the VPC and the internet
- C. To impose bandwidth constraints on internet traffic
- D. To load balance traffic from the internet across Amazon EC2 instances

Correct Answer: B

"An internet gateway is a horizontally scaled, redundant, and highly available VPC component that allows communication between your VPC and the internet."

Reference: https://docs.aws.amazon.com/vpc/latest/userguide/VPC_Internet_Gateway.html

25) A company is running a monolithic on-premises application that does not scale and is difficult to maintain. The company has a plan to migrate the application to AWS and divide the application into microservices. Which best practice of the AWS Well-Architected Framework is the company following with this plan?

- A. Integrate functional testing as part of AWS deployment.
- B. Use automation to deploy changes.
- C. Deploy the application to multiple locations.
- D. Implement loosely coupled dependencies.

Correct Answer: D

Dependencies such as queuing systems, streaming systems, workflows, and load balancers are loosely coupled. Loose coupling helps isolate behavior of a component from other components that depend on it, increasing resiliency and agility.

https://docs.aws.amazon.com/wellarchitected/latest/framework/rel_prevent_interaction_failure_loosely_coupled_system.html

26) A company has an AWS account. The company wants to audit its password and access key rotation details for compliance purposes.

Which AWS service or tool will meet this requirement?

- A. IAM Access Analyzer
- B. AWS Artifact
- C. IAM credential report
- D. AWS Audit Manager

Correct Answer: C

You can use credential reports to assist in your auditing and compliance efforts. You can use the report to audit the effects of credential lifecycle requirements, such as password and access key updates.

https://docs.aws.amazon.com/IAM/latest/UserGuide/id_credentials_getting-report.html

27) A company wants to receive a notification when a specific AWS cost threshold is reached. Which AWS services or tools can the company use to meet this requirement? (Choose two.)

- A. Amazon Simple Queue Service (Amazon SQS)
- B. AWS Budgets
- C. Cost Explorer
- D. Amazon CloudWatch
- E. AWS Cost and Usage Report

Correct Answer: BD

28) Which AWS service or resource provides answers to the most frequently asked security-related questions that AWS receives from its users?

- A. AWS Artifact
- B. Amazon Connect
- C. AWS Chatbot
- D. AWS Knowledge Center

Correct Answer: D

The AWS Knowledge Center is a comprehensive resource that provides answers to the most frequently asked security-related questions that AWS receives from its users. It is a central repository of security information and guidance, covering a wide range of topics

29) Which tasks are customer responsibilities, according to the AWS shared responsibility model? (Choose two.)

- A. Configure the AWS provided security group firewall.
- B. Classify company assets in the AWS Cloud.
- C. Determine which Availability Zones to use for Amazon S3 buckets.
- D. Patch or upgrade Amazon DynamoDB.
- E. Select Amazon EC2 instances to run AWS Lambda on.

Correct Answer: AB

30) Which of the following are pillars of the AWS Well-Architected Framework? (Choose two.)

- A. Availability
- B. Reliability
- C. Scalability
- D. Responsive design
- E. Operational excellence

Correct Answer: BE

AWS Well-Architected helps cloud architects build secure, high-performing, resilient, and efficient infrastructure for a variety of applications and workloads.

Built around six pillars—operational excellence, security, reliability, performance efficiency, cost optimization, and sustainability.

31) Which AWS service or feature is used to send both text and email messages from distributed applications?

- A. Amazon Simple Notification Service (Amazon SNS)
- B. Amazon Simple Email Service (Amazon SES)
- C. Amazon CloudWatch alerts
- D. Amazon Simple Queue Service (Amazon SQS)

Correct Answer: A

Amazon Simple Notification Service (Amazon SNS) sends notifications two ways, A2A and A2P. A2A provides high-throughput, push-based, many-to-many messaging between distributed systems, microservices, and event-driven serverless applications. These applications include Amazon Simple Queue Service (SQS), Amazon Kinesis Data Firehose, AWS Lambda, and other HTTPS endpoints. A2P functionality lets you send messages to your customers with SMS texts, push notifications, and email.

<https://aws.amazon.com/sns/>

32) A user needs programmatic access to AWS resources through the AWS CLI or the AWS API. Which option will provide the user with the appropriate access?

- A. Amazon Inspector
- B. Access keys
- C. SSH public keys
- D. AWS Key Management Service (AWS KMS) keys

Correct Answer: B

"Access keys are long-term credentials for an IAM user or the AWS account root user. You can use access keys to sign programmatic requests to the AWS CLI or AWS API (directly or using the AWS SDK)." https://docs.aws.amazon.com/IAM/latest/UserGuide/id_credentials_access-keys.html

33) A company runs thousands of simultaneous simulations using AWS Batch. Each simulation is stateless, is fault tolerant, and runs for up to 3 hours.

Which pricing model enables the company to optimize costs and meet these requirements?

- A. Reserved Instances
- B. Spot Instances
- C. On-Demand Instances
- D. Dedicated Instances

Correct Answer: B

Spot Instances are a good fit for stateless, fault-tolerant workloads that can be interrupted without any impact on the overall job.

34) What does the concept of agility mean in AWS Cloud computing? (Choose two.)

- A. The speed at which AWS resources are implemented
- B. The speed at which AWS creates new AWS Regions
- C. The ability to experiment quickly
- D. The elimination of wasted capacity
- E. The low cost of entry into cloud computing

Correct Answer: AC

<https://docs.aws.amazon.com/whitepapers/latest/aws-overview/six-advantages-of-cloud-computing.html>

35) A company needs to block SQL injection attacks.
Which AWS service or feature can meet this requirement?

- A. AWS WAF
- B. AWS Shield
- C. Network ACLs
- D. Security groups

Correct Answer: A

AWS WAF helps you protect against common web exploits and bots that can affect availability, compromise security, or consume excessive resources.

<https://aws.amazon.com/waf/>

36) Which AWS service or feature identifies whether an Amazon S3 bucket or an IAM role has been shared with an external entity?

- A. AWS Service Catalog
- B. AWS Systems Manager
- C. AWS IAM Access Analyzer
- D. AWS Organizations

Correct Answer: C

IAM Access Analyzer helps identify resources in your organization and accounts that are shared with an external entity.

<https://docs.aws.amazon.com/IAM/latest/UserGuide/what-is-access-analyzer.html>

37) A cloud practitioner needs to obtain AWS compliance reports before migrating an environment to the AWS Cloud. How can these reports be generated?

- A. Contact the AWS Compliance team.
- B. Download the reports from AWS Artifact.
- C. Open a case with AWS Support.
- D. Generate the reports with Amazon Macie.

Correct Answer: B

AWS Artifact is a portal that provides access to various compliance reports, including certifications, attestations, and other relevant documents. You can download these reports directly from AWS Artifact.

38) An ecommerce company has migrated its IT infrastructure from an on-premises data center to the AWS Cloud. Which cost is the company's direct responsibility?

- A. Cost of application software licenses
- B. Cost of the hardware infrastructure on AWS
- C. Cost of power for the AWS servers
- D. Cost of physical security for the AWS data center

Correct Answer: A

BCD are AWS responsibility

39) A company is setting up AWS Identity and Access Management (IAM) on an AWS account. Which recommendation complies with IAM security best practices?

- A. Use the account root user access keys for administrative tasks.
- B. Grant broad permissions so that all company employees can access the resources they need.
- C. Turn on multi-factor authentication (MFA) for added security during the login process.
- D. Avoid rotating credentials to prevent issues in production applications.

Correct Answer: C

Enabling multi-factor authentication (MFA) for user accounts, especially for users with administrative or high-privilege access, is a crucial security best practice. MFA adds an additional layer of security by requiring users to provide two or more verification factors (typically something they know, like a password, and something they have, like a temporary MFA code from a hardware token or mobile app) before gaining access. This significantly reduces the risk of unauthorized access, even if login credentials are compromised.

40) Elasticity in the AWS Cloud refers to which of the following? (Choose two.)

- A. How quickly an Amazon EC2 instance can be restarted
- B. The ability to rightsize resources as demand shifts
- C. The maximum amount of RAM an Amazon EC2 instance can use
- D. The pay-as-you-go billing model
- E. How easily resources can be procured when they are needed

Correct Answer: BE

Elasticity is the ability to add and release resources as business needs change. It has nothing to do with pricing.

41) Which service enables customers to audit API calls in their AWS accounts?

- A. AWS CloudTrail
- B. AWS Trusted Advisor
- C. Amazon Inspector
- D. AWS X-Ray

Correct Answer: A

AWS CloudTrail is a service that records all API calls made on your AWS account. It provides a detailed history of events, including who made the call, what actions were performed, and from which IP address the call originated. This audit trail is valuable for security, compliance, troubleshooting, and monitoring purposes, and it helps you maintain visibility into how your AWS resources are being used.

42) What is a customer responsibility when using AWS Lambda according to the AWS shared responsibility model?

- A. Managing the code within the Lambda function
- B. Confirming that the hardware is working in the data center
- C. Patching the operating system
- D. Shutting down Lambda functions when they are no longer in use

Correct Answer: A

In the AWS shared responsibility model, AWS is responsible for the infrastructure and security "of" the cloud, while customers are responsible for the security "in" the cloud. When using AWS Lambda, customers are responsible for managing the code and configuration within the Lambda function. This includes writing and updating the code, configuring the function's execution environment, and ensuring that the code complies with security best practices and is free from vulnerabilities.

43) A company has 5 TB of data stored in Amazon S3. The company plans to occasionally run queries on the data for analysis.

Which AWS service should the company use to run these queries in the MOST cost-effective manner?

- A. Amazon Redshift
- B. Amazon Athena
- C. Amazon Kinesis
- D. Amazon RDS

Correct Answer: B

Amazon Athena is a serverless, interactive analytics service built on open-source frameworks, supporting open-table and file formats. Athena provides a simplified, flexible way to analyze petabytes of data where it lives. Analyze data or build applications from an Amazon Simple Storage Service (S3) data lake and 30 data sources, including on-premises data sources or other cloud systems using SQL or Python. Athena is built on open-source Trino and Presto engines and Apache Spark frameworks, with no provisioning or configuration effort required.
<https://aws.amazon.com/athena/>

44) Which AWS service can be used at no additional cost?

- A. Amazon SageMaker
- B. AWS Config
- C. AWS Organizations
- D. Amazon CloudWatch

Correct Answer: C

AWS Organizations is an account management service that enables you to consolidate multiple AWS accounts into an organization that you create and centrally manage. AWS Organizations is offered at no additional charge. You are charged only for AWS resources that users and roles in your member accounts use. For example, you are charged the standard fees for Amazon EC2 instances that are used by users or roles in your member accounts. https://docs.aws.amazon.com/organizations/latest/userguide/orgs_introduction.html

45) Which AWS Cloud Adoption Framework (AWS CAF) capability belongs to the people perspective?

- A. Data architecture
- B. Event management
- C. Cloud fluency
- D. Strategic partnership

Correct Answer: C

46) A company wants to make an upfront commitment for continued use of its production Amazon EC2 instances in exchange for a reduced overall cost.

Which pricing options meet these requirements with the LOWEST cost? (Choose two.)

- A. Spot Instances
- B. On-Demand Instances
- C. Reserved Instances
- D. Savings Plans
- E. Dedicated Hosts

Correct Answer: CD

C. Reserved Instances: Reserved Instances provide a significant discount (up to 75%) compared to On-Demand pricing in exchange for a one-time upfront payment and/or a lower hourly rate. The more you commit, the greater the discount.

D. Savings Plans: Savings Plans offer flexible pricing and savings on your AWS usage, with discounts of up to 72% compared to On-Demand pricing. With Savings Plans, you commit to a certain amount of usage (measured in dollars per hour) for a one- or three-year term, and receive a lower rate for that usage.

47) A company wants to migrate its on-premises relational databases to the AWS Cloud. The company wants to use infrastructure as close to its current geographical location as possible.

Which AWS service or resource should the company use to select its Amazon RDS deployment area?

- A. Amazon Connect
- B. AWS Wavelength
- C. AWS Regions
- D. AWS Direct Connect

Correct Answer: C

Amazon Connect: Customer service cloud contact center

AWS Wavelength: 5G devices

AWS Direct Connect: Dedicated network connection bypass public network

48) A company is exploring the use of the AWS Cloud, and needs to create a cost estimate for a project before the infrastructure is provisioned.

Which AWS service or feature can be used to estimate costs before deployment?

- A. AWS Free Tier
- B. AWS Pricing Calculator
- C. AWS Billing and Cost Management
- D. AWS Cost and Usage Report

Correct Answer: B

AWS Pricing Calculator is a web-based planning tool that you can use to create estimates for your AWS use cases. You can use it to model your solutions before building them, explore the AWS service price points, and review the calculations behind your estimates.

<https://docs.aws.amazon.com/pricing-calculator/latest/userguide/what-is-pricing-calculator.html>

49) A company is building an application that needs to deliver images and videos globally with minimal latency. Which approach can the company use to accomplish this in a cost effective manner?

- A. Deliver the content through Amazon CloudFront.
- B. Store the content on Amazon S3 and enable S3 cross-region replication.
- C. Implement a VPN across multiple AWS Regions.
- D. Deliver the content through AWS PrivateLink.

Correct Answer: A

<https://aws.amazon.com/cloudfront/>

Securely deliver content with low latency and high transfer speeds

50) Which option is a benefit of the economies of scale based on the advantages of cloud computing?

- A. The ability to trade variable expense for fixed expense
- B. Increased speed and agility
- C. Lower variable costs over fixed costs
- D. Increased operational costs across data centers

Correct Answer: C

Benefit from massive economies of scale – By using cloud computing, you can achieve a lower variable cost than you can get on your own. Because usage from hundreds of thousands of customers is aggregated in the cloud, providers such as AWS can achieve higher economies of scale, which translates into lower pay as-you-go prices.

<https://docs.aws.amazon.com/whitepapers/latest/aws-overview/six-advantages-of-cloud-computing.html>

51) Which of the following is a software development framework that a company can use to define cloud resources as code and provision the resources through AWS CloudFormation?

- A. AWS CLI
- B. AWS Developer Center
- C. AWS Cloud Development Kit (AWS CDK)
- D. AWS CodeStar

Correct Answer: C

AWS CDK is a software development framework that enables developers to define infrastructure as code (IaC) using familiar programming languages like TypeScript, Python, Java, C#, and more. With AWS CDK, you can define cloud resources, their relationships, and provisioning logic in your preferred programming language. AWS CDK also generates CloudFormation templates based on your code, making it easier to manage and deploy infrastructure resources in AWS.

52) A company is developing an application that uses multiple AWS services. The application needs to use temporary, limited-privilege credentials for authentication with other AWS APIs. Which AWS service or feature should the company use to meet these authentication requirements?

- A. Amazon API Gateway
- B. IAM users
- C. AWS Security Token Service (AWS STS)
- D. IAM instance profiles

Correct Answer: C

AWS provides AWS Security Token Service (AWS STS) as a web service that enables you to request temporary, limited-privilege credentials for users.
<https://docs.aws.amazon.com/STS/latest/APIReference/welcome.html>

53) Which AWS service is a cloud security posture management (CSPM) service that aggregates alerts from various AWS services and partner products in a standardized format?

- A. AWS Security Hub
- B. AWS Trusted Advisor
- C. Amazon EventBridge
- D. Amazon GuardDuty

Correct Answer: A

"AWS Security Hub is a cloud security posture management (CSPM) service that performs automated, continuous security best practice checks against your AWS resources to help you identify misconfigurations, and aggregates your security alerts (i.e. findings) in a standardized format so that you can more easily enrich, investigate, and remediate them." Reference:
<https://aws.amazon.com/security-hub/features/?nc=sn&loc=2>

54) Which AWS service is always provided at no charge?

- A. Amazon S3
- B. AWS Identity and Access Management (IAM)
- C. Elastic Load Balancers
- D. AWS WAF

Correct Answer: B

"IAM is a feature of your AWS account and is offered at no additional charge." Reference:
<https://aws.amazon.com/iam/getting-started/?nc=sn&loc=3>

55) To reduce costs, a company is planning to migrate a NoSQL database to AWS. Which AWS service is fully managed and can automatically scale throughput capacity to meet database workload demands?

- A. Amazon Redshift
- B. Amazon Aurora
- C. Amazon DynamoDB
- D. Amazon RDS

Correct Answer: C

"Amazon DynamoDB is a fully managed NoSQL database service that provides fast and predictable performance with seamless scalability."

<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/Introduction.html>

56) A company is using Amazon DynamoDB. Which task is the company's responsibility, according to the AWS shared responsibility model?

- A. Patch the operating system.
- B. Provision hosts.
- C. Manage database access permissions.
- D. Secure the operating system.

Correct Answer: C

57) A company has a test AWS environment. A company is planning on testing an application within AWS. The application testing can be interrupted and does not need to run continuously. Which Amazon EC2 purchasing option will meet these requirements MOST cost-effectively?

- A. On-Demand Instances
- B. Dedicated Instances
- C. Spot Instances
- D. Reserved Instances

Correct Answer: C

58) Which AWS service gives users the ability to discover and protect sensitive data that is stored in Amazon S3 buckets?

- A. Amazon Macie
- B. Amazon Detective
- C. Amazon GuardDuty
- D. AWS IAM Access Analyzer

Correct Answer: A

A : Amazon Macie is a data security service that discovers sensitive data using machine learning and pattern matching, provides visibility into data security risks, and enables you to automate protection against those risks.

<https://aws.amazon.com/macie/features/#:~:text=Amazon%20Macie%20is%20a%20data,automate%20protection%20against%20those%20risks.>

59) Which of the following services can be used to block network traffic to an instance? (Choose two.)

- A. Security groups
- B. Amazon Virtual Private Cloud (Amazon VPC) flow logs
- C. Network ACLs
- D. Amazon CloudWatch
- E. AWS CloudTrail

Correct Answer: AC

Security groups are stateful firewalls that control inbound and outbound traffic at the instance level. You can configure security groups to allow or deny specific types of network traffic to and from your instances.

Network ACLs (Access Control Lists) are stateless firewalls that control traffic at the subnet level. Network ACLs define rules to allow or deny traffic based on source and destination IP addresses, ports, and protocols.

60) Which AWS service can identify when an Amazon EC2 instance was terminated?

- A. AWS Identity and Access Management (IAM)
- B. AWS CloudTrail
- C. AWS Compute Optimizer
- D. Amazon EventBridge

Correct Answer: B

AWS CloudTrail is a service that records all API activity in your AWS account, including the termination of EC2 instances. It creates log entries for various events, providing an audit trail of actions taken on resources. By reviewing CloudTrail logs, you can identify when an EC2 instance was terminated, who initiated the termination, and other relevant details about the event.

61) Which of the following is a fully managed MySQL-compatible database?

- A. Amazon S3
- B. Amazon DynamoDB
- C. Amazon Redshift
- D. Amazon Aurora

Correct Answer: D

Amazon Aurora is a relational database service that is compatible with MySQL and PostgreSQL. It is fully managed by AWS and is designed for high availability, performance, and scalability while maintaining MySQL compatibility. Aurora offers features like automated backups, read replicas, and seamless failover to ensure data durability and availability. It's a popular choice for applications that require a MySQL-compatible database with the benefits of a fully managed service.

62) Which AWS service supports a hybrid architecture that gives users the ability to extend AWS infrastructure, AWS services, APIs, and tools to data centers, co-location environments, or on-premises facilities?

- A. AWS Snowmobile
- B. AWS Local Zones
- C. AWS Outposts
- D. AWS Fargate

Correct Answer: C

AWS Outposts enables you to run AWS infrastructure and services on premises while seamlessly connecting to the AWS cloud. This service extends the AWS ecosystem to your on-premises locations, allowing you to take advantage of cloud benefits while addressing the requirements of data residency, low-latency applications, and specific regulatory needs in hybrid environments.

63) Which AWS service can run a managed PostgreSQL database that provides online transaction processing (OLTP)?

- A. Amazon DynamoDB
- B. Amazon Athena
- C. Amazon RDS
- D. Amazon EMR

Correct Answer: C

Amazon RDS supports various database engines, including PostgreSQL, and offers a managed database service suitable for OLTP workloads. With Amazon RDS for PostgreSQL, you can easily set up, operate, and scale a PostgreSQL database without the administrative overhead of managing the infrastructure.

64) A company wants to provide managed Windows virtual desktops and applications to its remote employees over secure network connections.

Which AWS services can the company use to meet these requirements? (Choose two.)

- A. Amazon Connect
- B. Amazon AppStream 2.0
- C. Amazon WorkSpaces
- D. AWS Site-to-Site VPN
- E. Amazon Elastic Container Service (Amazon ECS)

Correct Answer: BC

B. Amazon AppStream 2.0: Amazon AppStream 2.0 is a service that enables you to stream desktop applications to users through web browsers. You can deliver Windows applications securely to remote users without the need to provision and manage full virtual desktops.

C. Amazon WorkSpaces: Amazon WorkSpaces is a fully managed desktop-as-a-service (DaaS) solution that provides Windows desktops to users. You can configure and manage virtual desktops for remote employees using WorkSpaces.

65) A company wants to monitor for misconfigured security groups that are allowing unrestricted access to specific ports.

Which AWS service will meet this requirement?

- A. AWS Trusted Advisor
- B. Amazon CloudWatch
- C. Amazon GuardDuty
- D. AWS Health Dashboard

Correct Answer: A

Security Groups – Unrestricted Access Checks security groups for rules that allow unrestricted access to a resource. Unrestricted access increases opportunities for malicious activity (hacking, denial-of-service attacks, loss of data). For more information, see the Trusted Advisor FAQ <https://docs.aws.amazon.com/awssupport/latest/user/security-checks.html>

66) Which AWS service is a key-value database that provides sub-millisecond latency on a large scale?

- A. Amazon DynamoDB
- B. Amazon Aurora
- C. Amazon DocumentDB (with MongoDB compatibility)
- D. Amazon Neptune

Correct Answer: A

Because Amazon DynamoDB is a nonrelational database that delivers reliable performance at any scale. It's a fully managed, multi-region, multi-master database that provides consistent single-digit millisecond latency, and offers built-in security, backup and restore, and in-memory caching.

67) Which AWS services or features provide disaster recovery solutions for Amazon EC2 instances? (Choose two.)

- A. EC2 Reserved Instances
- B. EC2 Amazon Machine Images (AMIs)
- C. Amazon Elastic Block Store (Amazon EBS) snapshots
- D. AWS Shield
- E. Amazon GuardDuty

Correct Answer: BC

B. EC2 Amazon Machine Images (AMIs): AMIs are used to create backups of EC2 instances, and they can be used to launch replacement instances in the event of a disaster or data loss. AMIs are essential for creating recovery points for your EC2 instances.

C. Amazon Elastic Block Store (Amazon EBS) snapshots: EBS snapshots allow you to create point-in-time backups of your EBS volumes. These snapshots can be used to restore data or create new EBS volumes, making them a key component of disaster recovery for EC2 instances.

68) Which AWS service provides command line access to AWS tools and resources directly from a web browser?

- A. AWS CloudHSM
- B. AWS CloudShell
- C. Amazon WorkSpaces
- D. AWS Cloud Map

Correct Answer: B

"Using AWS CloudShell, a browser-based shell, you can quickly run scripts with the AWS Command Line Interface (CLI), experiment with service APIs using the AWS CLI, and use other tools to increase your productivity. The CloudShell icon appears in AWS Regions where CloudShell is available."

69) A network engineer needs to build a hybrid cloud architecture connecting on-premises networks to the AWS Cloud using AWS Direct Connect. The company has a few VPCs in a single AWS Region and expects to increase the number of VPCs to hundreds over time.

Which AWS service or feature should the engineer use to simplify and scale this connectivity as the VPCs increase in number?

- A. VPC endpoints
- B. AWS Transit Gateway
- C. Amazon Route 53
- D. AWS Secrets Manager

Correct Answer: B

AWS Transit Gateway connects your Amazon Virtual Private Clouds (VPCs) and on-premises networks through a central hub. This connection simplifies your network and puts an end to complex peering relationships. Transit Gateway acts as a highly scalable cloud router—each new connection is made only once.

<https://aws.amazon.com/transit-gateway/>

70) A company wants to establish a schedule for rotating database user credentials.

Which AWS service will support this requirement with the LEAST amount of operational overhead?

- A. AWS Systems Manager
- B. AWS Secrets Manager
- C. AWS License Manager
- D. AWS Managed Services

Correct Answer: B

<https://aws.amazon.com/secrets-manager/>

AWS Secrets Manager helps you manage, retrieve, and rotate database credentials, API keys, and other secrets throughout their lifecycles.

71) Which AWS service is used to provide encryption for Amazon EBS?

- A. AWS Certificate Manager
- B. AWS Systems Manager
- C. AWS KMS
- D. AWS Config

Correct Answer: C

AWS Key Management Service (AWS KMS) lets you create, manage, and control cryptographic keys across your applications and AWS services.

Ref: <https://aws.amazon.com/kms/>

72) A company wants to manage its AWS Cloud resources through a web interface. Which AWS service will meet this requirement?

- A. AWS Management Console
- B. AWS CLI
- C. AWS SDK
- D. AWS Cloud9

Correct Answer: A

The AWS Management Console is a web application that comprises and refers to a broad collection of service consoles for managing AWS resources.
<https://docs.aws.amazon.com/awsconsolehelpdocs/latest/gsg/learn-whats-new.html>

73) Which of the following are advantages of the AWS Cloud? (Choose two.)

- A. Trade variable expenses for capital expenses
- B. High economies of scale
- C. Launch globally in minutes
- D. Focus on managing hardware infrastructure
- E. Overprovision to ensure capacity

Correct Answer: BC

B. Ability to quickly change required capacity: With the AWS Cloud, users can easily scale their infrastructure up or down based on demand. This flexibility allows for rapid adjustment of resources to match application needs, enabling cost optimization and efficient resource utilization.

C. High economies of scale: AWS operates at a large scale, serving millions of customers globally. This scale allows AWS to achieve cost efficiencies and pass on the benefits to customers. By leveraging AWS services, users can access enterprise-grade infrastructure and services without the need for significant upfront investment in hardware or infrastructure.

74) Which AWS Cloud benefit is shown by an architecture's ability to withstand failures with minimal downtime?

- A. Agility
- B. Elasticity
- C. Scalability
- D. High availability

Correct Answer: D

The ability of an architecture to withstand failures with minimal downtime is a characteristic of high availability. High availability ensures that your system remains operational and accessible even in the face of component failures. This is critical for maintaining a reliable and responsive application or service.

75) A developer needs to maintain a development environment infrastructure and a production environment infrastructure in a repeatable fashion.

Which AWS service should the developer use to meet these requirements?

- A. AWS Ground Station
- B. AWS Shield
- C. AWS IoT Device Defender
- D. AWS CloudFormation

Correct Answer: D

"AWS CloudFormation lets you model, provision, and manage AWS and third-party resources by treating infrastructure as code."

76) Which task is the customer's responsibility, according to the AWS shared responsibility model?

- A. Maintain the security of the AWS Cloud.
- B. Configure firewalls and networks.
- C. Patch the operating system of Amazon RDS instances.
- D. Implement physical and environmental controls.

Correct Answer: B

77) Which AWS service helps deliver highly available applications with fast failover for multi-Region and Multi-AZ architectures?

- A. AWS WAF
- B. AWS Global Accelerator
- C. AWS Shield
- D. AWS Direct Connect

Correct Answer: B

"Deliver highly available applications with fast failover for multi-Region and multi-AZ architectures." Reference: <https://aws.amazon.com/global-accelerator/>

78) A company has a set of ecommerce applications. The applications need to be able to send messages to each other. Which AWS service meets this requirement?

- A. AWS Auto Scaling
- B. Elastic Load Balancing
- C. Amazon Simple Queue Service (Amazon SQS)
- D. Amazon Kinesis Data Streams

Correct Answer: C

Amazon Simple Queue Service (SQS): SQS is a fully managed message queuing service that enables you to decouple and scale microservices, distributed systems, and serverless applications. It allows one application to send messages to a queue, and another application to retrieve those messages from the queue. This can be helpful in scenarios where the sender and receiver are not required to interact with each other in real-time.

79) What are the benefits of consolidated billing for AWS Cloud services? (Choose two.)

- A. Volume discounts
- B. A minimal additional fee for use
- C. One bill for multiple accounts
- D. Installment payment options
- E. Custom cost and usage budget creation

Correct Answer: AC

Consolidated billing has the following benefits: One bill – You get one bill for multiple accounts. Easy tracking – You can track the charges across multiple accounts and download the combined cost and usage data. Combined usage – You can combine the usage across all accounts in the organization to share the volume pricing discounts, Reserved Instance discounts, and Savings Plans. This can result in a lower charge for your project, department, or company than with individual standalone accounts. For more information, see Volume discounts. No extra fee – Consolidated billing is offered at no additional cost.

80) A user wants to review all Amazon S3 buckets with ACLs and S3 bucket policies in the S3 console. Which AWS service or resource will meet this requirement?

- A. S3 Multi-Region Access Points
- B. S3 Storage Lens
- C. AWS IAM Identity Center (AWS Single Sign-On)
- D. Access Analyzer for S3

Correct Answer: D

Access Analyzer for S3 allows you to analyze and review access policies for your S3 buckets. It helps you identify and resolve unintended access to your S3 resources. With Access Analyzer for S3, you can review both bucket policies and bucket ACLs to ensure proper access controls.

81) What is the best resource for a user to find compliance-related information and reports about AWS?

- A. AWS Artifact
- B. AWS Marketplace
- C. Amazon Inspector
- D. AWS Support

Correct Answer: A

"AWS Artifact is your go-to, central resource for compliance-related information that matters to you. It provides on-demand access to security and compliance reports from AWS and ISVs who sell their products on AWS Marketplace."

Reference: <https://aws.amazon.com/artifact/>

82) Which AWS service enables companies to deploy an application close to end users?

- A. Amazon CloudFront
- B. AWS Auto Scaling
- C. AWS AppSync
- D. Amazon Route 53

Correct Answer: A

"Amazon CloudFront speeds up distribution of your static and dynamic web content, such as .html, .css, .php, image, and media files. When users request your content, CloudFront delivers it through a worldwide network of edge locations that provide low latency and high performance."

Reference: <https://docs.aws.amazon.com/cloudfront/>

83) Which AWS service or feature improves network performance by sending traffic through the AWS worldwide network infrastructure?

- A. Route table
- B. AWS Transit Gateway
- C. AWS Global Accelerator
- D. Amazon VPC

Correct Answer: C

"Improve application availability, performance, and security using the AWS global network"

Reference: <https://aws.amazon.com/global-accelerator/>

84) Which AWS service provides highly durable object storage?

- A. Amazon S3
- B. Amazon Elastic File System (Amazon EFS)
- C. Amazon Elastic Block Store (Amazon EBS)
- D. Amazon FSx

Correct Answer: A

"S3 Standard offers high durability, availability, and performance object storage for frequently accessed data."

Reference: <https://aws.amazon.com/s3/storage-classes/>

85) Which responsibility belongs to AWS when a company hosts its databases on Amazon EC2 instances?

- A. Database backups
- B. Database software patches
- C. Operating system patches
- D. Operating system installations

Correct Answer: D

D. AWS provides the infrastructure and services (like EC2) that include a range of Amazon Machine Images (AMIs) with pre-installed operating systems. This means AWS is responsible for ensuring that these AMIs are available and that the underlying infrastructure to run these instances is secure and reliable.

The other responsibilities listed - database backups (A), database software patches (B), and operating system patches (C) - are under the purview of the customer when using Amazon EC2 instances.

86) Which of the following are advantages of moving to the AWS Cloud? (Choose two.)

- A. The ability to turn over the responsibility for all security to AWS.
- B. The ability to use the pay-as-you-go model.
- C. The ability to have full control over the physical infrastructure.
- D. No longer having to guess what capacity will be required.
- E. No longer worrying about users access controls.

Correct Answer: BD

Stop guessing capacity Trade fixed expense for variable expense
<https://docs.aws.amazon.com/whitepapers/latest/aws-overview/six-advantages-of-cloud-computing.html>

87) Which AWS service is a hybrid cloud storage service that provides on-premises users access to virtually unlimited cloud storage?

- A. AWS DataSync
- B. Amazon S3 Glacier
- C. AWS Storage Gateway
- D. Amazon Elastic Block Store (Amazon EBS)

Correct Answer: C

<https://aws.amazon.com/storagegateway/>

AWS Storage Gateway is a set of hybrid cloud storage services that provide on-premises access to virtually unlimited cloud storage.

88) A company plans to migrate to AWS and wants to create cost estimates for its AWS use cases. Which AWS service or tool can the company use to meet these requirements?

- A. AWS Pricing Calculator
- B. Amazon CloudWatch
- C. AWS Cost Explorer
- D. AWS Budgets

Correct Answer: A

"AWS Pricing Calculator is a web-based planning tool that you can use to create estimates for your AWS use cases. You can use it to model your solutions before building them, explore the AWS service price points, and review the calculations behind your estimates. You can use it to help you plan how you spend, find cost saving opportunities, and make informed decisions when using Amazon Web Services."

Reference: <https://docs.aws.amazon.com/pricing-calculator/latest/userguide/what-is-pricing-calculator.html>

89) Which tool should a developer use to integrate AWS service features directly into an application?

- A. AWS Software Development Kit
- B. AWS CodeDeploy
- C. AWS Lambda
- D. AWS Batch

Correct Answer: A

<https://aws.amazon.com/developer/tools/>

90) Which of the following is a recommended design principle of the AWS Well-Architected Framework?

- A. Reduce downtime by making infrastructure changes infrequently and in large increments.
- B. Invest the time to configure infrastructure manually.
- C. Learn to improve from operational failures.
- D. Use monolithic application design for centralization.

Correct Answer: C

Learn from all operational failures: Drive improvement through lessons learned from all operational events and failures. Share what is learned across teams and through the entire organization.

<https://docs.aws.amazon.com/whitepapers/latest/aws-overview/six-advantages-of-cloud-computing.html>

91) Using AWS Identity and Access Management (IAM) to grant access only to the resources needed to perform a task is a concept known as:

- A. restricted access.
- B. as-needed access.
- C. least privilege access.
- D. token access.

Correct Answer: C

access only to the resources needed = least

92) A company wants to operate a data warehouse to analyze data without managing the data warehouse infrastructure.

Which AWS service will meet this requirement?

- A. Amazon Aurora
- B. Amazon Redshift Serverless
- C. AWS Lambda
- D. Amazon RDS

Correct Answer: B

"Amazon Redshift Serverless makes it easy to run analytics workloads of any size without having to manage data warehouse infrastructure."

Reference: <https://aws.amazon.com/redshift/redshift-serverless/>

93) How does AWS Cloud computing help businesses reduce costs? (Choose two.)

- A. AWS charges the same prices for services in every AWS Region.
- B. AWS enables capacity to be adjusted on demand.
- C. AWS offers discounts for Amazon EC2 instances that remain idle for more than 1 week.
- D. AWS does not charge for data sent from the AWS Cloud to the internet.
- E. AWS eliminates many of the costs of building and maintaining on-premises data centers.

Correct Answer: BE

Stop spending money running and maintaining data centers Benefit from massive economies of scale

<https://docs.aws.amazon.com/whitepapers/latest/aws-overview/six-advantages-of-cloud-computing.html>

94) A company wants to grant users in one AWS account access to resources in another AWS account. The users do not currently have permission to access the resources. Which AWS service will meet this requirement?

- A. IAM group
- B. IAM role
- C. IAM tag
- D. IAM Access Analyzer

Correct Answer:B

https://docs.aws.amazon.com/IAM/latest/UserGuide/id_roles.html

95) Which task is the responsibility of AWS when using AWS services?

- A. Management of IAM user permissions
- B. Creation of security group rules for outbound access
- C. Maintenance of physical and environmental controls
- D. Application of Amazon EC2 operating system patches

Correct Answer: C

AWS is responsible for maintaining the physical and environmental controls of its data centers, including the security and reliability of the infrastructure. This includes aspects such as power, cooling, and physical security

96) A company wants to automate infrastructure deployment by using infrastructure as code (IaC). The company wants to scale production stacks so the stacks can be deployed in multiple AWS Regions. Which AWS service will meet these requirements?

- A. Amazon CloudWatch
- B. AWS Config
- C. AWS Trusted Advisor
- D. AWS CloudFormation

Correct Answer: D

"AWS CloudFormation gives you an easy way to model a collection of related AWS and third-party resources, provision them quickly and consistently, and manage them throughout their lifecycles, by treating infrastructure as code. A CloudFormation template describes your desired resources and their dependencies so you can launch and configure them together as a stack. You can use a template to create, update, and delete an entire stack as a single unit, as often as you need to, instead of managing resources individually. You can manage and provision stacks across multiple AWS accounts and AWS Regions."

Reference: <https://aws.amazon.com/cloudformation/features/>

97) Which option is an AWS Cloud Adoption Framework (AWS CAF) platform perspective capability?

- A. Data architecture
- B. Data protection
- C. Data governance
- D. Data science

Correct Answer: A

AWS CAF: 7 Platform Perspective : Platform Architecture, Data Architecture, Platform Engineering, Data Engineering, Provisioning and Orchestration, Modern App Development, CI/CD

98) A company is running a workload in the AWS Cloud.
Which AWS best practice ensures the MOST cost-effective architecture for the workload?

- A. Loose coupling
- B. Rightsizing
- C. Caching
- D. Redundancy

Correct Answer: B

Right sizing is the process of matching instance types and sizes to your workload performance and capacity requirements at the lowest possible cost.

Ref link: <https://aws.amazon.com/aws-cost-management/aws-cost-optimization/right-sizing/#:~:text=Right%20sizing%20is%20the%20process,at%20the%20lowest%20possible%20cost.>

99) A company is using a third-party service to back up 10 TB of data to a tape library. The on-premises backup server is running out of space. The company wants to use AWS services for the backups without changing its existing backup workflows.
Which AWS service should the company use to meet these requirements?

- A. Amazon Elastic Block Store (Amazon EBS)
- B. AWS Storage Gateway
- C. Amazon Elastic Container Service (Amazon ECS)
- D. AWS Lambda

Correct Answer: B

AWS Storage Gateway is a hybrid cloud storage service that gives you on-premises access to virtually unlimited cloud storage. Storage Gateway provides a standard set of storage protocols such as iSCSI, SMB, and NFS, which allow you to use AWS storage without rewriting your existing applications.

100) Which tasks are the customer's responsibility, according to the AWS shared responsibility model? (Choose two.)

- A. Establish the global infrastructure.
- B. Perform client-side data encryption.
- C. Configure IAM credentials.
- D. Secure edge locations.
- E. Patch Amazon RDS DB instances.

Correct Answer: BC

101) A developer has been hired by a large company and needs AWS credentials. Which are security best practices that should be followed? (Choose two.)

- A. Grant the developer access to only the AWS resources needed to perform the job.
- B. Share the AWS account root user credentials with the developer.
- C. Add the developer to the administrator's group in AWS IAM.
- D. Configure a password policy that ensures the developer's password cannot be changed.
- E. Ensure the account password policy requires a minimum length.

Correct Answer: AE

102) A company has multiple AWS accounts that include compute workloads that cannot be interrupted. The company wants to obtain billing discounts that are based on the company's use of AWS services. Which AWS feature or purchasing option will meet these requirements?

- A. Resource tagging
- B. Consolidated billing
- C. Pay-as-you-go pricing
- D. Spot Instances

Correct Answer: B

Consolidated billing allows you to combine multiple AWS accounts and aggregate the usage and spending across those accounts. This simplifies billing and enables you to take advantage of volume discounts, which can lead to cost savings. Each account can still maintain its own resource tagging and pricing options, so it's possible to manage compute workloads in individual accounts as needed.

103) A user wants to allow applications running on an Amazon EC2 instance to make calls to other AWS services. The access granted must be secure. Which AWS service or feature should be used?

- A. Security groups
- B. AWS Firewall Manager
- C. IAM roles
- D. IAM user SSH keys

Correct Answer: C

"You can use roles to delegate access to users, applications, or services that don't normally have access to your AWS resources."

Reference: https://docs.aws.amazon.com/IAM/latest/UserGuide/id_roles.html

104) A company wants a fully managed Windows file server for its Windows-based applications. Which AWS service will meet this requirement?

- A. Amazon FSx
- B. Amazon Elastic Kubernetes Service (Amazon EKS)
- C. Amazon Elastic Container Service (Amazon ECS)
- D. Amazon EMR

Correct Answer: A

Amazon FSx lets you easily and securely backup, archive, or replicate your on-premises file storage to AWS in order to meet regulatory, data retention, or disaster recovery requirements. Learn how to get started. Choosing an Amazon FSx File System.

105) A company wants to migrate its NFS on-premises workload to AWS. Which AWS Storage Gateway type should the company use to meet this requirement?

- A. Tape Gateway
- B. Volume Gateway
- C. Amazon FSx File Gateway
- D. Amazon S3 File Gateway

Correct Answer: D

Amazon S3 File Gateway provides a seamless way to connect to the cloud in order to store application data files and backup images as durable objects in Amazon S3 cloud storage. Amazon S3 File Gateway offers SMB or NFS-based access to data in Amazon S3 with local caching. <https://aws.amazon.com/storagegateway/file/s3/>

106) A company needs to track the activity in its AWS accounts, and needs to know when an API call is made against its AWS resources.

Which AWS tool or service can be used to meet these requirements?

- A. Amazon CloudWatch
- B. Amazon Inspector
- C. AWS CloudTrail
- D. AWS IAM

Correct Answer: C

"CloudTrail enables auditing, security monitoring, and operational troubleshooting by tracking user activity and API usage. CloudTrail logs, continuously monitors, and retains account activity related to actions across your AWS infrastructure, giving you control over storage, analysis, and remediation actions."

Reference: <https://aws.amazon.com/cloudtrail/faqs/>

107) A company has an uninterruptible application that runs on Amazon EC2 instances. The application constantly processes a backlog of files in an Amazon Simple Queue Service (Amazon SQS) queue. This usage is expected to continue to grow for years.

What is the MOST cost-effective EC2 instance purchasing model to meet these requirements?

- A. Spot Instances
- B. On-Demand Instances
- C. Savings Plans
- D. Dedicated Hosts

Correct Answer: C

Savings Plans are a flexible pricing model that offer low prices on EC2 usage, in exchange for a commitment to a consistent amount of usage, measured in dollars per hour, for a term of 1 or 3 years. Savings Plans can reduce your EC2 costs by up to 72% compared to On-Demand prices.

108) A company wants an AWS service to provide product recommendations based on its customer data. Which AWS service will meet this requirement?

- A. Amazon Polly
- B. Amazon Personalize
- C. Amazon Comprehend
- D. Amazon Rekognition

Correct Answer: B

"Amazon Personalize accelerates your digital transformation with ML, making it easier to integrate personalized recommendations into existing websites, applications, email marketing systems, and more."

Reference: <https://aws.amazon.com/personalize/>

109) A company is planning its migration to the AWS Cloud. The company is identifying its capability gaps by using the AWS Cloud Adoption Framework (AWS CAF) perspectives. Which phase of the cloud transformation journey includes these identification activities?

- A. Envision
- B. Align
- C. Scale
- D. Launch

Correct Answer: B

"Align phase focuses on identifying capability gaps across the six AWS CAF perspectives, identifying cross-organizational dependencies, and surfacing stakeholder concerns and challenges."

Reference: <https://docs.aws.amazon.com/whitepapers/latest/overview-aws-cloud-adoption-framework/your-cloud-transformation-journey.html>

110) A social media company wants to protect its web application from common web exploits such as SQL injections and cross-site scripting.

Which AWS service will meet these requirements?

- A. Amazon Inspector
- B. AWS WAF
- C. Amazon GuardDuty
- D. Amazon CloudWatch

Correct Answer: B

"AWS WAF is a web application firewall that helps protect web applications from attacks by allowing you to configure rules that allow, block, or monitor (count) web requests based on conditions that you define. These conditions include IP addresses, HTTP headers, HTTP body, URI strings, SQL injection and cross-site scripting."

References: <https://aws.amazon.com/waf/faqs/>

111) Which fully managed AWS service assists with the creation, testing, and management of custom Amazon EC2 images?

- A. EC2 Image Builder
- B. Amazon Machine Image (AMI)
- C. AWS Launch Wizard
- D. AWS Elastic Beanstalk

Correct Answer: A

"EC2 Image Builder simplifies the creation, maintenance, validation, sharing, and deployment of Linux or Windows images for use with Amazon EC2 and on-premises."

Reference: <https://aws.amazon.com/image-builder/faqs/>

112) A company wants an automated process to continuously scan its Amazon EC2 instances for software vulnerabilities.

Which AWS service will meet these requirements?

- A. Amazon GuardDuty
- B. Amazon Inspector
- C. Amazon Detective
- D. Amazon Cognito

Correct Answer: B

"Amazon Inspector is an automated vulnerability management service that continually scans Amazon Elastic Compute Cloud (EC2), AWS Lambda functions, and container workloads for software vulnerabilities and unintended network exposure."

Reference: <https://aws.amazon.com/inspector/faqs/>

113) A company needs to perform data processing once a week that typically takes about 5 hours to complete. Which AWS service should the company use for this workload?

- A. AWS Lambda
- B. Amazon EC2
- C. AWS CodeDeploy
- D. AWS Wavelength

Correct Answer: B

The best AWS service for a workload that needs to be processed once a week for about 5 hours is Amazon EC2.

AWS Lambda is a serverless computing service that runs your code without provisioning or managing servers. However, Lambda functions have a maximum execution time of 15 minutes. Therefore, Lambda is not suitable for workloads that need to run for longer than 15 minutes.

114) Which AWS service or feature provides log information of the inbound and outbound traffic on network interfaces in a VPC?

- A. Amazon CloudWatch Logs
- B. AWS CloudTrail
- C. VPC Flow Logs
- D. AWS Identity and Access Management (IAM)

Correct Answer: C

VPC flow logs is a feature that enables you to capture information about the IP traffic going to and from network interfaces in your VPC. Flow logs data can be published to either Amazon CloudWatch Logs or Amazon S3. You can monitor your VPC flow logs to gain operational visibility about your network dependencies and traffic patterns, detect anomalies and prevent data leakage, or troubleshoot network connectivity and configuration issues.

<https://aws.amazon.com/vpc/faqs/#:~:text=VPC%20flow%20logs%20is%20a,network%20interfaces%20in%20your%20VPC.>

115) A company wants to design a centralized storage system to manage the configuration data and passwords for its critical business applications.

Which AWS service or capability will meet these requirements MOST cost-effectively?

- A. AWS Systems Manager Parameter Store
- B. AWS Secrets Manager
- C. AWS Config
- D. Amazon S3

Correct Answer: A

"manage your configuration data, whether plain-text data such as database strings or secrets such as passwords"

<https://aws.amazon.com/systems-manager/features/>

116) A company plans to deploy containers on AWS. The company wants full control of the compute resources that host the containers. Which AWS service will meet these requirements?

- A. Amazon Elastic Kubernetes Service (Amazon EKS)
- B. AWS Fargate
- C. Amazon EC2
- D. Amazon Elastic Container Service (Amazon ECS)

Correct Answer: C

Amazon EC2: Amazon Elastic Compute Cloud (EC2) provides scalable computing capacity in the AWS cloud. By using Amazon EC2, the company can have full control over the virtual servers (instances), including the choice of instance type, configuration, and the underlying host system. EC2 allows for the deployment of containerized applications, where the company can manage the orchestration itself, either manually or by using container orchestration tools like Kubernetes or Docker Swarm.

117) Which AWS service or feature allows users to create new AWS accounts, group multiple accounts to organize workflows, and apply policies to groups of accounts?

- A. AWS Identity and Access Management (IAM)
- B. AWS Trusted Advisor
- C. AWS CloudFormation
- D. AWS Organizations

Correct Answer: D

AWS Organizations lets you create new AWS accounts at no additional charge. With accounts in an organization, you can easily allocate resources, group accounts, and apply governance policies to accounts or groups.

118) A company wants to store and retrieve files in Amazon S3 for its existing on-premises applications by using industry-standard file system protocols.

Which AWS service will meet these requirements?

- A. AWS DataSync
- B. AWS Snowball Edge
- C. Amazon S3 File Gateway
- D. AWS Transfer Family

Correct Answer: C

"Amazon S3 File Gateway provides a seamless way to connect to the cloud in order to store application data files and backup images as durable objects in Amazon S3 cloud storage. Amazon S3 File Gateway offers SMB or NFS-based access to data in Amazon S3 with local caching. It can be used for on-premises data-intensive Amazon EC2-based applications that need file protocol access to S3 object storage." Reference: <https://aws.amazon.com/storagegateway/file/s3/>

119) A company wants to block SQL injection attacks.

Which AWS service or feature should the company use to meet this requirement?

- A. AWS WAF
- B. Network ACLs
- C. Security groups
- D. AWS Certificate Manager (ACM)

Correct Answer: A

"AWS WAF is a web application firewall that helps protect web applications from attacks by allowing you to configure rules that allow, block, or monitor (count) web requests based on conditions that you define. These conditions include IP addresses, HTTP headers, HTTP body, URI strings, SQL injection and cross-site scripting."

References: <https://aws.amazon.com/waf/faqs/>

120) A company wants a unified tool to provide a consistent method to interact with AWS services. Which AWS service or tool will meet this requirement?

- A. AWS CLI
- B. Amazon Elastic Container Service (Amazon ECS)
- C. AWS Cloud9
- D. AWS Virtual Private Network (AWS VPN)

Correct Answer: A

"The AWS Command Line Interface (AWS CLI) is a unified tool to manage your AWS services. With just one tool to download and configure, you can control multiple AWS services from the command line and automate them through scripts."

Reference: <https://aws.amazon.com/cli/>

101) A company is using the AWS Free Tier for several AWS services for an application. What will happen if the Free Tier usage period expires or if the application use exceeds the Free Tier usage limits?

-  A. The company will be charged the standard pay-as-you-go service rates for the usage that exceeds the Free Tier usage.
- B. AWS Support will contact the company to set up standard service charges.
- C. The company will be charged for the services it consumed during the Free Tier period, plus additional charges for service consumption after the Free Tier period.
- D. The company's AWS account will be frozen and can be restarted after a payment plan is established.

Correct Answer: A

When your free tier expires or if your application use exceeds the free tier limits, you simply pay standard, pay-as-you-go service rates (see each service page for full pricing details)

102) Which AWS service uses machine learning to help discover, monitor, and protect sensitive data that is stored in Amazon S3 buckets?

- A. AWS Shield
-  B. Amazon Macie
- C. AWS Network Firewall
- D. Amazon Cognito

Correct Answer: B

Amazon Macie is a data security and data privacy service that uses machine learning (ML) and pattern matching to discover and protect your sensitive data.

103) According to the AWS shared responsibility model, which of the following is exclusively the responsibility of AWS?

- A. Patching of the guest operating system
- B. Security awareness and training
-  C. Physical and environmental controls
- D. Development of an IAM password policy

Correct Answer: C

According to AWS Shared Responsibility Model, AWS is responsible for the **Security of the Cloud** and the customer is responsible for the Security in the Cloud. AWS Responsibility: AWS is responsible for protecting the infrastructure that runs all the AWS services.

104) What can users do using AWS Marketplace? (Select two.)

- ☐ A. Sell unused Amazon EC2 Spot Instances.
- ☒ B. Sell solutions to other AWS users.
- ☒ C. Buy third-party software that runs on AWS.
- ☐ D. Purchase AWS security and compliance documents.
- ☐ E. Order AWS Snowball.

Correct Answer: BC

The AWS Marketplace enables qualified partners to market and sell their software to AWS Customers. AWS Marketplace is an online software store that helps customers find, buy, and immediately start using the software and services that run on AWS.

105) What are the possible uses for AWS edge locations? (Select two.)

- ☐ A. Hosting applications
- ☒ B. Delivering content closer to users
- ☐ C. Running NoSQL database caching services
- ☒ D. Reducing traffic on the server by caching responses
- ☐ E. Sending notification messages to end users

Correct Answer: BD

CloudFront delivers your content through a worldwide network of data centers called edge locations. When a user requests content that you're serving with CloudFront, the user is routed to the edge location that provides the lowest latency (time delay), so that content is delivered with the best possible performance.

106) Amazon Elastic Container Service (Amazon ECS) and Amazon DynamoDB are used by a firm to execute its mission-critical web application. Multiple times during the day, the workload jumps to up to ten times the regular level.

Which AWS Cloud feature helps the business to adapt to these demand changes?

- A. Agility
- B. Global reach
-  C. Scalability
- D. Security

Correct Answer: C

AWS EC2 Auto Scaling Servers can get more requests than they can handle. Too many requests can cause timeouts and outages. AWS EC2 Auto Scaling allows you to **add or remove EC2 instances automatically**. It automates the capacity to the demand.

107) A company wants to improve the overall availability and performance of its applications that are hosted on AWS. Which AWS service should the company use?

- A. Amazon Connect
- B. Amazon Lightsail
-  C. AWS Global Accelerator
- D. AWS Storage Gateway

Correct Answer: C

AWS Global Accelerator is a networking service that improves the performance of your users' traffic by up to 60% using Amazon Web Services' global network infrastructure. When the internet is congested, AWS Global Accelerator optimizes the path to your application to keep packet loss, jitter, and latency consistently low.

108) Which of the following tasks is the customer's duty under the shared responsibility model? (Select two.)

- ☐ A. Maintaining the underlying Amazon EC2 hardware.
- ☒ B. Managing the VPC network access control lists.
- ☒ C. Encrypting data in transit and at rest.
- ☐ D. Replacing failed hard disk drives.
- ☐ E. Deploying hardware in different Availability Zones.

Correct Answer: BC



The hardware related jobs is the prime responsibility of AWS. VPC network access control lists is something a customer has to do himself to secure the applications. Encrypting data in transit and at rest is a shared responsibility in which AWS plays a part. All hardware related jobs have nothing to do with the customer.

109) Which of the following are AWS obligations, according to the AWS shared responsibility model? (Select two.)

- ☒ A. Network infrastructure and virtualization of infrastructure
- ☐ B. Security of application data
- ☐ C. Guest operating systems
- ☒ D. Physical security of hardware
- ☐ E. Credentials and policies

Correct Answer: AD

Physical security is one area where the [AWS Shared Responsibility Model](#) comes into play. AWS is responsible for protecting “security of the cloud” which includes the infrastructure of hardware, software, networking, and facilities that run AWS services.

110) What charges are included in the comparison of AWS vs on-premises Total Cost of Ownership (TCO)?

-  A. Data center security
- B. Business analysis
- C. Project management
- D. Operating system administration

Correct Answer: A

Cloud TCO involves calculating the costs required to host, run, integrate, secure and manage workloads in the cloud over their lifetime. These include fees associated with the underlying infrastructure, such as compute, data transfer and storage. It also includes the cost of supporting cloud services, ranging from security and management tools to data analytics. Manpower costs for cloud engineers should also be part of a cloud TCO equation.

111) Which AWS service or feature identifies whether an Amazon S3 bucket or an IAM role has been shared with an external entity?

- A. AWS Service Catalog
- B. AWS Systems Manager
-  C. AWS IAM Access Analyzer
- D. AWS Organizations

Correct Answer: C

Access Analyzer helps you identify the resources in your organization and accounts, such as Amazon S3 buckets or IAM roles, shared with an external entity. This lets you identify unintended access to your resources and data, which is a security risk.

112) Which AWS service supports MySQL and PostgreSQL as relational databases?

- A. Amazon Redshift
- B. Amazon DynamoDB
-  C. Amazon Aurora
- D. Amazon Neptune

Correct Answer: C

AWS Aurora is a fully managed relational database optimized for use in AWS. It supports MySQL and PostgreSQL. With this service, you can combine the cost-effectiveness of open-source databases with increased availability and performance.

113) Which AWS services is natively supported by AWS Snowball Edge?

- A. AWS Server Migration Service (AWS SMS)
- B. Amazon Aurora
- C. AWS Trusted Advisor
-  D. Amazon EC2

Correct Answer: D

•Snowball Edge devices have Amazon S3 and Amazon EC2 compatible endpoints available, enabling programmatic use cases.

114) Which AWS shared responsibility controls are shared? (Select two.)

-  A. Awareness and training
- B. Patching of Amazon RDS
-  C. Configuration management
- D. Physical and environmental controls
- E. Service and communications protection or security

Correct Answer: AC

Shared Controls – Controls which apply to both the infrastructure layer and customer layers, but in completely separate contexts or perspectives. In a shared control, AWS provides the requirements for the infrastructure and the customer must provide their own control implementation within their use of AWS services. Examples include:

- Patch Management – AWS is responsible for patching and fixing flaws within the infrastructure, but customers are responsible for patching their guest OS and applications.
- Configuration Management – AWS maintains the configuration of its infrastructure devices, but a customer is responsible for configuring their own guest operating systems, databases, and applications.
- Awareness & Training - AWS trains AWS employees, but a customer must train their own employees.

115) A company does not want to rely on elaborate forecasting to determine its usage of compute resources. Instead, the company wants to pay only for the resources that it uses. The company also needs the ability to increase or decrease its resource usage to meet business requirements. Which pillar of the AWS Well-Architected Framework aligns with these requirements?

- A. Operational excellence
- B. Security
- C. Reliability
-  D. Cost optimization

Correct Answer: D

Cost optimization is a continual process of refinement and improvement over the span of a workload's lifecycle. The practices in this paper help you build and operate cost-aware workloads that achieve business outcomes while minimizing costs and allowing your organization to maximize its return on investment.

116) After a single Availability Zone service disruption, a corporation must guarantee that the endpoint for a database instance stays the same. The program must continue database operations without human intervention from an administrator.

How are these stipulations to be met?

- A. Use multiple Amazon Route 53 routes to the standby database instance endpoint hosted on AWS Storage Gateway.
-  B. Configure Amazon RDS Multi-Availability Zone deployments with automatic failover to the standby.
- C. Add multiple Application Load Balancers and deploy the database instance with AWS Elastic Beanstalk.
- D. Deploy a single Network Load Balancer to distribute incoming traffic across multiple Amazon CloudFront origins.

Correct Answer: B

Multi-AZ deployments can have one standby or two standby DB instances. When the deployment has one standby DB instance, it's called a *Multi-AZ DB instance deployment*. A Multi-AZ DB instance deployment has one standby DB instance that provides failover support, but doesn't serve read traffic. When the deployment has two standby DB instances, it's called a *Multi-AZ DB cluster deployment*. A Multi-AZ DB cluster deployment has standby DB instances that provide failover support and can also serve read traffic.

117) Which costs must be addressed when comparing AWS Cloud vs on-premises Total Cost of Ownership? (Select two.)

- A. Software development
- B. Project management
-  C. Storage hardware
-  D. Physical servers
- E. Antivirus software license

Correct Answer: CD

To get the most out of your estimates, you should have a good idea of your basic requirements. For example, if you're going to try Amazon Elastic Compute Cloud (Amazon EC2), it might help if you know what kind of operating system you need, what your memory requirements are, and how much I/O you need. You should also decide whether you need storage, such as if you're going to run a database and how long you intend to use the servers. You don't need to make these decisions before generating an estimate, though. You can play around with the service configuration and parameters to see which options fit your use case and budget best.

118) A company is migrating to the AWS Cloud. The company requires consultative review and guidance for its applications during the migration. After the migration is complete, the company requires a response within 30 minutes if business-critical systems go down.

Which AWS Support plans meet these requirements? (Choose two.)

-  A. AWS Enterprise Support
-  B. AWS Enterprise On-Ramp Support
- C. AWS Developer Support
- D. AWS Basic Support
- E. AWS Business Support

Correct Answer: AB

With Enterprise On-Ramp, you get 24x7 technical support from high-quality engineers, tools and technology to automatically manage health of your environment, consultative architectural guidance delivered in the context of your applications and use-cases, and a pool of Technical Account Managers (TAMs) to coordinate access to proactive / preventative programs and AWS subject matter experts.

119) Which AWS products anticipate future AWS expenses automatically?

- A. AWS Support Center
- B. AWS Total Cost of Ownership (TCO) Calculator
- C. AWS Simple Monthly Calculator
-  D. Cost Explorer

Correct Answer: D

AWS Cost Explorer has an easy-to-use interface that lets you visualize, understand, and manage your AWS costs and usage over time.

120) Which functionality may be utilized to prevent inadvertent overwrites or deletions of Amazon S3 buckets?

- ☐ A. Lifecycle policy
- ☒ B. Object versioning
- ☐ C. Server-side encryption
- ☐ D. Bucket ACL

Correct Answer: B

S3 Object Versioning is one of the most salient features in Amazon S3. Object versioning is used to avoid unintended overwrite and deletion. Versioning is not enabled by default and this feature is used to keep multiple versions of objects at the same time in the bucket.

If we enable versioning on the bucket, we cannot delete an object directly. All versions remain in the bucket and a delete marker is introduced which becomes the current version. So if you need to delete an object, you need to remove that delete marker also. Existing objects in your bucket do not change and only future requests behavior changes. If we put an object retrieval request, the current version of the object will always return.

121) Amazon DynamoDB is used by a business in its AWS Cloud architecture.

Which of the following is a duty of the organization, according to the AWS shared responsibility model? (Select two.)

- ☐ A. Operating system patching and upgrades
- ☒ B. Application of appropriate permissions with IAM tools
- ☒ C. Configuration of data encryption options
- ☐ D. Creation of DynamoDB endpoints
- ☐ E. Infrastructure provisioning and maintenance

Correct Answer: BC

Cloud security at AWS is the highest priority. As an AWS customer, you benefit from a data center and network architecture that is built to meet the requirements of the most security-sensitive organizations.

Security is a shared responsibility between AWS and AWS customer.

122) A company wants to launch its workload on AWS and requires the system to automatically recover from failure. Which pillar of the AWS Well-Architected Framework includes this requirement?

- A. Cost optimization
- B. Operational excellence
- C. Performance efficiency
-  D. Reliability

Correct Answer: D

The reliability pillar includes the ability of a system to recover from infrastructure or service disruptions, dynamically acquire computing resources to meet demand, and mitigate disruptions such as misconfigurations or transient network issues.

123) Which AWS service should a company use to create a NoSQL database?

- A. Amazon Aurora
-  B. Amazon DynamoDB
- C. Amazon Redshift
- D. Amazon Neptune

Correct Answer: B

AWS DynamoDB is a fully managed NoSQL database service that is designed to provide fast and predictable performance with seamless scalability. It is a good choice for companies looking to create a NoSQL database in AWS.

124) Which AWS service or functionality is utilized by distributed applications to send text and email messages?

-  A. Amazon Simple Notification Service (Amazon SNS)
- B. Amazon Simple Email Service (Amazon SES)
- C. Amazon CloudWatch alerts
- D. Amazon Simple Queue Service (Amazon SQS)

Correct Answer: A

Amazon Simple Notification Service (Amazon SNS) is a fully managed messaging service for both application-to-application (A2A) and application-to-person (A2P) communication. The A2A pub/sub functionality provides topics for high-throughput, push-based, many-to-many messaging between distributed systems, microservices, and event-driven serverless applications. Using Amazon SNS topics, your publisher systems can fanout messages to a large number of subscriber systems, including Amazon SQS queues, AWS Lambda functions, HTTPS endpoints, and Amazon Kinesis Data Firehose, for parallel processing. The A2P functionality enables you to send messages to users at scale via SMS, mobile push, and email.

125) To achieve high availability, how many Availability Zones should computing resources be provided across?

- A. A minimum of one
-  B. A minimum of two
- C. A minimum of three
- D. A minimum of four or more

Correct Answer: B

High availability requires at least **two availability zones**. The idea is that only one zone will go down at a time: the proverbial backhoe cutting power and network cables. Since Amazon isolates the data centers for each availability zone, that backhoe won't take out more than one AZ.

126) Which of the following is AWS's obligation under the AWS shared responsibility model?

- A. Data encryption in transit
-  B. Firmware updates on hardware
- C. Operating system patching on Amazon EC2 instances
- D. Data encryption at rest

Correct Answer: B

Under the **shared responsibility model**, AWS is responsible for the hardware and software that run AWS services. This applies to AWS Outposts, just as it does to an AWS Region. For example, AWS manages security patches, updates firmware, and maintains the Outpost equipment.

127) To boost availability, a user intends to create two more Amazon EC2 instances.

What should the user do?

-  A. Launch the instances across multiple Availability Zones in a single AWS Region.
- B. Launch the instances as EC2 Reserved Instances in the same AWS Region and the same Availability Zone.
- C. Launch the instances in multiple AWS Regions, but in the same Availability Zone.
- D. Launch the instances as EC2 Spot Instances in the same AWS Region, but in different Availability Zones.

Correct Answer: A

AWS operates state-of-the-art, highly available data centers. Although rare, failures can occur that affect the availability of instances that are in the same location. If you host all of your instances in a single location that is affected by a failure, none of your instances would be available.

128) What are the customer's duties under the AWS shared responsibility model?
(Select two.)

- A. Physical and environmental security
- B. Physical network devices including firewalls
- C. Storage device decommissioning
-  D. Security of data in transit
-  E. Data integrity authentication

Correct Answer: DE

Customers' responsibility is the **security of everything they make in AWS Cloud**. Customers (you) have complete control over your content. Customer manages AWS services, software, and access to the data.

129) A large enterprise with multiple VPCs in several AWS Regions around the world needs to connect and centrally manage network connectivity between its VPCs.
Which AWS service or feature meets these requirements?

- A. AWS Direct Connect
-  B. AWS Transit Gateway
- C. AWS Site-to-Site VPN
- D. VPC endpoints

Correct Answer: B

AWS Transit Gateway connects your Amazon Virtual Private Clouds (VPCs) and on-premises networks through a central hub. This connection simplifies your network and puts an end to complex peering relationships. Transit Gateway acts as a highly scalable cloud router—each new connection is made only once.

Deliver applications around the world

Build, deploy, and manage applications across thousands of Amazon VPCs without having to manage peering connections or update routing tables.

130) Which AWS service should be used to monitor Amazon EC2 instances for CPU and network utilization?

- A. Amazon Inspector
- B. AWS CloudTrail
-  C. Amazon CloudWatch
- D. AWS Config

Correct Answer: C

Using Amazon CloudWatch, which is a monitoring service provided by AWS. CloudWatch provides metrics and logs for ECS, such as CPU and memory usage for the container instances, the number of tasks and services running, and the number of containers that are being deployed or stopped.

131) How can customers minimize the amount of time they spend patching their operating systems by migrating to the AWS Cloud? (Select two.)

-  A. Users can take advantage of managed services on AWS.
- B. Users can outsource operating system patching to the AWS Support team.
- C. AWS Professional Services will upgrade instances to the latest operating system versions.
- D. Users have the ability to use license-included Amazon EC2 instances.
-  E. Users can take advantage of AWS Systems Manager features.

Correct Answer: AE

Patch Manager, a capability of AWS Systems Manager, automates the process of patching managed nodes with both security-related updates and other types of updates.

132) In the case of an environmental disruption, a company needs to make sure its infrastructure is structured for fault tolerance and business continuity. Which parts of the AWS architecture should the organization replicate?

- A. Edge locations
- B. Availability Zones
-  C. Regions
- D. Amazon Route 53

Correct Answer: C

AWS has the concept of a Region, which is a physical location around the world where we cluster data centers. We call each group of logical data centers an Availability Zone. Each AWS Region consists of a minimum of three, isolated, and physically separate AZs within a geographic area. Unlike other cloud providers, who often define a region as a single data center, the multiple AZ design of every AWS Region offers advantages for customers. Each AZ has independent power, cooling, and physical security and is connected via redundant, ultra-low-latency networks.

133) According to which AWS cloud design guideline, systems should minimize their interdependence?

- A. Scalability
- B. Services, not servers
- C. Removing single points of failure
-  D. Loose coupling

Correct Answer: D

AWS helps loose coupled architecture that reduces interdependencies, a change or failure in a component does not cascade to other components

134) Which tasks need the root user credentials for an AWS account? (Select two.)

- A. Creating an Amazon EC2 key pair
- B. Removing an IAM user from the administrators group
-  C. Changing the AWS Support plan
- D. Creating an Amazon CloudFront key pair
-  E. Granting an IAM user full administrative access

Correct Answer: CE

There are only a few tasks that require you to use the root user:

- Change your account settings. This includes the account name, email address, root user password, and root user access keys.
- View certain tax invoices.
- Close your AWS account.
- Restore IAM user permissions.
- Change your AWS Support plan or Cancel your AWS Support plan.

135) A user wants to deploy a service to the AWS Cloud by using infrastructure-as-code (IaC) principles. Which AWS service can be used to meet this requirement?

- A. AWS Systems Manager
-  B. AWS CloudFormation
- C. AWS CodeCommit
- D. AWS Config

Correct Answer: B

Use **AWS CloudFormation** templates: **AWS CloudFormation** is a service that helps you model and set up your Amazon Web Services resources, making it easier to manage and automate your infrastructure. With CloudFormation, you can define your infrastructure as code using templates written in JSON or YAML.

136) A company that has multiple business units wants to centrally manage and govern its AWS Cloud environments. The company wants to automate the creation of AWS accounts, apply service control policies (SCPs), and simplify billing processes. Which AWS service or tool should the company use to meet these requirements?

-  A. AWS Organizations
- B. Cost Explorer
- C. AWS Budgets
- D. AWS Trusted Advisor

Correct Answer: A

AWS Organizations provides you with the capability to centrally manage and govern your cloud environment. You can manage and organize your accounts under a single bill, set central policies and configuration requirements for your entire organization, create custom permissions or capabilities within the organization, and delegate responsibilities to other accounts so they can manage on behalf of the organization.

In addition, AWS Organizations is integrated with other AWS services so you can define central configurations, security mechanisms, audit requirements, and resource sharing across accounts in your organization.

137) An administrator must first install and begin utilizing a popular IT product.

What resources are available to the administrator?

- A. AWS Well-Architected Framework documentation
- B. Amazon CloudFront
- C. AWS CodeCommit
-  D. AWS Quick Start reference deployments

Correct Answer: D

Quick Starts are built by AWS solutions architects and partners to help you deploy popular technologies on AWS, based on AWS best practices for security and high availability. These accelerators reduce hundreds of manual procedures into just a few steps, so you can build your production environment quickly and start using it immediately.

138) Which solution enables users in various AWS Regions to have the FASTEST application response times for frequently requested data?

-  A. AWS CloudTrail across multiple Availability Zones
-  B. Amazon CloudFront to edge locations
- C. AWS CloudFormation in multiple regions
- D. A virtual private gateway over AWS Direct Connect

Correct Answer: B

You can deliver content and decrease end-user latency of your web application using Amazon CloudFront. CloudFront speeds up content delivery by leveraging its global network of data centers, known as edge locations, to reduce delivery time by caching your content close to your end users. CloudFront fetches your content from an origin, such as an Amazon S3 bucket, an Amazon EC2 instance, an Amazon Elastic Load Balancing load balancer or your own web server, when it's not already in an edge location. CloudFront can be used to deliver your entire website or application, including dynamic, static, streaming, and interactive content.

139) Which qualities make AWS Cloud computing advantageous? (Select two.)

- A. A 100% service level agreement (SLA) for all AWS services
-  B. Compute capacity that is adjusted on demand
- C. Availability of AWS Support for code development
-  D. Enhanced security
- E. Increases in cost and complexity

Correct Answer: BD

Application availability is crucial for providing an error-free experience and for minimizing application latency. Availability depends on having resources that are accessible and have enough capacity to meet demand. AWS provides several mechanisms to manage availability. For applications hosted on Amazon ECS, these include autoscaling and Availability Zones (AZs). Autoscaling manages the number of tasks or instances based on metrics you define, while Availability Zones allow you to host your application in isolated but geographically-close locations.

140) A corporation anticipates a brief increase in internet traffic for their application. The program cannot be interrupted during the traffic spike. In addition, the organization must reduce costs while increasing flexibility. To achieve these needs, which Amazon EC2 instance type should the organization use?

- ▶ A. On-Demand Instances
- B. Spot Instances
- C. Reserved Instances
- D. Dedicated Hosts

Correct Answer: A

AWS On-Demand Instances (**Amazon Web Services On-Demand Instances**) are virtual servers that run in AWS Elastic Compute Cloud (EC2) or AWS Relational Database Service (RDS) and are purchased at a fixed rate per hour. AWS recommends using On-Demand Instances for applications with short-term, irregular workloads that cannot be interrupted.

141) Which IT controls do AWS and the customer share, according to the AWS shared responsibility model? (Choose two.)

-  A. Physical and environmental controls
- B. Patch management
-  C. Cloud awareness and training
- D. Zone security
- E. Application data encryption

Correct Answer: BC

Examples of shared controls include:

- Patch Management** – AWS is responsible for patching and fixing flaws within the infrastructure, but customers are responsible for patching their guest OS and applications.
- Configuration Management** – AWS maintains the configuration of its infrastructure devices, but a customer is responsible for configuring their own guest operating systems, databases, and applications.
- Awareness & Training** – AWS trains AWS employees, but a customer must train their own employees.

142) A company is launching an application in the AWS Cloud. The application will use Amazon S3 storage. A large team of researchers will have shared access to the data. The company must be able to recover data that is accidentally overwritten or deleted.

Which S3 feature should the company turn on to meet this requirement?

- ☐ A. Server access logging
- ☒ B. S3 Versioning
- ☐ C. S3 Lifecycle rules
- ☐ D. Encryption in transit and at rest

Correct Answer: B

Versioning-enabled buckets can help you recover objects from accidental deletion or overwrite. For example, if you delete an object, Amazon S3 inserts a delete marker instead of removing the object permanently. The delete marker becomes the current object version. If you overwrite an object, it results in a new object version in the bucket. You can always restore the previous version.

143) An Amazon RDS database instance is deployed across several Availability Zones.

Which pillar of the AWS Well-Architected Framework is included in this strategy?

- ☐ A. Performance efficiency
- ☒ B. Reliability
- ☐ C. Cost optimization
- ☐ D. Security

Correct Answer: B

The Reliability pillar includes the ability of a workload to perform its intended function correctly and consistently when it's expected to. This includes the ability to operate and test the workload through its total lifecycle.

144) Amazon EC2, an Elastic Load Balancer, and Amazon RDS are all components of an architectural design.

What is the BEST method for estimating the monthly cost of this architecture?

- A. Open an AWS Support case, provide the architecture proposal, and ask for a monthly cost estimation.
-  B. Collect the published prices of the AWS services and calculate the monthly estimate.
- C. Use the AWS Simple Monthly Calculator to estimate the monthly cost.
- D. Use the AWS Total Cost of Ownership (TCO) Calculator to estimate the monthly cost.

Correct Answer: C

- To estimate a bill, use the [AWS Pricing Calculator](#). Choose Create estimate, and then choose your planned resources by service. The AWS Pricing Calculator provides an estimated cost per month.
- To forecast your costs, use the [AWS Cost Explorer](#). Use [cost allocation tags](#) to divide your resources into groups, and then estimate the costs for each group.

145) According to the AWS shared responsibility model, which job is the customer's duty?

- A. Maintain the security of the AWS Cloud.
-  B. Configure firewalls and networks.
- C. Patch the operating system of Amazon RDS instances.
- D. Implement physical and environmental controls.

Correct Answer: B

The customer assumes responsibility and management of the guest operating system (including updates and security patches), other associated application software as well as the configuration of the AWS provided security group firewall. Customers should carefully consider the services they choose as their responsibilities vary depending on the services used, the integration of those services into their IT environment, and applicable laws and regulations. The nature of this shared responsibility also provides the flexibility and customer control that permits the deployment.

146) On Amazon EC2, a business hosts a web application in a Docker container.

Which of the following duties is AWS in charge of?

- A. Scaling the web application and services developed with Docker
- B. Provisioning or scheduling containers to run on clusters and maintain their availability
-  C. Performing hardware maintenance in the AWS facilities that run the AWS Cloud
- D. Managing the guest operating system, including updates and security patches

Correct Answer: C

AWS is responsible for protecting the infrastructure that runs all of the services offered in the AWS Cloud. This infrastructure is composed of the hardware, software, networking, and facilities that run AWS Cloud services.

147) What are the AWS Cloud's advantages? (Select two.)

- A. Fixed rate monthly cost
-  B. No need to guess capacity requirements
-  C. Increased speed to market
- D. Increased upfront capital expenditure
- E. Physical access to cloud data centers

Correct Answer: BC

Eliminate guessing on your infrastructure capacity needs. When you make a capacity decision prior to deploying an application, you often end up either sitting on expensive idle resources or dealing with limited capacity. With cloud computing, these problems go away. You can access as much or as little capacity as you need, and scale up and down as required with only a few minutes' notice.

In a cloud computing environment, new IT resources are only a click away, which means that you reduce the time to make those resources available to your developers from weeks to just minutes. This results in a dramatic increase in agility for the organization, since the cost and time it takes to experiment and develop is significantly lower.

148) An Elastic Load Balancer, numerous Amazon EC2 instances, and Amazon RDS are used to run a web application on AWS.

Which security measures are AWS's responsibility? (Select two.)

- A. Running a virus scan on EC2 instances
- ➡ B. Protecting against IP spoofing and packet sniffing
- ➡ C. Installing the latest security patches on the RDS instance
- D. Encrypting communication between the EC2 instances and the Elastic Load Balancer
- E. Configuring a security group and a network access control list (NACL) for EC2 instances

Correct Answer: BC

AWS is responsible for protecting the infrastructure that runs AWS services in the AWS Cloud. AWS also provides you with services that you can use securely. Third-party auditors regularly test and verify the effectiveness of our security as part of the AWS compliance programs.

149) A manufacturing company has a critical application that runs at a remote site that has a slow internet connection. The company wants to migrate the workload to AWS. The application is sensitive to latency and interruptions in connectivity. The company wants a solution that can host this application with minimum latency. Which AWS service or feature should the company use to meet these requirements?

- A. Availability Zones
- ➡ B. AWS Local Zones
- C. AWS Wavelength
- D. AWS Outposts

Correct Answer: B

AWS Local Zones are a type of infrastructure deployment that places compute, storage, database, and other select AWS services close to large population and industry centers.

Run low-latency applications at the edge

Build and deploy applications close to end users to enable real-time gaming, live streaming, augmented and virtual reality (AR/VR), virtual workstations, and more.

150) Which AWS service or feature facilitates the purchase and deployment of third-party software by providing an online, managed software catalog?

- A. AWS Support
-  B. AWS Marketplace
- C. Amazon EC2 private Amazon Machine Images (AMIs)
- D. AWS reseller programs

Correct Answer: B

AWS Marketplace is a curated digital catalog that makes it easy for customers to find, buy, deploy, and manage the third-party software.

151) A business wishes to improve its capacity for infrastructure recovery in the event of a natural catastrophe.

This capability corresponds to which pillar of the AWS Well-Architected Framework?

- A. Cost optimization
- B. Performance efficiency
-  C. Reliability
- D. Security

Correct Answer: C

There are five design principles for reliability in the cloud:

- **Automatically recover from failure**
- **Test recovery procedures**
- **Scale horizontally to increase aggregate workload availability**
- **Stop guessing capacity**
- **Manage change in automation**

152) A business may be required to operate its workload exclusively in its on-premises data center due to performance and regulatory limitations.

Which Amazon Web Services (AWS) services or resources should the business utilize? (Select two.)

- A. Amazon Pinpoint
- B. Amazon WorkLink
-  C. AWS Outposts
-  D. AWS Snowball Edge
- E. AWS AppSync

Correct Answer: CD

AWS Outposts is a family of fully managed solutions delivering AWS infrastructure and services to virtually any on-premises or edge location for a truly consistent hybrid experience. Outposts solutions allow you to extend and run native AWS services on premises, and is available in a variety of form factors, from 1U and 2U Outposts servers to 42U Outposts racks, and multiple rack deployments.

AWS Snowball Edge is a type of Snowball device with on-board storage and compute power for select AWS capabilities. Snowball Edge can do local processing and edge-computing workloads in addition to transferring data between your local environment and the AWS Cloud.

153) For security reasons, a business demands an isolated environment inside AWS.

Which course of action is necessary to achieve this?

- A. Create a separate Availability Zone to host the resources.
-  B. Create a separate VPC to host the resources.
- C. Create a placement group to host the resources.
- D. Create an AWS Direct Connect connection between the company and AWS.

Correct Answer: B

Network isolation

A virtual private cloud (VPC) is a virtual network in your own logically isolated area in the AWS Cloud. Use separate VPCs to isolate infrastructure by workload or organizational entity.

A subnet is a range of IP addresses in a VPC. When you launch an instance, you launch it into a subnet in your VPC. Use subnets to isolate the tiers of your application (for example, web, application, and database) within a single VPC. Use private subnets for your instances if they should not be accessed directly from the internet.

To call the Amazon EC2 API from your VPC without sending traffic over the public internet, use AWS PrivateLink.

154) How do Amazon's massive economies of scale help customers?

-  A. Periodic price reductions as the result of Amazon's operational efficiencies
- B. New Amazon EC2 instance types providing the latest hardware
- C. The ability to scale up and down when needed
- D. Increased reliability in the underlying hardware of Amazon EC2 instances

Correct Answer: A

•**Benefit from massive economies of scale** – By using cloud computing, you can achieve a lower variable cost than you can get on your own. Because usage from hundreds of thousands of customers is aggregated in the cloud, providers such as AWS can achieve higher economies of scale, which translates into lower pay as-you-go prices.

155) A company wants to implement threat detection on its AWS infrastructure. However, the company does not want to deploy additional software.

Which AWS service should the company use to meet these requirements?

- A. Amazon VPC
- B. Amazon EC2
-  C. Amazon GuardDuty
- D. AWS Direct Connect

Correct Answer: C

Amazon GuardDuty

1. Continuously monitor your AWS accounts, instances, container workloads, users, and storage for potential threats.
2. Expose threats quickly using anomaly detection, machine learning, behavioral modeling, and threat intelligence feeds from AWS and leading third-parties.
3. Mitigate threats early by initiating automated responses.

156) Which AWS service uses edge locations?

- A. Amazon Aurora
-  B. AWS Global Accelerator
- C. Amazon Connect
- D. AWS Outposts

Correct Answer: B

AWS Global Accelerator and Amazon CloudFront are separate services that use the AWS global network and its edge locations around the world.

157) When utilizing the AWS Command Line Interface (AWS CLI), which of the following Identity and Access Management (IAM) entities is connected with an access key ID and secret access key?

- A. IAM group
-  B. IAM user
- C. IAM role
- D. IAM policy

Correct Answer: B

Access keys are long-term credentials for an IAM user or the AWS account root user. You can use access keys to sign programmatic requests to the AWS CLI or AWS API (directly or using the AWS SDK).

158) What is AWS's sole obligation under the AWS shared responsibility model?

- A. Application security
-  B. Edge location management
- C. Patch management
- D. Client-side data

Correct Answer: B

Client-side data, application security is the sole responsibility of the customer. Patch management is a shared responsibility. That leaves us with edge location management and since this out of the control of the customer, AWS is the one responsible for it.

159) Which component of the AWS architecture permits global computing and storage deployment?

- A. Availability Zones
-  B. Regions
- C. Tags
- D. Resource groups

Correct Answer: B

The AWS Cloud spans 99 Availability Zones within 31 geographic regions around the world, with announced plans for 12 more Availability Zones and 4 more AWS Regions in Canada, Israel, New Zealand, and Thailand.

160) Which of the following is a design concept associated with dependability in the AWS Well-Architected Framework?

-  A. Deployment to a single Availability Zone
- B. Ability to recover from failure
- C. Design for cost optimization
- D. Perform operations as code

Correct Answer: B

3. Reliability

The Reliability pillar encompasses the ability of a workload to perform its intended function correctly and consistently when it's expected to. This includes the ability to operate and test the workload through its total lifecycle. You can find prescriptive guidance on implementation in the [Reliability Pillar whitepaper](#).

Design Principles

There are five design principles for reliability in the cloud:

- Automatically recover from failure
- Test recovery procedures
- Scale horizontally to increase aggregate workload availability
- Stop guessing capacity
- Manage change in automation

161) Which activity is entirely the user's responsibility while executing workloads on AWS?

-  A. Patching the infrastructure components
- B. Implementing controls to route application traffic
- C. Maintaining physical and environmental controls
- D. Maintaining the underlying infrastructure components

Correct Answer: B

Customer responsibility will be determined by the AWS Cloud services that a customer selects. This determines the amount of configuration work the customer must perform as part of their security responsibilities. For example, a service such as Amazon Elastic Compute Cloud (Amazon EC2) is categorized as Infrastructure as a Service (IaaS) and, as such, requires the customer to perform all of the necessary security configuration and management tasks. Customers that deploy an Amazon EC2 instance are responsible for management of the guest operating system (including updates and security patches), any application software or utilities installed by the customer on the instances, and the configuration of the AWS-provided firewall (called a security group) on each instance. For abstracted services, such as Amazon S3 and Amazon DynamoDB, AWS operates the infrastructure layer, the operating system, and platforms, and customers access the endpoints to store and retrieve data. Customers are responsible for managing their data (including encryption options), classifying their assets, and using IAM tools to apply the appropriate permissions.

162) Which statement best describes the AWS Cloud's agility?

- A. Agility gives users the ability to host applications in multiple AWS Regions around the world.
- B. Agility gives users the ability to pay upfront to reduce cost.
- C. Agility provides customizable physical hardware at the lowest possible cost.
-  D. Agility provides the means for users to provision resources in minutes.

Correct Answer: D

In a cloud computing environment, new IT resources are only a click away, which means that you reduce the time to make those resources available to your developers from weeks to just minutes. This results in a dramatic increase in agility for the organization, since the cost and time it takes to experiment and develop is significantly lower.

163) Which AWS product or service enables businesses to monitor and classify their expenditure at a precise level?

-  A. Cost allocation tags
- B. Consolidated billing
- C. AWS Budgets
- D. AWS Marketplace

Correct Answer: A

AWS cost allocation tags are **labels for classifying, organizing, and identifying your resources**. These tags, which are key-value pairs, make it easier to manage them and track their usage. By assigning tags to your resources, you can track your overall AWS costs, either through Cost Explorer or through the AWS API.

165) Which AWS service is responsible for monitoring the health of your applications automatically?

- A. Amazon API Gateway
- ➔ B. AWS Elastic Beanstalk
- C. AWS Lambda
- D. AWS Config

Correct Answer: B

With Elastic Beanstalk, you can quickly deploy and manage applications in the AWS Cloud without having to learn about the infrastructure that runs those applications. Elastic Beanstalk reduces management complexity without restricting choice or control. You simply upload your application, and Elastic Beanstalk automatically handles the details of capacity provisioning, load balancing, scaling, and application health monitoring.

166) A user is developing a service that adheres to the AWS Well-Architected Framework's operational excellence pillar.

- Which design concept is the user to adhere to?
- ➔ A. Anticipate failure
 - B. Make large-scale changes
 - C. Perform manual operations
 - D. Create static operational procedures

Correct Answer: A

1. Operational Excellence

The Operational Excellence pillar includes the ability to support development and run workloads effectively, gain insight into their operation, and continuously improve supporting processes and procedures to deliver business value. You can find prescriptive guidance on implementation in the [Operational Excellence Pillar whitepaper](#).

Design Principles

There are five design principles for operational excellence in the cloud:

- Perform operations as code
- Make frequent, small, reversible changes
- Refine operations procedures frequently
- Anticipate failure
- Learn from all operational failures

168) Which AWS service or feature checks access policies and offers actionable recommendations to help users set secure and functional policies?

-  A. AWS Systems Manager
- B. AWS IAM Access Analyzer
- C. AWS Trusted Advisor
- D. Amazon GuardDuty

Correct Answer: B

IAM Access Analyzer generates a finding for each instance of a resource-based policy that grants access to a resource within your zone of trust to a principal that is not within your zone of trust.

IAM Access Analyzer reports a security warning when your policy grants access to pass any role to any service, which is overly permissive. The security warning includes a recommendation that you scope down the permissions to pass specific role(s) instead.

169) Which design concept is fulfilled by adhering to the AWS Well-Architected Framework's dependability pillar?

- A. Vertical scaling
- B. Manual failure recovery
-  C. Testing recovery procedures
- D. Changing infrastructure manually

Correct Answer: C

•**Test recovery procedures:** In an on-premises environment, testing is often conducted to prove that the workload works in a particular scenario. Testing is not typically used to validate recovery strategies. In the cloud, you can test how your workload fails, and you can validate your recovery procedures. You can use automation to simulate different failures or to recreate scenarios that led to failures before. This approach exposes failure pathways that you can test and fix before a real failure scenario occurs, thus reducing risk.

170) Which AWS service or functionality can assist a business in determining if it has publicly accessible Amazon S3 buckets?

- A. AWS Service Health Dashboard
- B. Amazon CloudWatch Logs
-  C. AWS Trusted Advisor
- D. AWS Service Catalog

Correct Answer: C

Trusted Advisor can help improve the security of your AWS environment by suggesting foundational security best practices curated by security experts. Examples include identifying RDS security group access risk, exposed access keys, and unnecessary S3 bucket permissions.

171) Seasonal sales surges occur many times a year for an online retailer, most notably during the holidays. At other times of year, demand is lower. The corporation has difficulty forecasting the seasonal increase in infrastructure demand.

Which benefits of migrating to the AWS Cloud would be the most beneficial to the business? (Select two.)

- A. Global footprint
-  B. Elasticity
- C. AWS service quotas
- D. AWS shared responsibility model
-  E. Pay-as-you-go pricing

Correct Answer: BE

Benefits of migrating to the AWS Cloud

- Trade fixed expense for variable expense –Benefit from massive economies of scale –
- Stop guessing capacity
- Increase speed and agility
- Stop spending money running and maintaining data centers -
- Go global in minutes –

172) Which concepts of AWS Cloud architecture may assist boost reliability?
(Select two.)

- A. Using monolithic architecture
- B. Measuring overall efficiency
-  C. Testing recovery procedures
- D. Adopting a consumption model
-  E. Automatically recovering from failure

Correct Answer: CE

- **Automatically recover from failure:** By monitoring a workload for key performance indicators (KPIs), you can trigger automation when a threshold is breached. These KPIs should be a measure of business value, not of the technical aspects of the operation of the service. This allows for automatic notification and tracking of failures, and for automated recovery processes that work around or repair the failure. With more sophisticated automation, it's possible to anticipate and remediate failures before they occur.
- **Test recovery procedures:** In an on-premises environment, testing is often conducted to prove that the workload works in a particular scenario. Testing is not typically used to validate recovery strategies. In the cloud, you can test how your workload fails, and you can validate your recovery procedures. You can use automation to simulate different failures or to recreate scenarios that led to failures before. This approach exposes failure pathways that you can test and fix before a real failure scenario occurs, thus reducing risk.

173) Which sort of storage does Amazon Elastic File System (Amazon EFS) and Amazon FSx provide?

-  A. File storage
- B. Object storage
- C. Block storage
- D. Instance store

Correct Answer: A

Both these services provide file storage, the major difference being that FSx integrates with Windows environments.

174) Which AWS service enables customers to view AWS compliance control reports on-demand and self-service?

- A. AWS Config
- B. Amazon GuardDuty
- C. AWS Trusted Advisor
-  D. AWS Artifact

Correct Answer: D

AWS Artifact provides a central resource for AWS security and compliance reports. The artifacts available in AWS Artifact include Service Organization Control (SOC) reports, Payment Card Industry (PCI) reports, and certifications from accreditation bodies that validate the implementation and operating effectiveness of AWS security controls. Additionally, AWS Artifact provides on-demand access to the security and compliance documents such as ISO certifications, and Service Organization Control (SOC) reports of the Independent Software Vendors (ISVs) who sell their products on AWS Marketplace.

175) What may aid in the evaluation of a cloud-based application? (Select two.)

- A. AWS Trusted Advisor
-  B. AWS Professional Services
- C. AWS Systems Manager
-  D. AWS Partner Network (APN)
- E. AWS Secrets Manager

Correct Answer: BD

Professional Services available in AWS Marketplace enables you to find and buy assessments, implementation, support, managed services, and training for third-party software and building on AWS. AWS Marketplace helps you find the software and associated services you need to innovate all in one place, simplifying procurement. You can discover complete business solutions and curated service offerings from independent software vendors and consulting partners, and select payment options and contract terms that fit your needs.

The AWS Partner Network (APN) is a global community of partners that leverages programs, expertise, and resources to build, market, and sell customer offerings.

176) Which AWS service enables expense control across numerous AWS accounts effectively?

-  A. AWS Organizations
- B. AWS Trusted Advisor
- C. AWS Direct Connect
- D. Amazon Connect

Correct Answer: A

AWS Organizations provides you with the capability to centrally manage and govern your cloud environment. You can manage and organize your accounts under a single bill, set central policies and configuration requirements for your entire organization, create custom permissions or capabilities within the organization, and delegate responsibilities to other accounts so they can manage on behalf of the organization.

In addition, AWS Organizations is integrated with other AWS services so you can define central configurations, security mechanisms, audit requirements, and resource sharing across accounts in your organization.

177) A company's on-premises servers and the AWS Cloud need a dedicated network connection.

Which Amazon Web Services (AWS) service should be used?

- A. AWS VPN
-  B. AWS Direct Connect
- C. Amazon API Gateway
- D. Amazon Connect

Correct Answer: B

You can use AWS Direct Connect to establish a private virtual interface from your on-premise network directly to your Amazon VPC, providing you with a private, high bandwidth network connection between your network and your VPC. With multiple virtual interfaces, you can even establish private connectivity to multiple VPCs while maintaining network isolation.

178) On-premises resources have been underused by a user.

Which AWS Cloud idea is optimal for resolving this issue?

- A. High availability
-  B. Elasticity
- C. Security
- D. Loose coupling

Correct Answer: B

Most people, when thinking of cloud computing, think of the ease with which they can procure resources when needed. This is only one aspect to elasticity. The other aspect is to contract when they no longer need resources. Scale out and scale in. Scale up and scale down.

179) Which of the following advantages does Amazon Relational Database Service (Amazon RDS) provide over conventional database management?

- A. AWS manages the data stored in Amazon RDS tables.
-  B. AWS manages the maintenance of the operating system.
- C. AWS automatically scales up instance types on demand.
- D. AWS manages the database type.

Correct Answer: B

Amazon RDS will make sure that the relational database software powering your deployment stays up to date with the latest patches. You can exert optional control over when and if your database instance is patched.

180) Service control policies (SCPs) manage permissions for which of the following?

- A. Availability Zones
- B. AWS Regions
-  C. AWS Organizations
- D. Edge locations

Correct Answer: C

AWS Organizations provides you with the capability to centrally manage and govern your cloud environment. You can manage and organize your accounts under a single bill, set central policies and configuration requirements for your entire organization, create custom permissions or capabilities within the organization, and delegate responsibilities to other accounts so they can manage on behalf of the organization.

181) Which AWS service can be used to encrypt data at rest?

- A. Amazon GuardDuty
- B. AWS Shield
- C. AWS Security Hub
-  D. AWS Key Management Service (AWS KMS)

Correct Answer: D

[AWS Key Management Service](#) (KMS) – AWS KMS is a managed service that enables easy creation and control of encryption keys used to encrypt data. KMS uses envelope encryption in which data is encrypted using a data key that is then encrypted using a master key. Master keys can also be used to encrypt and decrypt up to 4 kilobytes of data. In our solution, I use KMS encrypt/decrypt APIs to encrypt the encrypted file system's password. See more information about [envelope encryption](#).

182) A cloud practitioner has a seldom run data analysis job that can be stopped without causing damage.

Which Amazon EC2 purchase option should be utilized to maximize cost savings?

- A. On-Demand Instances
- B. Reserved Instances
-  C. Spot Instances
- D. Dedicated Hosts

Correct Answer: C

AWS allows you to scale services for short-term usage and helping you reduce your bill. As Spot Instances can be reclaimed with a two minute warning when EC2 needs capacity back, Spot Instances are a great fit for fault-tolerant, stateless applications. Use Spot Instances to significantly lower your cost and increase the elasticity of your application.

183) What are the advantages of AWS Cloud service billing consolidation? (Select two.)

-  A. Volume discounts
- B. A minimal additional fee for use
-  C. One bill for multiple accounts
- D. Installment payment options
- E. Custom cost and usage budget creation

Correct Answer: AC

Consolidated billing has the following benefits:

- **One bill** – You get one bill for multiple accounts.
- **Easy tracking** – You can track the charges across multiple accounts and download the combined cost and usage data.
- **Combined usage** – You can combine the usage across all accounts in the organization to share the volume pricing discounts, Reserved Instance discounts, and Savings Plans. This can result in a lower charge for your project, department, or company than with individual standalone accounts. For more information, see [Volume discounts](#).
- **No extra fee** – Consolidated billing is offered at no additional cost.

184) Which Amazon EC2 instance type is necessary when a user wishes to use their current per-socket, per-core, or per-virtual machine software licenses on a Microsoft Windows server operating on AWS?

- A. Spot Instances
- B. Dedicated Instances
-  C. Dedicated Hosts
- D. Reserved Instances

Correct Answer: C

Dedicated Hosts

[PDF](#) [Kindle](#) [RSS](#)

An Amazon EC2 Dedicated Host is a physical server with EC2 instance capacity fully dedicated to your use. Dedicated Hosts allow you to use your existing per-socket, per-core, or per-VM software licenses, including Windows Server, Microsoft SQL Server, SUSE, and Linux Enterprise Server.

For information about the configurations supported on Dedicated Hosts, see [Dedicated Hosts Configuration](#).

185) When users connect to a website with a worldwide consumer base, they report experiencing delay.

Which Amazon Web Services (AWS) offering will enhance the user experience by lowering latency?

-  A. Amazon CloudFront
- B. AWS Direct Connect
- C. Amazon EC2 Auto Scaling
- D. AWS Transit Gateway

Correct Answer: A

CloudFront delivers your content through a worldwide network of data centers called edge locations. When a user requests content that you're serving with CloudFront, the request is routed to the edge location that provides the lowest latency (time delay), so that content is delivered with the best possible performance.

186) Which capabilities or services are available for monitoring an AWS account's charges and expenses? (Select two.)

- ☒ A. AWS Cost and Usage report
- B. AWS product pages
- C. AWS Simple Monthly Calculator
- ☒ D. Billing alerts and Amazon CloudWatch alarms
- E. AWS Price List API

Correct Answer: AD

With AWS Cost and Usage Reports (CUR), you can review, itemize, and organize the most comprehensive cost and usage data for your account.

You can monitor your estimated AWS charges by using Amazon CloudWatch. When you enable the monitoring of estimated charges for your AWS account, the estimated charges are calculated and sent several times daily to CloudWatch as metric data.

187) Which service enables users to store data in the Amazon Web Services cloud?

- ☒ A. Amazon EFS
- B. Amazon Redshift
- C. Amazon RDS
- D. Amazon VPC

Correct Answer: A

Amazon EFS is a fully managed service providing NFS shared file system storage for Linux workloads. Amazon EFS makes it quick and efficient to create and configure file systems. You needn't worry about managing file servers or storage, updating hardware, configuring software, or performing backups. In seconds, create a fully managed file system using the AWS Management Console, the AWS Command Line Interface (CLI), or an AWS SDK.

188) Which AWS service or product enables an organization to automate the delivery of application changes?

- A. Amazon AppFlow
-  B. AWS CodeDeploy
- C. AWS PrivateLink
- D. Amazon EKS Distro

Correct Answer: B

AWS CodeDeploy is a fully managed deployment service that automates software deployments to various compute services, such as Amazon Elastic Compute Cloud (EC2), Amazon Elastic Container Service (ECS), AWS Lambda, and your on-premises servers. Use CodeDeploy to automate software deployments, eliminating the need for error-prone manual operations.

189) A corporation wishes to minimize the physical footprint of the computing resources used by developers to execute programs.

Which service would enable serverless architectures to address this need?

- A. Amazon Elastic Compute Cloud (Amazon EC2)
-  B. AWS Lambda
- C. Amazon DynamoDB
- D. AWS CodeCommit

Correct Answer: B

AWS Lambda is a serverless, event-driven compute service that lets you run code for virtually any type of application or backend service without provisioning or managing servers. You can trigger Lambda from over 200 AWS services and software as a service (SaaS) applications, and only pay for what you use.

190) A Load Balancer Elastic enables online traffic to be distributed across multiple:

- A. AWS Regions.
-  B. Availability Zones.
- C. Dedicated Hosts.
- D. Amazon S3 buckets.

Correct Answer: B

AWS Elastic Load Balancing (ELB) distributes incoming application **traffic** automatically across **multiple** targets such as containers, EC2 instances, and IP addresses in one or more **availability zones**. This distributes and balances how frontend traffic reaches backend servers and increases the fault tolerance and availability of user applications.

191) What is AWS Storage Gateway's purpose?

- A. It ensures on-premises data storage is 99.999999999% durable.
- B. It transports petabytes of data to and from AWS.
- C. It connects to multiple Amazon EC2 instances.
-  D. It connects on-premises data storage to the AWS Cloud.

Correct Answer: D

Moving data to the cloud is not quite as simple as flipping a switch. For companies that have managed their own data centers or server rooms for decades, there are a few steps to consider -- and it's not always wise to pull the plug on an internal infrastructure quite so quickly. If a startup uses on-premise business servers and then experiences unexpected growth, abandoning those servers doesn't make sense (even if the long-term plan is to do exactly that). AWS Storage Gateway is a way to bridge this gap for companies of any size. It's a hybrid storage option that connects on-premise storage including age-old tape backup systems to the cloud in a way that also provides one console to access all storage configurations.

192) How can a client anticipate future expenses associated with the operation of a new web application?

- A. Amazon Aurora Backtrack
- B. Amazon CloudWatch Billing Alarms
- C. AWS Simple Monthly Calculator
-  D. AWS Cost and Usage report

Correct Answer: D

You can use Cost explorer which is part of Cost and Usage report to forecast future costs of running an application.

193) The term "fault tolerance" relates to the following:

- A. the ability of an application to accommodate growth without changing design
- B. how well and how quickly an application's environment can have lost data restored
- C. how secure your application is
-  D. the built-in redundancy of an application's components

Correct Answer: D

AWS recommends connecting from **multiple data centers** for physical location redundancy. When designing remote connections, consider using redundant hardware and telecommunications providers.

194) Which AWS service enables conventional SQL queries against stored datasets straight from Amazon S3?

- A. AWS Glue
- B. AWS Data Pipeline
- C. Amazon CloudSearch
-  D. Amazon Athena

Correct Answer: D

Amazon Athena is defined as an interactive query service that makes it easy to analyse data directly in Amazon Simple Storage Service (Amazon S3) using standard SQL. So, it's another SQL query engine for large data sets stored in S3. This is very similar to other SQL query engines, such as Apache Drill. But unlike Apache Drill, Athena is limited to data only from Amazon's own S3 storage service. However, Athena is able to query a variety of file formats, including, but not limited to CSV, Parquet, JSON, etc.

195) A startup is developing a new application that must be launched immediately. In the near future, the application criteria may need to be changed.

Which of the following is an AWS Cloud feature that would fulfill this particular requirement?

- A. Elasticity
- B. Reliability
- C. Performance
-  D. Agility

Correct Answer: D

Cloud infrastructure provides more agility and responsiveness than traditional IT environments. This requires organizations to think differently about how they design, build, and manage applications.

196) Which characteristics are advantages of using the AWS Cloud? (Choose two.)

- ☐ A. A 100% service level agreement (SLA) for all AWS services
- ☒ B. Compute capacity that is adjusted on demand
- ☐ C. Availability of AWS Support for code development
- ☒ D. Enhanced security
- ☐ E. Increases in cost and complexity

Correct Answer: BD

Six advantages of cloud computing

- Trade fixed expense for variable expense –
- Benefit from massive economies of scale –
- Stop guessing capacity –
- Increase speed and agility –
- Stop spending money running and maintaining data centers –
- Go global in minutes –

197) A company wants to convert video files and audio files from their source format into a format that will play on smartphones, tablets, and web browsers.

Which AWS service will meet these requirements?

- ☒ A. Amazon Elastic Transcoder
- ☐ B. Amazon Comprehend
- ☐ C. AWS Glue
- ☐ D. Amazon Rekognition

Correct Answer: A

Amazon Elastic Transcoder lets you convert media files that you have stored in Amazon Simple Storage Service (Amazon S3) into media files in the formats required by consumer playback devices. For example, you can convert large, high-quality digital media files into formats that users can play back on mobile devices, tablets, web browsers, and connected televisions.

198) A company has several departments. Each department has its own AWS accounts for its applications. The company wants all AWS costs on a single invoice to simplify payment, but the company wants to know the costs that each department is incurring. Which AWS tool or feature will provide this functionality?

-  A. AWS Cost and Usage Reports
- B. Consolidated billing
- C. Savings Plans
- D. AWS Budgets

Correct Answer: B

Consolidated billing is a feature of AWS Organizations that allows a single AWS account to pay the bills for multiple AWS accounts. This can be useful for companies that have multiple AWS accounts, as it allows them to see all of their costs on a single invoice, while still being able to track the costs of each department separately.

199) A company wants to eliminate the need to guess infrastructure capacity before deployments. The company also wants to spend its budget on cloud resources only as the company uses the resources. Which advantage of the AWS Cloud matches the company's requirements?

- A. Reliability
- B. Global reach
- C. Economies of scale
-  D. Pay-as-you-go pricing

Correct Answer: D

By using cloud computing, you can achieve a lower variable cost than you can get on your own. Because usage from hundreds of thousands of customers is aggregated in the cloud, providers such as AWS can achieve higher economies of scale, which translates into lower pay as-you-go prices.

200) Which AWS service must be enabled in order for the AWS Management Console to monitor all user account changes?

-  A. AWS CloudTrail
- B. Amazon Simple Notification Service (Amazon SNS)
- C. VPC Flow Logs
- D. AWS CloudHSM

Correct Answer: A

AWS CloudTrail is a service that enables governance, compliance, operational auditing, and risk auditing of your AWS account. With CloudTrail, you can log, continuously monitor, and retain account activity related to actions across your AWS infrastructure. CloudTrail provides event history of your AWS account activity, including actions taken through the AWS Management Console, AWS SDKs, command line tools, and other AWS services. This event history simplifies security analysis, resource change tracking, and troubleshooting. In addition, you can use CloudTrail to detect unusual activity in your AWS accounts. These capabilities help simplify operational analysis and troubleshooting.

201) Recently, an ecommerce firm began using the AWS Cloud. Which security-related responsibilities fall within the purview of the business? (Select two.)

- A. Restrict who is allowed physical access to the hosts that run the company's Amazon EC2 instances.
-  B. Install security patches on Amazon EC2 Linux instances.
-  C. Choose to encrypt data at rest that is stored on Amazon S3.
- D. Wipe Amazon Elastic Block Store (Amazon EBS) volumes clean before they are decommissioned.
- E. Conduct database patching for Amazon RDS instances.

Correct Answer: BC

For abstracted services, such as Amazon S3 and Amazon DynamoDB, AWS operates the infrastructure layer, the operating system, and platforms, and customers access the endpoints to store and retrieve data. Customers are responsible for managing their data (including encryption options), classifying their assets, and using IAM tools to apply the appropriate permissions.

202) A business needs to guarantee that users of the AWS Management Console adhere to password complexity guidelines.

How can a business customize the difficulty of its passwords?

- A. Using an AWS IAM user policy
- B. Using an AWS Organizations service control policy (SCP)
-  C. Using an AWS IAM account password policy
- D. Using an AWS Security Hub managed insight

Correct Answer: C

You can set a custom password policy on your Amazon Web Services account to specify complexity requirements and mandatory rotation periods for your IAM users' passwords. If you don't set a custom password policy, IAM user passwords must meet the default Amazon password policy.

203) Which AWS service would determine if a security group has granted unlimited access to a resource?

-  A. AWS Trusted Advisor
- B. Amazon CloudWatch
- C. VPC Flow Logs
- D. AWS CloudTrail

Correct Answer: A

In the AWS Trusted Advisor, there are lots of controls for security configurations of your AWS resources, some examples are the following:

•**Security Groups — Specific Ports Unrestricted:** This control checks security groups for rules that allow unrestricted access to specific ports such as SSH and RDP. Unrestricted access increases opportunities for malicious activity (hacking, denial-of-service attacks, loss of data).

204) Which features are available to users while use AWS KMS?

- A. Create and manage AWS access keys for the AWS account root user
- B. Create and manage AWS access keys for an AWS account IAM user
-  C. Create and manage keys for encryption and decryption of data
- D. Create and manage keys for multi-factor authentication

Correct Answer: C

AWS Key Management Service (AWS KMS) is an encryption and key management service scaled for the cloud. AWS KMS keys and functionality are used by other AWS services, and you can use them to protect data in your own applications that use AWS.

205) On Amazon EC2 instances, a business has installed various relational databases. Each month, the database software manufacturer publishes new security updates for databases that must be deployed.

Which method is the MOST EFFECTIVE for applying security patches?

- A. Connect to each database instance on a monthly basis, and download and apply the necessary security patches from the vendor.
- B. Enable automatic patching for the instances using the Amazon RDS console.
- C. In AWS Config, configure a rule for the instances and the required patch level.
-  D. Use AWS Systems Manager to automate database patching according to a schedule.

Correct Answer: D

Patch Manager, a capability of AWS Systems Manager, automates the process of patching managed nodes with both security related and other types of updates. You can use Patch Manager to apply patches for both operating systems and applications. (On Windows Server, application support is limited to updates for applications released by Microsoft.) You can use Patch Manager to install Service Packs on Windows nodes and perform minor version upgrades on Linux nodes. You can patch fleets of Amazon Elastic Compute Cloud (Amazon EC2) instances, edge devices, or your on-premises servers and virtual machines (VMs) by operating system type.

206) A business is releasing a new application on AWS. The application will be hosted on an Amazon Elastic Compute Cloud (EC2) instance. Additional EC2 instances will be required as the demand grows.

Which AWS service or technology can the business utilize to deploy the required number of EC2 instances?

- A. Elastic Load Balancing
-  B. Amazon EC2 Auto Scaling
- C. AWS App2Container (A2C)
- D. AWS Systems Manager

Correct Answer: B

Amazon EC2 Auto Scaling helps you ensure that you have the correct number of Amazon EC2 instances available to handle the load for your application. You create collections of EC2 instances, called *Auto Scaling groups*. You can specify the minimum number of instances in each Auto Scaling group, and Amazon EC2 Auto Scaling ensures that your group never goes below this size. You can specify the maximum number of instances in each Auto Scaling group, and Amazon EC2 Auto Scaling ensures that your group never goes above this size.

207) What does it mean to provide AWS IAM users the fewest possible privileges?

- A. It is granting permissions to a single user only.
- B. It is granting permissions using AWS IAM policies only.
- C. It is granting AdministratorAccess policy permissions to trustworthy users.
-  D. It is granting only the permissions required to perform a given task.

Correct Answer: D

When you create IAM policies, follow the standard security advice of granting least privilege, or granting only the permissions required to perform a task. Determine what users (and roles) need to do and then craft policies that allow them to perform only those tasks.

208) When building an Amazon Relational Database Service (Amazon RDS) instance in Multiple Availability Zone mode, which architectural concept is followed?

- A. Implement loose coupling.
-  B. Design for failure.
- C. Automate everything that can be automated.
- D. Use services, not servers.

Correct Answer: B

Amazon RDS Multi-AZ deployments provide enhanced availability and durability for Database (DB) Instances, making them a natural fit for production database workloads. When you provision a Multi-AZ DB Instance, Amazon RDS automatically creates a primary DB Instance and synchronously replicates the data to a standby instance in a different Availability Zone (AZ). Each AZ runs on its own physically distinct, independent infrastructure, and is engineered to be highly reliable. In case of an infrastructure failure, Amazon RDS performs an automatic failover to the standby (or to a read replica in the case of Amazon Aurora), so that you can resume database operations as soon as the failover is complete. Since the endpoint for your DB Instance remains the same after a failover, your application can resume database operation without the need for manual administrative intervention.

209) What is a user's responsibility while using the AWS Cloud to execute an application?

- A. Managing physical hardware
- B. Updating the underlying hypervisor
- C. Providing a list of users approved for data center access
-  D. Managing application software updates

Correct Answer: D

The customer assumes responsibility and management of the guest operating system (including updates and security patches), other associated application software as well as the configuration of the AWS provided security group firewall. Customers should carefully consider the services they choose as their responsibilities vary depending on the services used, the integration of those services into their IT environment, and applicable laws and regulations.

210) Which of the following statements concerning AWS's worldwide infrastructure is true?

- A. Availability Zones can span multiple AWS Regions.
- B. A VPC can have different subnets in different AWS Regions.
-  C. AWS Regions consist of multiple Availability Zones.
- D. A single subnet can span multiple Availability Zones.

Correct Answer: C

AWS provides a more extensive global footprint than any other cloud provider, and to support its global footprint and ensure customers are served across the world, AWS opens new Regions rapidly. AWS maintains multiple geographic Regions, including Regions in North America, South America, Europe, China, Asia Pacific, South Africa, and the Middle East.

211) A business wishes to link AWS to its corporate network through a private network connection.

Which Amazon Web Services (AWS) service or functionality will satisfy this requirement?

- A. Amazon Connect
- B. Amazon Route 53
-  C. AWS Direct Connect
- D. VPC peering

Correct Answer: C

•AWS Direct Connect is a **cloud service solution** that makes it easy to establish a dedicated network connection from your premises to AWS. Using AWS Direct Connect, you can establish private connectivity between AWS and your datacenter, office, or colocation environment, which in many cases can reduce your network costs, increase bandwidth throughput, and provide a more consistent network experience than Internet-based connections.

212) Which of the following may be used to restrict certain users' access to Amazon Simple Storage Service (Amazon S3) buckets?

- A. A public and private key-pair
- B. Amazon Inspector
-  C. AWS Identity and Access Management (IAM) policies
- D. Security Groups

Correct Answer: C

To allow users to perform S3 actions on the bucket from the VPC endpoints or IP addresses, you must explicitly grant those user-level permissions. You can grant user-level permissions on either an AWS Identity and Access Management (IAM) policy or another statement in the bucket policy.

213) Which of the following is an AWS Well-Architected Framework design principle?

- A. Reduce downtime by making infrastructure changes infrequently and in large increments.
- B. Invest the time to configure infrastructure manually.
-  C. Learn to improve from operational failures.
- D. Use monolithic application design for centralization.

Correct Answer: C

Learn from all operational failures: Drive improvement through lessons learned from all operational events and failures. Share what is learned across teams and through the entire organization.

214) Which AWS service enables a business to identify and reroute customers to other servers in the event of a website server outage?

- A. Amazon CloudFront
- B. Amazon GuardDuty
-  C. Amazon Route 53
- D. AWS Trusted Advisor

Correct Answer: C

Amazon Route 53 - Set routing policies to pre-determine and automate responses in case of failure, like redirecting traffic to alternative Availability Zones or Regions.

215) A organization with an AWS Support plan for developers established an Amazon RDS database but is unable to connect to it.

To get this degree of help, who should the developer contact?

-  A. AWS Support using a support case
- B. AWS Professional Services
- C. AWS technical account manager
- D. AWS consulting partners

Correct Answer: A

Customers with a Developer Support plan have access to these additional features:

- Best practice guidance
- Client-side diagnostic tools
- Building-block architecture support: guidance on how to use AWS products, features, and services together
- Supports an unlimited number of support cases that can be opened by one primary contact, which is the AWS account *root user*.

216) Which AWS service enables you to monitor and debug distributed applications end-to-end?

- A. AWS Cloud9
- B. AWS CodeStar
- C. AWS Cloud Map
-  D. AWS X-Ray

Correct Answer: D

AWS X-Ray is a **powerful tool offered by Amazon that enables developers to debug production and distributed applications**, especially in the microservices architecture. By analyzing the performance of the application and its underlying services, developers can identify root causes of performance issues to quickly resolve them.

217) Which AWS shared responsibility model duties are the customer's responsibility? (Select two.)

- A. Infrastructure facilities access management
- B. Cloud infrastructure hardware lifecycle management
-  C. Configuration management of user's applications
- D. Networking infrastructure protection
-  E. Security groups configuration

Correct Answer: CE

The customer is responsible for the security configuration or firewall (like security groups), Identity and Access Management (IAM), client and server-side encryption and customer data. Now, apart from AWS services, AWS Shared Responsibility also extends to IT controls.

218) Which AWS Cloud service gives tips on how to optimize an AWS account's performance?

- A. Amazon Inspector
-  B. AWS Trusted Advisor
- C. Amazon CloudWatch
- D. AWS CloudTrail

Correct Answer: *B*

AWS Trusted Advisor provides recommendations that help you follow AWS best practices. Trusted Advisor evaluates your account by using checks. These checks identify ways to optimize your AWS infrastructure, improve security and performance, reduce costs, and monitor service quotas.

219) Which variables impact AWS Cloud costs? (Select two.)

- A. The number of unused AWS Lambda functions
- B. The number of configured Amazon S3 buckets
- C. Inbound data transfers without acceleration
-  D. Outbound data transfers without acceleration
-  E. Compute resources that are currently in use

Correct Answer: *DE*

Charges may apply if there is data transfer between different components of your workload. These charges vary depending on where the components are deployed.

220) A corporation is required by law to track and assess configuration changes to AWS resources, as well as to conduct corrective steps.

Which Amazon Web Services (AWS) service should the business use?



- A. AWS Config
- B. AWS Secrets Manager
- C. AWS CloudTrail
- D. AWS Trusted Advisor

Correct Answer: A

AWS Config continuously monitors and records your AWS resource configurations. You can use the service to automate the evaluation and remediation of recorded configurations against desired configurations. You also can review changes in configurations and relationships between AWS resources and dive into the history of a resource configuration.

221) Which AWS service should be utilized to store data backups for an extended period of time at a reasonable cost?



- A. Amazon RDS
- B. Amazon Glacier
- C. AWS Snowball
- D. AWS EBS

Correct Answer: B

Explanation -

Amazon S3 Glacier is a secure, durable, and low-cost storage class of S3 for data archiving and long-term backup. Customers can store large or small amounts of data for as little as \$0.004 per gigabyte per month. The S3 Glacier storage class is ideal for archives where data is regularly retrieved and some of the data may be needed in minutes.

Amazon RDS is a relational database service that hosts databases. It helps you create and manage databases.

Amazon Snowball is a petabyte-scale data transfer service that provides cost efficient data transfer to AWS from tamper proof physical devices. Similarly, Elastic block storage offers persistent block storage volumes for EC2 instances.

222) What does the AWS Cloud bring clients in terms of increased execution speed and agility? (Select two.)

- ☒ A. Readily available resources with low provisioning times
- ☒ B. Scalable compute capacity
- ☐ C. Free Tier services usage
- ☐ D. Access to AWS data centers
- ☐ E. Lower resource provisioning cost

Correct Answer: AB

In a cloud computing environment, new IT resources are only a click away, which means that you reduce the time to make those resources available to your developers from weeks to just minutes. This results in a dramatic increase in agility for the organization, since the cost and time it takes to experiment and develop is significantly lower.

223) A retailer wishes to supply just the resources required to meet current demand.

Which cloud advantage is the organization attempting to accomplish with this objective?

- ☐ A. Reliability
- ☐ B. Global reach
- ☒ C. Scalability
- ☐ D. High availability

Correct Answer: C

AWS Auto Scaling lets you build scaling plans that automate how groups of different resources respond to changes in demand.

224) What storage capabilities does Amazon S3 Intelligent-Tiering provide?

- A. Payment flexibility by reserving storage capacity
- B. Long-term retention of data by copying the data to an encrypted Amazon Elastic Block Store (Amazon EBS) volume
-  C. Automatic cost savings by moving objects between tiers based on access pattern changes
- D. Secure, durable, and lowest cost storage for data archival

Correct Answer: C

The S3 Intelligent-Tiering storage class delivers automatic storage cost savings in three low latency and high throughput access tiers. It also offers optional archive capabilities to help you get the lowest storage costs in the cloud for data that can be accessed in minutes to hours.

225) A business wishes to transfer its apps to an AWS VPC. These apps will need access to resources located on-premises.

Which combination of activities will allow the business to achieve this objective? (Select two.)

- A. Use the AWS Service Catalog to identify a list of on-premises resources that can be migrated.
-  B. Build a VPN connection between an on-premises device and a virtual private gateway in the new VPC.
- C. Use Amazon Athena to query data from the on-premises database servers.
-  D. Connect the company's on-premises data center to AWS using AWS Direct Connect.
- E. Leverage Amazon CloudFront to restrict access to static web content provided through the company's on-premises web servers.

Correct Answer: BD

AWS Direct Connect - Link your AWS and on-premises networks to build applications that span environments without compromising performance.

226) What is the most effective approach to link an on-premises network to numerous VPCs located in separate AWS Regions?

- A. Use AWS Direct Connect
- B. Use AWS VPN
- C. Use AWS Client VPN
-  D. Use an AWS Transit Gateway

Correct Answer: *D*

AWS Transit Gateway connects your Amazon Virtual Private Clouds (VPCs) and on-premises networks through a central hub. This connection simplifies your network and puts an end to complex peering relationships. Transit Gateway acts as a highly scalable cloud router—each new connection is made only once.

227) Multiple Regions of the AWS Cloud are an example of:

- A. agility.
-  B. global infrastructure.
- C. elasticity.
- D. pay-as-you-go pricing.

Correct Answer: *B*

The AWS Cloud spans 99 Availability Zones within 31 geographic regions around the world, with announced plans for 12 more Availability Zones and 4 more AWS Regions in Canada, Israel, New Zealand, and Thailand.

228) Which AWS service does Chef and Puppet utilize to automate configuration management?

- A. AWS Config
- ▶ B. AWS OpsWorks
- C. AWS CloudFormation
- D. AWS Systems Manager

Correct Answer: B

AWS OpsWorks for Chef Automate Chef Automate is an **enterprise level-platform** that provides actionable insights with enterprise scale and performance across your cloud architecture. AWS OpsWorks for Chef Automate is a managed way of launching a Chef Automate server in OpsWorks AWS.

OpsWorks for Puppet Enterprise lets you launch a Puppet Enterprise master in minutes, and lets AWS OpsWorks handle its operations, backups, restorations, and software upgrades. OpsWorks for Puppet Enterprise frees you to focus on core configuration management tasks, instead of managing a Puppet master.

229) A corporation want to remove the necessity for pre-deployment estimation of infrastructure capacity. Additionally, the corporation want to spend its money on cloud resources only when the resources are used.

Which AWS Cloud feature best meets your business's needs?

- A. Reliability
- B. Global reach
- C. Economies of scale
- ▶ D. Pay-as-you-go pricing

Correct Answer: D

The **on-demand pricing model** is the true embodiment of Amazon's **pay-as-you-go payment philosophy**. You have no upfront payments, which means you're not committing yourself over a long-term period. Amazon will charge you for the compute capacity by the hour, and you can increase or decrease usage depending on your application.

230) SQL injection attacks are being launched against an application from a variety of external locations.

Which AWS service or functionality can assist in automating response to these attacks?

-  A. AWS WAF
- B. Security groups
- C. Elastic Load Balancer
- D. Network ACL

Correct Answer: A

AWS WAF makes it easy to **create rules that block common web exploits like SQL injection and cross site scripting**. AWS WAF allows you to create a centralized set of rules that you can deploy across multiple websites.

231) Which feature enables Amazon EC2 instances to be more elastic in response to changing workload demand?

- A. Resource groups
- B. Lifecycle policies
- C. Application Load Balancer
-  D. Amazon EC2 Auto Scaling

Correct Answer: D

Support for monitoring the health of each service independently, as health checks are defined at the target group level and many CloudWatch metrics are reported at the target group level. Attaching a target group to an Auto Scaling group enables you to scale each service dynamically based on demand.

232) What time-saving benefits can Amazon Rekognition provide?

-  A. Amazon Rekognition provides automatic watermarking of images.
- B. Amazon Rekognition provides automatic detection of objects appearing in pictures.
- C. Amazon Rekognition provides the ability to resize millions of images automatically.
- D. Amazon Rekognition uses Amazon Mechanical Turk to allow humans to bid on object detection jobs.

Correct Answer: B

Rekognition Image is an image recognition service that detects objects, scenes, and faces; extracts text; recognizes celebrities; and identifies inappropriate content in images. It also allows you to search and compare faces. Rekognition Image is based on the same proven, highly scalable, deep learning technology developed by Amazon's computer vision scientists to analyze billions of images daily for Prime Photos.

233) A corporation want to migrate petabytes of data from on-premises sites to the AWS Cloud as rapidly as feasible.
Which Amazon Web Services (AWS) service should the business use?

-  A. AWS Snowball
- B. AWS Global Accelerator
- C. Amazon S3 Transfer Acceleration
- D. Amazon Connect

Correct Answer: A

The Snowball appliance allows you to move archives, data lakes, and whatever data you have at faster-than Internet speeds right into Amazon S3 buckets. From Amazon S3, data can be archived into Glacier or analyzed by other services such as AWS Redshift or EMR. This implementation guide will get you started with AWS Snowball's quick, simple, and secure process for migrating large amounts of data into AWS.

234) Where should a business go to locate, test, purchase, and deploy software that works on AWS?

-  A. AWS Marketplace
- B. Amazon Lumberyard
- C. AWS Artifact
- D. Amazon CloudSearch

Correct Answer: A

AWS Marketplace is a digital catalog with thousands of software listings from independent software vendors that make it easy to find, test, buy, and deploy software that runs on AWS.

235) A business's user base is worldwide in scope. The organization need a highly available application with reduced latency for end users.

Which AWS architecture approach will meet these criteria the MOST EFFECTIVELY?

- A. Single-Region, Multi-AZ architecture
-  B. Multi-Region, active-active architecture
- C. Multi-Region, active-passive architecture
- D. Single-Region, Single-AZ architecture

Correct Answer: B

Simply put, a multi-region, active-active architecture **gets all the services on the client request path deployed across multiple AWS Regions**. In order to do so, several requirements have to be fulfilled. Data replication between regions must be fast and reliable.

236) Which duty is the customer's responsibility under the AWS shared responsibility model?

- A. Maintaining the infrastructure needed to run AWS Lambda
- B. Updating the operating system of Amazon DynamoDB instances
- C. Maintaining Amazon S3 infrastructure
-  D. Updating the guest operating system on Amazon EC2 instances

Correct Answer: D

The customer assumes responsibility and management of the guest operating system (including updates and security patches), other associated application software as well as the configuration of the AWS provided security group firewall.

237) How can a user safeguard against AWS service outages in the event of a widespread natural disaster?

- A. Deploy applications across multiple Availability Zones within an AWS Region.
- B. Use a hybrid cloud computing deployment model within the geographic area.
-  C. Deploy applications across multiple AWS Regions.
- D. Store application artifacts using AWS Artifact and replicate them across multiple AWS Regions.

Correct Answer: C

An AWS Region is a geographic location where AWS provides multiple, physically separated and isolated Availability Zones which are connected with low latency, high throughput, and highly redundant networking.

238) What is an example of a cloud-based application that is decoupled and scalable?

- A. A mail and log application that runs on a single Amazon EC2 instance
-  B. A webpage that is hosted on Amazon S3 and uses AWS Lambda to update an Amazon DynamoDB database
- C. An Application Load Balancer, web server, and database server that support a monolithic application
- D. A legacy database server that is running on the maximum instance size supported by its license

Correct Answer: B

Many applications start to grow in complexity as they mature, making it harder for developers to maintain code or add new features. This can lead to monolithic applications, where developers must know more about the entire architecture to make changes. Typically, this causes code to become more fragile, and the rate of development slows down.

239) A client has many AWS accounts, each with its own billing.

How can the client benefit from bulk savings while minimizing the effect on AWS resources?

- A. Create one global AWS account and move all AWS resources to that account.
- B. Sign up for three years of Reserved Instance pricing up front.
-  C. Use the consolidated billing feature from AWS Organizations.
- D. Sign up for the AWS Enterprise support plan to get volume discounts.

Correct Answer: C

Consolidated billing is a feature of AWS Organizations. You can use the management account of your organization to consolidate and pay for all member accounts. In consolidated billing, management accounts can also access the billing information, account information, and account activity of member accounts in their organization. This information may be used for services such as Cost Explorer, which can help management accounts improve their organization's cost performance.

240) A business must keep its data near to its core consumers.

Which AWS Cloud advantage satisfies this requirement?

- A. Security
- B. High availability
- C. Elasticity
-  D. Global footprint

Correct Answer: D

A successful global footprint depends on how you use Regions and their Availability Zones. The AWS Global Infrastructure is comprised of **69 Availability Zones within 22 geographic Regions.**

241) Who is responsible for the virtualization layer down to the physical security of the facilities in which AWS services operate under the AWS shared responsibility model?

- A. It is the sole responsibility of the customer.
-  B. It is the sole responsibility of AWS.
- C. It is a shared responsibility between AWS and the customer.
- D. The customer's AWS Support plan tier determines who manages the configuration.

Correct Answer: B

AWS is responsible for protecting the infrastructure that runs all of the services offered in the AWS Cloud. This infrastructure is composed of the hardware, software, networking, and facilities that run AWS Cloud services.

242) A company's managed IAM policy does not allow users the rights essential to do needed activities.

How is this situation to be resolved?

- A. Enable AWS Shield Advanced
-  B. Create a custom IAM policy
- C. Use a third-party web application firewall (WAF) managed rule from the AWS Marketplace
- D. Use AWS Key Management Service (AWS KMS) to create a customer-managed key

Correct Answer: B

When you create or edit IAM policies, AWS can automatically perform policy validation to help you create an effective policy with least privilege in mind. In the AWS Management Console, IAM identifies JSON syntax errors, while IAM Access Analyzer provides additional policy checks with recommendations to help you further refine your policies.

243) A business wishes to strengthen its security and audit posture by restricting incoming access to Amazon EC2.

What should the organization use instead of opening incoming SSH ports and handling SSH keys to remotely access instances?

- A. EC2 key pairs
-  B. AWS Systems Manager Session Manager
- C. AWS Identity and Access Management (IAM)
- D. Network ACLs

Correct Answer: B

You can use the AWS Systems Manager console to start a session with a managed node in your account.

244) Which scenarios warrant the utilization of Amazon EC2 Spot Instances?

- A. A company wants to move its main website to AWS from an on-premises web server.
- B. A company has a number of application services whose Service Level Agreement (SLA) requires 99.999% uptime.
- C. A company's heavily used legacy database is currently running on-premises.
-  D. A company has a number of infrequent, interruptible jobs that are currently using On-Demand Instances.

Correct Answer: D

A Spot Instance is an **unused Amazon EC2 instance that is available for less than the On-Demand price**. Because Spot Instances enable you to request unused EC2 instances at steep discounts, you can lower your Amazon EC2 costs significantly. The hourly price for a Spot Instance is called a Spot price.

245) A business must transmit time-sensitive communications to a large number of subscribers using a push technique.

Which Amazon Web Services (AWS) service should the business use?

- A. Amazon Kinesis
- B. Amazon MQ
- C. Amazon Simple Queue Service (Amazon SQS)
-  D. Amazon Simple Notification Service (Amazon SNS)

Correct Answer: D

Amazon SNS allows applications to send time-critical messages to multiple subscribers through a "Push" mechanism

246) Which tasks need access to the root user of the AWS account? (Select two.)

-  A. Changing an AWS Support plan
- B. Modifying an Amazon EC2 instance type
- C. Grouping resources in AWS Systems Manager
- D. Running applications in Amazon Elastic Kubernetes Service (Amazon EKS)
-  E. Closing an AWS account

Correct Answer: AE

Tasks

- [Change your account settings](#). This includes the account name, email address, root user password, and root user access keys. Other account settings, such as contact information, payment currency preference, and Regions, do not require root user credentials.
- [Restore IAM user permissions](#). If the only IAM administrator accidentally revokes their own permissions, you can sign in as the root user to edit policies and restore those permissions.
- [Activate IAM access to the Billing and Cost Management console](#).
- [View certain tax invoices](#). An IAM user with the [awsportal-viewBilling](#) permission can view and download VAT invoices from AWS Europe, but not AWS Inc or Amazon Internet Services Pvt. Ltd (AISPL).
- [Close your AWS account](#).
- [Change your AWS Support plan ID](#) or [cancel your AWS Support plan ID](#). For more information, see [IAM for AWS Support ID](#).
- [Register as a seller in the Reserved Instance Marketplace](#).
- [Configure an Amazon S3 bucket to enable MFA \(multi factor authentication\)](#) Delete.
- [Edit or delete an Amazon S3 bucket policy that includes an invalid VPC ID or VPC endpoint ID](#).
- [Sign up for GovCloud](#).

247) Which AWS Cloud best practice makes advantage of cloud computing's flexibility and agility?

-  A. Provision capacity based on past usage and theoretical peaks
-  B. Dynamically and predictively scale to meet usage demands
- C. Build the application and infrastructure in a data center that grants physical access
- D. Break apart the application into loosely coupled components

Correct Answer: B

In a traditional computing environment, you provision capacity based on an estimate of a theoretical maximum peak. This can result in periods where expensive resources are sitting idle or occasions of insufficient capacity. With cloud computing, you can access as much or as little capacity as you need and dynamically scale to meet actual demand, while only paying for what you use.

248) Users of Amazon Route 53 are able to:

-  A. encrypt data in transit
- B. register DNS domain names
- C. generate and manage SSL certificates
- D. establish a dedicated network connection to AWS

Correct Answer: B

Amazon Route 53 is a highly available and scalable [Domain Name System \(DNS\)](#) web service. Route 53 connects user requests to internet applications running on AWS or on-premises.

249) Which characteristic of the Amazon Virtual Private Cloud (Amazon VPC) allows customers to link two VPCs?

- A. Amazon VPC endpoints
- B. Amazon Elastic Compute Cloud (Amazon EC2) ClassicLink
-  C. Amazon VPC peering
- D. AWS Direct Connect

Correct Answer: C

A VPC peering connection is a networking connection between two VPCs that enables you to route traffic between them using private IPv4 addresses or IPv6 addresses. Instances in either VPC can communicate with each other as if they are within the same network. You can create a VPC peering connection between your own VPCs, or with a VPC in another AWS account. The VPCs can be in different regions (also known as an inter-region VPC peering connection).

250) Multiple Amazon EC2 instances are used to host an application. The program sends messages using Amazon Simple Notification Service (Amazon SNS).

Which AWS service or feature grants authorization for the application to access needed AWS services?

-  A. AWS Certificate Manager (ACM)
- B. IAM roles
- C. AWS Security Hub
- D. Amazon GuardDuty

Correct Answer: B

An IAM role is an IAM entity that defines a set of permissions for making AWS service requests. IAM roles are not associated with a specific user or group. Instead, trusted entities assume roles, such as IAM users, applications, or AWS services such as EC2.

251) A business is contemplating migrating to the AWS Cloud. The firm wishes to be able to scale its computing capacity in response to changing demand conditions.

Which AWS Cloud advantage does this case illustrate?

- A. Global deployments in minutes
- B. Cost savings
- C. Agility
-  D. Elasticity

Correct Answer: D

In AWS, **the process of getting the resources dynamically when you actually require them and then release the resources when you are done and do not need them** is known as elasticity. In another way, growing or shrinking the resources dynamically when needed is known as Elasticity.

252) Which of the following is a NoSQL database service that is both quick and dependable?

- A. Amazon Redshift
- B. Amazon RDS
-  C. Amazon DynamoDB
- D. Amazon S3

Correct Answer: C

Amazon DynamoDB is a fully managed NoSQL database service that provides fast and predictable performance with seamless scalability.

253) A client wants to develop and construct a new workload on AWS Cloud but lacks the necessary technical skills in AWS-related technologies. Which of the following AWS programs may a client use to accomplish that goal?

- A. AWS Partner Network Technology Partners
- B. AWS Marketplace
-  C. AWS Partner Network Consulting Partners
- D. AWS Service Catalog

Correct Answer: C

benefits of becoming an APN Consulting Partner include:

- Access to 'Go-to-market' resources
- Training & Certification,
- Market Development Funding (MDF),
- Innovation Sandbox Credits
- Increased visibility to AWS customers and AWS field teams

254) Which AWS solution enables rapid setup and management of a new multi-account AWS environment?

- A. AWS Trusted Advisor
- B. AWS Security Hub
-  C. AWS Control Tower
- D. AWS Resource Access Manager

Correct Answer: C

AWS Control Tower is a service for organizations with multiple accounts and teams who are looking for the easiest way to set up their new or existing multi-account AWS environment and govern at scale. With AWS Control Tower, cloud administrators get peace of mind knowing accounts in their organization are compliant with their established policies. Builders benefit from being able to provision new AWS accounts quickly.

255) Which operation needs the usage of the root account user credentials for the AWS account?

-  A. Closing an AWS account
- B. Creating a log file
- C. Modifying IAM user permissions
- D. Deleting IAM users

Correct Answer: A

To close your AWS account, do the following: **Sign in to the AWS Management Console as the root user of the account. From the navigation bar, choose your account name, and then choose Account.**

256) On Amazon EC2 instances, a business is operating and controlling its own Docker environment. Alternate is desired to assist in managing cluster size, scheduling, and environment management.

Which AWS service satisfies these criteria?

- A. AWS Lambda
- B. Amazon RDS
-  C. AWS Fargate
- D. Amazon Athena

Correct Answer: C

With AWS Fargate, you no longer have to provision, configure, or scale clusters of virtual machines to run containers. This removes the need to choose server types, decide when to scale your clusters, or optimize cluster packing.

257) Which AWS situations exemplify the idea of elasticity? (Select two.)

-  A. Scaling the number of Amazon EC2 instances based on traffic.
-  B. Resizing Amazon RDS instances as business needs change.
- C. Automatically directing traffic to less-utilized Amazon EC2 instances.
- D. Using AWS compliance documents to accelerate the compliance process.
- E. Having the ability to create and govern environments using code.

Correct Answer: AB

Implement elasticity

- Identify the workloads that have variable load.
- Identify the workload load range. That is, is there enough variability to warrant adding or removing resources?
- Identify the application limitations (sessions, long initialization, licensing, etc.) that may limit elasticity.
- Identify if the increase in demand can be met by automatic scaling, or if it needs to be in place before (for events, launches, etc.).
- Identify applications that can use Amazon Athena or Amazon Aurora Serverless
- Implement elasticity using AWS Auto Scaling or Application Auto Scaling for the aspects of your service that are not elastic by design.
- Test elasticity both up and down, ensuring it will meet requirements for load variance
- Iterate on implementation and testing until you can meet requirements. You may want to investigate golden Amazon Machine Images, docker containers, etc. to speed launch.

258) Which of the following are advantages of using the AWS Cloud to host infrastructure? (Select two.)

-  A. There are no upfront commitments.
-  B. AWS manages all security in the cloud.
-  C. Users have the ability to provision resources on demand.
- D. Users have access to free and unlimited storage.
- E. Users have control over the physical infrastructure.

Correct Answer: AC

The No Upfront option does not require any upfront payment, and your commitment will be charged purely on a monthly basis.

Savings Plans offers significant savings over On Demand, just like EC2 RIs, but automatically reduce your bills on compute usage across any AWS region, even as usage changes.

259) Which of the following are benefits of Amazon Web Services' cloud computing platform? (Select two.)

-  A. AWS manages the maintenance of the cloud infrastructure
-  B. AWS manages the security of applications built on AWS
-  C. AWS manages capacity planning for physical servers
- D. AWS manages the development of applications on AWS
- E. AWS manages cost planning for virtual servers

Correct Answer: AC

AWS manages all infrastructure layers.
Some of the infrastructure layers are:

- Data centers
- Hardware and software
- Virtualization
- Networking

260) Which AWS service enables infrastructure as code management?

- A. AWS CodePipeline
- B. AWS CodeDeploy
- C. AWS Direct Connect
-  D. AWS CloudFormation

Correct Answer: *D*

AWS CloudFormation provides a common language for you to describe and provision all the infrastructure resources in your cloud environment. CloudFormation allows you to use a simple text file to model and provision, in an automated and secure manner, all the resources needed for your applications across all regions and accounts. This file serves as the single source of truth for your cloud environment.

261) How might AWS help a business manage costs when an application's consumption varies unpredictably?

- A. AWS will refund the cost difference if a customer moves to larger servers.
-  B. The application can be built to scale up or down automatically as resources are needed
- C. Spot instances will automatically be used if the price is lower than on-demand instances.
- D. Amazon CloudWatch will automatically predict what resources are needed.

Correct Answer: *B*

AWS Auto Scaling can help us to optimize our utilization and cost performances when consuming the AWS services, so you only pay for the resources you need to run the application. When demand falls, AWS Auto Scaling will automatically eliminate any excess resource capacity, so you avoid any overspending.

262) What are the benefits of deploying an application across various Availability Zones using Amazon EC2 instances? (Select two.)

- ☒ A. Preventing a single point of failure
- ☐ B. Reducing the operational costs of the application
- ☐ C. Allowing the application to serve cross-region users with low latency
- ☒ D. Increasing the availability of the application
- ☐ E. Increasing the load of the application

Correct Answer: AD

The application will have higher availability because it can withstand a service disruption in one Availability Zone

263) Which AWS service supports a hybrid architecture that gives users the ability to extend AWS infrastructure, AWS services, APIs, and tools to data centers, co- location environments, or on-premises facilities?

- ☐ A. AWS Snowmobile
- ☐ B. AWS Local Zones
- ☒ C. AWS Outposts
- ☐ D. AWS Fargate

Correct Answer: C

AWS Outposts is a service that supports a hybrid architecture that gives users the ability to extend AWS infrastructure, AWS services, APIs, and tools to data centers, co-location environments, or on-premises facilities.

264) A company has a physical tape library to store data backups. The tape library is running out of space. The company needs to extend the tape library's capacity to the AWS Cloud. Which AWS service should the company use to meet this requirement?

- A. Amazon Elastic Block Store (Amazon EBS)
- B. Amazon S3
- C. Amazon Elastic File System (Amazon EFS)
-  D. AWS Storage Gateway

Correct Answer: D

AWS Storage Gateway is a service that can be used to extend the tape library's capacity to the AWS Cloud.

AWS Storage Gateway is a hybrid storage service that allows users to connect their on-premises data centers to the AWS Cloud. It provides a range of storage options, including file-based, block-based, and tape-based storage, which can be used to store data backups and other types of data.

265) A business wishes to downsize its infrastructure in order to save money.

At what stages should a business downsize? (Select two.)

-  A. Rightsize before a migration occurs to the cloud.
-  B. Rightsize continuously after the cloud onboarding process.
- C. Rightsize when AWS Support calls and explains that rightsizing is needed.
- D. Rightsize when seasonal workloads are at their peak.
- E. Rightsize after purchasing all Reserved Instances.

Correct Answer: AB

Right sizing is the process of matching instance types and sizes to your workload performance and capacity requirements at the lowest possible cost. It's also the process of looking at deployed instances and identifying opportunities to eliminate or downsize without compromising capacity or other requirements, which results in lower costs.

266) AWS Budgets may be used for the following purposes:

- A. prevent a given user from creating a resource
-  B. send an alert when the utilization of Reserved Instances drops below a certain percentage
- C. set resource limits in AWS accounts to prevent overspending
- D. split an AWS bill across multiple forms of payment

Correct Answer: B

AWS Budgets allows you to set custom budgets to track your cost and usage from the simplest to the most complex use cases. With AWS Budgets, you can choose to be alerted by email or SNS notification when actual or forecasted cost and usage exceed your budget threshold, or when your actual RI and Savings Plans' utilization or coverage drops below your desired threshold.

267) An online retail company has seasonal sales spikes several times a year, primarily around holidays. Demand is lower at other times. The company finds it difficult to predict the increasing infrastructure demand for each season.

Which advantages of moving to the AWS Cloud would MOST benefit the company? (Choose two.)

- A. Global footprint
-  B. Elasticity
- C. AWS service quotas
- D. AWS shared responsibility model
-  E. Pay-as-you-go pricing

Correct Answer: BE

Cloud elasticity is **the ability to rapidly and dynamically allocate cloud resources**, including compute, storage, and memory resources, in response to changing demands. The goal of cloud elasticity is to avoid either overprovisioning or underprovisioning a particular service or application.

268) What are the advantages of unified billing for Amazon Web Services accounts?

- A. Access to AWS Personal Health Dashboard
- B. Combined usage volume discounts
- C. Improved account security
- D. Centralized AWS IAM

Correct Answer: B

Consolidated Billing Benefits

- One Bill

- Consolidated Billing

- Easy Tracking

- Consolidated Billing

- Combined Usage & Volume Discounts

- Consolidated Billing

- Free Tier

- Consolidated Billing

269) A corporation want to connect to AWS from a distant office through a private, low-latency connection.
Which strategy is advised for meeting these requirements?

- A. Create a VPN tunnel
- B. Connect across the public internet
- C. Use VPC peering to create a connection.
- D. Use AWS Direct Connect.

Correct Answer: D

The AWS Direct Connect cloud service is the shortest path to your AWS resources. While in transit, your network traffic remains on the AWS global network and never touches the public internet. This reduces the chance of hitting bottlenecks or unexpected increases in latency.

270) Which AWS service can be used to turn text into lifelike speech?

-  A. Amazon Polly
- B. Amazon Kendra
- C. Amazon Rekognition
- D. Amazon Connect

Correct Answer: A

Amazon Polly is a service that can be used to turn text into lifelike speech. Amazon Polly uses advanced deep learning technologies to synthesize speech that sounds natural and lifelike, allowing users to convert written content into spoken language.

271) Which Amazon EC2 pricing model is dynamically adjusted in response to EC2 instance availability and demand?

- A. On-Demand Instances
- B. Reserved Instances
-  C. Spot Instances
- D. Convertible Reserved Instances

Correct Answer: C

In the new model, the Spot prices are more predictable, updated less frequently, and are determined by supply and demand for Amazon EC2 spare capacity, not bid prices.

272) Which service should a client utilize to integrate and manage several Amazon Web Services accounts centrally?

-  A. AWS IAM
- B. AWS Organizations
- C. AWS Schema Conversion Tool
- D. AWS Config

Correct Answer: B

Integrate multiple AWS services with **multiple unique AWS accounts**. Manage the user environments based on organizational, legal, or project-based policies. The accounts can also share resources, security mechanisms, audit requirements, configurations, and policies between multiple AWS organizations.

273) Which AWS service or tool can be used to capture information about inbound and outbound traffic in an Amazon VPC?

-  A. VPC Flow Logs
- B. Amazon Inspector
- C. VPC endpoint services
- D. NAT gateway

Correct Answer: A

VPC Flow Logs is a feature that enables you to capture information about the IP traffic going to and from network interfaces in your VPC.

274) Which choices do users have when contacting AWS Support?

- A. Create an email case in the AWS Support Center.
- B. Visit a local AWS Support Center.
-  C. Use live chat functionality.
- D. Call the customer service phone number.
- E. Use the video conference functionality of the AWS Support console.

Correct Answer: C

AWS Support offers a range of plans that provide access to tools and expertise that support the success and operational health of your AWS solutions. All support plans provide 24/7 access to customer service, AWS documentation, technical papers, and support forums. For technical support and more resources to plan, deploy, and improve your AWS environment, you can choose a support plan for your AWS use case.

275) A business must track and anticipate AWS expenditures and use. Additionally, the organization must configure event-driven alert alerts that are triggered when expenditure limitations are exceeded.

Which Amazon Web Services (AWS) offering or technology should the business employ to achieve these requirements?

-  A. AWS Budgets
- B. Amazon CloudWatch
- C. AWS Config
- D. AWS Service Catalog

Correct Answer: A

Within **AWS Budgets**, a budget is a single, regularly updating cost or usage metric relating to your AWS resources. The rich structure of this AWS data allows you to set up multiple budgets to track this data across different departments or drilling down into different systems and resource types.

276) Which of the following IT functions does AWS do in order to relieve a business of its responsibility for managing its IT resources? (Select two.)

- A. Configuring operating system firewalls
- B. Setting up access controls for data
-  C. Backing up databases
- D. Configuring database user accounts
-  E. Installing operating systems

Correct Answer: CE

AWS maintains the configuration of its infrastructure devices, but a customer is responsible for configuring their own guest operating systems, databases, and applications.

277) Which AWS technologies aid in cost estimation? (Select three.)

- A. Detailed billing report
-  B. Cost allocation tags
-  C. AWS Simple Monthly Calculator
-  D. AWS Total Cost of Ownership (TCO) Calculator
- E. Cost Estimator

Correct Answer: BCD

AWS cost allocation tags are **labels for classifying, organizing, and identifying your resources**. These tags, which are key-value pairs, make it easier to manage them and track their usage. By assigning tags to your resources, you can track your overall AWS costs, either through Cost Explorer or through the AWS API.

278) A company wants to ensure that two Amazon EC2 instances are in separate data centers with minimal communication latency between the data centers. How can the company meet this requirement?

- A. Place the EC2 instances in two separate AWS Regions connected with a VPC peering connection.
-  B. Place the EC2 instances in two separate Availability Zones within the same AWS Region.
- C. Place one EC2 instance on premises and the other in an AWS Region. Then connect them by using an AWS VPN connection.
- D. Place both EC2 instances in a placement group for dedicated bandwidth.

Correct Answer: B

easily **deploy EC2 instances** in **separate availability zones within the same region**, which is enough to meet the SLA (service-level agreement) for service credits from outages. If you haven't already, you can see which zone each instance is assigned to by viewing the details of each on the Instances page.

279) In comparison to conventional and virtualized data center prices, AWS offers the following:

- A. greater variable costs and greater upfront costs.
- B. fixed usage costs and lower upfront costs.
- C. lower variable costs and greater upfront costs.
-  D. lower variable costs and lower upfront costs.

Correct Answer: D

AWS does not require minimum spend commitments or long-term contracts. You replace large upfront expenses with **low variable payments** that only apply to what you use. With AWS you are not bound to multi-year agreements or complicated licensing models.

280) Which situations should the AWS Abuse team be notified about?

- A. In Availability Zone has a service disruption
-  B. An intrusion attempt is made from an AWS IP address
- C. A user has trouble accessing an Amazon S3 bucket from an AWS IP address
- D. A user needs to change payment methods due to a compromise

Correct Answer: B

The AWS Trust & Safety team can assist you when AWS resources are used to engage in the following types of abusive behavior:

- **Spam:** You are receiving unwanted emails from an AWS-owned IP address, or AWS resources are used to spam websites or forums.
- **Port scanning:** Your logs show that one or more AWS-owned IP addresses are sending packets to multiple ports on your server. You also believe this is an attempt to discover unsecured ports.
- **Denial-of-service (DoS) attacks:** Your logs show that one or more AWS-owned IP addresses are used to flood ports on your resources with packets. You also believe that this is an attempt to overwhelm or crash your server or the software running on your server.
- **Intrusion attempts:** Your logs show that one or more AWS-owned IP addresses are used to attempt to log in to your resources.

281) Which AWS feature should a client exploit to ensure an application's high availability?

- A. AWS Direct Connect
-  B. Availability Zones
- C. Data centers
- D. Amazon Virtual Private Cloud (Amazon VPC)

Correct Answer: B

This is to achieve High Availability for any web application (in this case SwiftCode) deployed in AWS. The following features will be present:

- ☞ High availability across multiple instances/multiple availability zones.
- ☞ Auto Scaling of instances (scale up and scale down) based on number of requests coming in
- ☞ Additional Security to the instances/database that are in production
- ☞ No impact to end users during newer version of code deployment
- ☞ No Impact during patching the instances

282) In which situations should a company create an IAM user instead of an IAM role? (Choose two.)

- A. When an application that runs on Amazon EC2 instances requires access to other AWS services
-  B. When the company creates AWS access credentials for individuals
- C. When the company creates an application that runs on a mobile phone that makes requests to AWS
-  D. When the company needs to add users to IAM groups
- E. When users are authenticated in the corporate network and want to be able to use AWS without having to sign in a second time

Correct Answer: BD

An AWS Identity and Access Management (IAM) *user* is an entity that you create in AWS. The IAM user represents the human user or workload who uses the IAM user to interact with AWS. A user in AWS consists of a name and credentials.

An IAM user with administrator permissions is not the same thing as the AWS account root user.

283) How might an AWS client implement common access restrictions to a big group of users easily?

-  A. Apply an IAM policy to an IAM group.
- B. Apply an IAM policy to an IAM role.
- C. Apply the same IAM policy to all IAM users with access to the same workload.
- D. Apply an IAM policy to an Amazon Cognito user pool.

Correct Answer: A

Instead of defining permissions for individual IAM users, it's usually more convenient to create groups that relate to job functions (administrators, developers, accounting, etc.). Next, define the relevant permissions for each group. Finally, assign IAM users to those groups. All the users in an IAM group inherit the permissions assigned to the group. That way, you can make changes for everyone in a group in just one place. As people move around in your company, you can simply change what IAM group their IAM user belongs to.

284) Which AWS service is used to provide encryption for Amazon EBS?

- A. AWS Certificate Manager
- B. AWS Systems Manager
-  C. AWS KMS
- D. AWS Config

Correct Answer: C

AWS Key Management Service (KMS) makes it easy for you to create and manage cryptographic keys and control their use across a wide range of AWS services and in your applications. AWS KMS is a secure and resilient service that uses hardware security modules that have been validated under FIPS 140-2, or are in the process of being validated, to protect your keys. AWS KMS is integrated with AWS CloudTrail to provide you with logs of all key usage to help meet your regulatory and compliance needs.

285) A business need software solutions that are either hosted on the AWS platform or are linked with it. Independent software providers, as well as management and security vendors, are required to provide solutions.

Which organization or team is capable of providing these solutions?

- A. AWS technical account managers (TAMs)
- B. AWS Partner Network (APN) Consulting Partners
- C. AWS Concierge Support
-  D. AWS Partner Network (APN) Technology Partners

Correct Answer: D

AWS Partner Network (APN) is a global partner program that vendors and other service providers offer cloud software services and build solutions for **Amazon Web Services customers**. The partners are professionals in software and services who have earned an endorsement from AWS

286) Which AWS services are available for application deployment?
(Select two.)

- ☒ •A. AWS Elastic Beanstalk
- ☐ •B. AWS Config
- ☒ •C. AWS OpsWorks
- ☐ •D. AWS Application Discovery Service
- ☐ •E. Amazon Kinesis

Correct Answer: AC

AWS Elastic Beanstalk deploys web applications so that you can focus on your business.

AWS OpsWorks provides a simple and flexible way to create and manage stacks and applications. With AWS OpsWorks, you can provision AWS resources, manage their configuration, deploy applications to those resources, and monitor their health.

287) Which AWS services make use of global edge locations? (Choose two.)

- ☐ A. AWS Fargate
- ☒ B. Amazon CloudFront
- ☒ C. AWS Global Accelerator
- ☐ D. AWS Wavelength
- ☐ E. Amazon VPC

Correct Answer: BC

- CloudFront is the most commonly discussed use of edge locations. It's a content delivery network that caches content in edge locations. Content can be served directly from the cache, so it gets to users faster. CloudFront is often used to serve static assets, speed up websites, and stream video.
- Route 53 is purportedly a managed DNS service with `Route53::DNS::get_record` and `Route53::DNS::set_record` . DNS responses come directly from the edge locations, so they're as fast as possible.
- Web Application Firewall and AWS Shield provide a firewall and DDoS protection, respectively. These services filter traffic in edge locations so malicious or unwanted traffic can be discarded as close to source as possible. This, in turn, reduces congestion on Amazon's global network and the public internet.
- AWS Global Accelerator allows you to route requests for key resources through Amazon's global network—even if the request is going halfway round the world. The request is initially routed to the closest edge location and then travels through Amazon's network—often with lower latency and higher throughput than the public internet.

288) Which responsibility do customers bear while using Amazon EC2?
(Select two.)

- ☐ A. Underlying hardware maintenance
- ☒ B. File-system-level encryption
- ☒ C. Guest operating system firewall configuration
- ☐ D. Hypervisor-level software patching
- ☐ E. Physical security at data center facilities

Correct Answer: BC

Customers that deploy an Amazon EC2 instance are responsible for **management of the guest operating system** (including updates and security patches), any application software or utilities installed by the customer on the instances, and the configuration of the AWS-provided firewall (called a security group) on each instance.

289) A company is operating several factories where it builds products. The company needs the ability to process data, store data, and run applications with local system interdependencies that require low latency. Which AWS service should the company use to meet these requirements?

- ☐ A. AWS IoT Greengrass
- ☐ B. AWS Lambda
- ☒ C. AWS Outposts
- ☐ D. AWS Snowball Edge

Correct Answer: C

Outposts is designed to provide the same APIs, control plane, and hardware as the AWS cloud, but with the ability to run workloads on-premises. This can be useful for applications that require low latency or local system interdependencies, as it allows you to keep your data and applications close to your users and processes.

290) A business needs handle a huge volume of data from social network accounts using high-throughput graphical queries.

Which AWS service will assist the business in developing a cloud architecture that satisfies these criteria?

- A. Amazon RDS
- B. Amazon DynamoDB
-  C. Amazon Neptune
- D. Amazon Redshift

Correct Answer: C

Amazon Neptune is a fast, reliable, fully managed graph database service that makes it easy to build and run applications.

291) Which AWS service enables the use of the AWS Cloud to host a NoSQL database?

- A. Amazon Aurora
-  B. Amazon DynamoDB
- C. Amazon RDS
- D. Amazon Redshift

Correct Answer: B

Amazon DynamoDB is a fully managed NoSQL database service that provides fast and predictable performance with seamless scalability.

292) Which of the following acts as an instance-level firewall to control inbound and outbound access?

-  A. Network access control list
- B. Security groups
- C. AWS Trusted Advisor
- D. Virtual private gateways

Correct Answer: B

A **security group** acts as a virtual firewall for your instance to control inbound and outbound traffic. When you launch an instance in a VPC, you can assign the instance to up to five security groups. Security groups act at the instance level, not the subnet level.

293) A business is consolidating many apps into a single Amazon Web Services (AWS) account. The organization wishes to keep track of the AWS Cloud fees paid by individual applications.

What can the business do to ensure compliance with this requirement?

- A. Set up invoiced billing.
- B. Use AWS Artifact.
-  C. Set the budgets in Cost Explorer.
- D. Create cost allocation tags.

Correct Answer: D

Using Cost Allocation Tags

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A tag is a label that you or AWS assigns to an AWS resource. Each tag consists of a *key* and a *value*. For each resource, each tag key must be unique, and each tag key can have only one value. You can use tags to organize your resources, and cost allocation tags to track your AWS costs on a detailed level. After you activate cost allocation tags, AWS uses the cost allocation tags to organize your resource costs on your cost allocation report, to make it easier for you to categorize and track your AWS costs. AWS provides two types of cost allocation tags, *AWS generated tags* and *user-defined tags*. AWS, or AWS Marketplace ISV defines, creates, and applies the AWS generated tags for you, and you define, create, and apply user-defined tags. You must activate both types of tags separately before they can appear in Cost Explorer or on a cost allocation report.

294) On an Amazon EC2 instance, a batch job takes 5 hours to complete. Monthly, the quantity of data to be processed doubles, and the time required to process it is proportionate.

What is the optimal cloud architecture for meeting this escalating demand?

- A. Run the application on a bigger EC2 instance size.
- B. Switch to an EC2 instance family that better matches batch requirements.
-  C. Distribute the application across multiple EC2 instances and run the workload in parallel.
- D. Run the application on a bare metal EC2 instance.

Correct Answer: C

295) A company has a workload that will run continuously for 1 year. The workload cannot tolerate service interruptions.

Which Amazon EC2 purchasing option will be MOST cost-effective?

-  A. All Upfront Reserved Instances
- B. Partial Upfront Reserved Instances
- C. Dedicated Instances
- D. On-Demand Instances

Correct Answer: A

With the All Upfront option, you pay for the entire Reserved Instance term with one upfront payment. This option provides you with the largest discount compared to On-Demand Instance pricing.

296) How can AWS Trusted Advisor assist AWS Cloud users? (Select two.)

- ☐ A. It identifies software vulnerabilities in applications running on AWS
- ☒ B. It provides a list of cost optimization recommendations based on current AWS usage
- ☒ C. It detects potential security vulnerabilities caused by permissions settings on account resources
- ☐ D. It automatically corrects potential security issues caused by permissions settings on account resources
- ☐ E. It provides proactive alerting whenever an Amazon EC2 instance has been compromised

Correct Answer: BC

AWS Trusted Advisor provides recommendations that help you follow AWS best practices. Trusted Advisor evaluates your account by using checks. These checks identify ways to optimize your AWS infrastructure, improve security and performance, reduce costs, and monitor service quotas. You can then follow the recommendations to optimize your services and resources.

297) Which AWS function will assist users in determining the CPU capacity of an application running on an Amazon EC2 instance?

- ☒ A. Amazon CloudWatch
- ☐ B. AWS Config
- ☐ C. AWS CloudTrail
- ☐ D. Amazon Inspector

Correct Answer: A

You can see the CPU Credit balance for each T2 instance in EC2 per-Instance metrics in Amazon CloudWatch. T2 instances have four metrics, CPUCreditUsage, CPUCreditBalance, CPUSurplusCreditBalance and CPUSurplusCreditsCharged.

298) A user may create a master payer account in order to access aggregated billing data via:

- A. AWS Budgets.
- B. Amazon Macie.
- C. Amazon QuickSight.
-  D. AWS Organizations.

Correct Answer: D

As part of AWS account creation, AWS Organizations creates an IAM role with full administrative permissions in the new account. IAM users and IAM roles with appropriate permissions in the master account can assume this IAM role to gain access to the newly created account.

299) Every few years, a business invests several months in modernizing its on-premises infrastructure. The organization wishes to shorten the time required for infrastructure purchase by shifting to the AWS Cloud.

What is the primary advantage of transferring this use case to the AWS Cloud?

- A. AWS will help move the existing hardware to the AWS data centers.
-  B. The company will have increased agility with on-demand access to IT resources.
- C. Enterprise support will be available to help with recurring application installation and setup.
- D. The company will experience less downtime with Multi-AZ deployments.

Correct Answer: B

300) Which AWS service helps protect against DDoS attacks?

-  A. AWS Shield
- B. Amazon Inspector
- C. Amazon GuardDuty
- D. Amazon Detective

Correct Answer: A

AWS Shield is a managed **Distributed Denial of Service (DDoS)** protection service that safeguards applications running on AWS. AWS Shield provides always-on detection and automatic inline mitigations that minimize application downtime and latency, so there is no need to engage AWS Support to benefit from DDoS protection.

301) Which Amazon Web Services (AWS) offering allows customers to aggregate billing for many accounts?

-  A. Amazon QuickSight
- B. AWS Organizations
- C. AWS Budgets
- D. Amazon Forecast

Correct Answer: B

You can use the consolidated billing feature in AWS Organizations to consolidate billing and payment for multiple AWS accounts or multiple Amazon Internet Services Pvt. Ltd (AISPL) accounts. Every organization in AWS Organizations has a master (payer) account that pays the charges of all the member (linked) accounts.

302) Which of the following are advantages of Amazon Web Services' cloud computing platform? (Select two.)

- ☐ A. Unlimited uptime
- ☒ B. Elasticity
- ☒ C. Agility
- ☐ D. Colocation
- ☐ E. Capital expenses

Correct Answer: BC

The most celebrated benefit of AWS cloud is elasticity since you can expand the services when you experience more traffic.

Agile developments in AWS Cloud through strategies are day by day becoming more established within the enterprises across the world. With so much improvement and call for optimization in the cloud, it is necessary that these strategies get established from the ground up within the organizations. It is highly important as already enterprises have a lot of bequest, politics and hierarchies which act as barriers in their businesses.

303) An application installed in the AWS Cloud exhibits irregular consumption patterns and is responsible for non-stop workloads.

Which Amazon EC2 pricing plan is the MOST cost-effective for this application?

- ☐ A. Dedicated Instances
- ☐ B. Spot Instances
- ☐ C. Reserved Instances
- ☒ D. On-Demand Instances

Correct Answer: D

On-demand instances are **virtual servers that run in Amazon EC2** and are available on demand, with organizations paying for them by the hour or second with no long-term commitments. An instance refers to a virtual server in the AWS Cloud, specifically in the Amazon EC2 compute platform.

304) Which service's primary aim is to manage software versions?

- A. Amazon CodeStar
- B. AWS Command Line Interface (AWS CLI)
- C. Amazon Cognito
-  D. AWS CodeCommit

Correct Answer: D

AWS CodeCommit is a version control service hosted by Amazon Web Services that you can use to privately store and manage assets (such as documents, source code, and binary files) in the cloud.

305) A business is now undergoing a security examination. The audit includes a security and compliance review of the company's AWS infrastructure and services. The auditor must identify and download AWS security and compliance materials. The System and Organization Control (SOC) reports are one of these documents.

Which AWS service or collection of services is capable of providing these documents?

- A. AWS Abuse team
-  B. AWS Artifact
- C. AWS Support
- D. AWS Config

Correct Answer: B

AWS Artifact is a portal that provides an enterprise with access to security and compliance reports that apply to the Amazon Web Services (AWS) public cloud.

306) Using AWS Config to record, audit, and evaluate changes to AWS resources to enable traceability is an example of which AWS Well-Architected Framework pillar?

-  A. Security
- B. Operational excellence
- C. Performance efficiency
- D. Cost optimization

307) Which of the following is a design concept for AWS Cloud architecture?

- A. Implement single points of failure.
-  B. Implement loose coupling.
- C. Implement monolithic design.
- D. Implement vertical scaling.

Correct Answer: B

Loose coupling between services can also be done through asynchronous integration. It involves one component that generates events and another that consumes them. The two components do not integrate through direct point-to-point interaction, but usually through an intermediate durable storage layer. This approach decouples the two components and introduces additional resiliency. So, for example, if a process that is reading messages from the queue fails, messages can still be added to the queue to be processed when the system recovers.

308) The continuous lowering in AWS Cloud price is a result of the following:

- A. pay-as-you go pricing
- B. the AWS global infrastructure
-  C. economies of scale
- D. reserved storage pricing

Correct Answer: C

•More customers lead to higher **AWS** usage which leads to more infrastructure.
The **economies of scale** effect occurs because costs can now be spread over a larger number of customers. The infrastructure becomes cheaper, prices can be reduced and new customers benefit from the lower prices.

309) Which phase discusses agility as an advantage of AWS Cloud-based development?

- A. The ability to pay only when computing resources are consumed, based on the volume of resources that are consumed
- B. The ability to eliminate guessing about infrastructure capacity needs
-  C. The ability to support innovation through a reduction in the time that is required to make IT resources available to developers
- D. The ability to deploy an application in multiple AWS Regions around the world in minutes

Correct Answer: C

310) Which AWS tool or feature acts as a VPC firewall at the subnet level?

- A. Security group
-  B. Network ACL
- C. Traffic Mirroring
- D. Internet gateway

Correct Answer: B

"A network access control list (ACL) allows or denies specific inbound or outbound traffic at the SUBNET LEVEL"

311) A business wishes to deliver managed Windows virtual desktops and programs through secure network connections to its distant workers.

Which AWS services does the business have access to in order to achieve these requirements? (Select two.)

- ☐ A. Amazon Connect
- ☒ B. Amazon AppStream 2.0
- ☒ C. Amazon WorkSpaces
- ☐ D. AWS Site-to-Site VPN
- ☐ E. Amazon Elastic Container Service (Amazon ECS)

Correct Answer: BC

With AppStream 2.0, you can easily add your existing desktop applications to AWS and enable your users to instantly stream them. Windows users can use either the AppStream 2.0 client or an HTML5-capable web browser for application streaming.

Amazon WorkSpaces enables you to provision virtual, cloud-based Microsoft Windows, Amazon Linux, or Ubuntu Linux desktops for your users, known as *WorkSpaces*.

312) Which of the following is an AWS shared responsibility?

- ☐ A. Identity and access management (IAM)
- ☐ B. Server-side encryption (SSE)
- ☐ C. Firewall configuration
- ☒ D. Maintaining physical hardware

Correct Answer: D

AWS is responsible for protecting the infrastructure that runs all of the services offered in the AWS Cloud. This infrastructure is composed of the hardware, software, networking, and facilities that run AWS Cloud services.

313) Which storage service can be utilized to host static webpages at a minimal cost?

- A. Amazon Glacier
- B. Amazon DynamoDB
- C. Amazon Elastic File System (Amazon EFS)
-  D. Amazon Simple Storage Service (Amazon S3)

Correct Answer: D

You can use **Amazon S3** to host a static website. On a static website, individual webpages include static content. They might also contain client-side scripts. By contrast, a dynamic website relies on server-side processing, including server-side scripts, such as PHP, JSP, or ASP.NET. Amazon S3 does not support server-side scripting, but AWS has other resources for hosting dynamic websites.

314) Which AWS service can be used to decouple applications?

- A. AWS Config
-  B. Amazon Simple Queue Service (Amazon SQS)
- C. AWS Batch
- D. Amazon Simple Email Service (Amazon SES)

Correct Answer: B

Amazon Simple Queue Service (Amazon SQS) is a fully managed message queuing service that makes it easy to decouple and scale microservices, distributed systems, and serverless applications. Amazon SQS moves data between distributed application components and helps you decouple these components.

315) Which AWS services are globally specified rather than regionally defined? (Select two.)

- ➡ A. Amazon Route 53
- B. Amazon EC2
- C. Amazon S3
- ➡ D. Amazon CloudFront
- E. Amazon DynamoDB

Correct Answer: AD

AWS Global, Regional, AZ resource Availability

AWS provides a lot of services and these services are either Global, Regional or specific to the Availability Zone and cannot be accessed outside. Most of the AWS managed services are regional based services (except for IAM, Route53, CloudFront, WAF etc).

316) Currently, a company's online program is highly dependent on underlying components, which means that when one component fails, the whole web site crashes.

Which AWS Cloud design concept is most appropriate for resolving the present design issue?

- A. Implementing elasticity, enabling the application to scale up or scale down as demand changes.
- B. Enabling several EC2 instances to run in parallel to achieve better performance.
- ▶ C. Focusing on decoupling components by isolating them and ensuring individual components can function when other components fail.
- D. Doubling EC2 computing resources to increase system fault tolerance.

Correct Answer: C

317) A user must locate, categorize, and safeguard sensitive data stored in Amazon S3 automatically.

Which AWS service satisfies these criteria?

- A. Amazon Inspector
-  B. Amazon Macie
- C. Amazon GuardDuty
- D. AWS Secrets Manager

Correct Answer: B

To help you manage the security posture of your organization's Amazon Simple Storage Service (Amazon S3) data estate, Macie provides you with an inventory of your S3 buckets, and automatically evaluates and monitors the buckets for security and access control. If Macie detects a potential issue with the security or privacy of your data, such as a bucket that becomes publicly accessible, Macie generates a finding for you to review and remediate as necessary.

318) On an Amazon EC2 instance, a business hosts an application. Several AWS resources, including Amazon S3 and Amazon DynamoDB, are required by the EC2 instance.

What is the MOST OPTIMAL way to delegate permissions?

-  A. Create an IAM role with the required permissions. Attach the role to the EC2 instance.
- B. Create an IAM user and use its access key and secret access key in the application.
- C. Create an IAM user and use its access key and secret access key to create a CLI profile in the EC2 instance
- D. Create an IAM role with the required permissions. Attach the role to the administrative IAM user.

Correct Answer: A

319) Which of the following are support categories for AWS Trusted Advisor?
(Select two.)

-  A. Operational excellence
-  B. Cost optimization
-  C. Security
- D. Well-Architected Framework
- E. Rightsizing

Correct Answer: *BC*

AWS Trusted Advisor -
Reduce costs, improve performance, improve security

320) Which disaster recovery option is the LEAST expensive?

- A. Warm standby
- B. Multisite
-  C. Backup and restore
- D. Pilot light

Correct Answer: *C*

Sending backups to AWS can be the first step in reducing total cost of ownership and your data center footprint.

321) Which type of AWS storage is ephemeral and is deleted when an Amazon EC2 instance is stopped or terminated?

- ☐ A. Amazon Elastic Block Store (Amazon EBS)
- ☒ B. Amazon EC2 instance store
- ☐ C. Amazon Elastic File System (Amazon EFS)
- ☐ D. Amazon S3

Correct Answer: B

When you stop or terminate an instance, every block of storage in the instance store is reset. Therefore, your data cannot be accessed through the instance store of another instance.

322) A company needs fully managed, highly reliable, and scalable file storage that is accessible over the Server Message Block (SMB) protocol.

Which AWS service will meet these requirements?

- ☐ A. Amazon S3
- ☐ B. Amazon Elastic File System (Amazon EFS)
- ☒ C. Amazon FSx for Windows File Server
- ☐ D. Amazon Elastic Block Store (Amazon EBS)

Correct Answer: C

Amazon FSx for Windows File Server is a fully managed, highly reliable, and scalable file storage service that is accessible over the Server Message Block (SMB) protocol. It provides fully managed, highly reliable, and scalable file storage that is accessible over the SMB protocol. It is designed to work with the Microsoft Windows operating system and it supports the SMB protocol which allows to access the file storage from Windows servers and clients.

323) Which of the following is a component of the AWS Global Infrastructure?

-  A. Amazon Alexa
- B. AWS Regions
- C. Amazon Lightsail
- D. AWS Organizations

Correct Answer: B

The AWS Cloud spans 99 Availability Zones within 31 geographic regions around the world, with announced plans for 12 more Availability Zones and 4 more AWS Regions in Canada, Israel, New Zealand, and Thailand.

324) What is the purpose of having an internet gateway within a VPC?

-  A. To create a VPN connection to the VPC
- B. To allow communication between the VPC and the internet
- C. To impose bandwidth constraints on internet traffic
- D. To load balance traffic from the internet across Amazon EC2 instances

Correct Answer: B

An internet gateway is a horizontally scaled, redundant, and highly available VPC component that allows communication between your VPC and the internet.

325) Which AWS service allows users to download security and compliance reports about the AWS infrastructure on demand?

- A. Amazon GuardDuty
- B. AWS Security Hub
-  C. AWS Artifact
- D. AWS Shield

Correct Answer: C

AWS Artifact is your go-to, central resource for compliance-related information that matters to you. It provides on-demand access to security and compliance reports from AWS and ISVs who sell their products on AWS Marketplace.

326) A company is planning an infrastructure deployment to the AWS Cloud. Before the deployment, the company wants a cost estimate for running the infrastructure. Which AWS service or feature can provide this information?

- A. Cost Explorer
- B. AWS Trusted Advisor
- C. AWS Cost and Usage Report
-  D. AWS Pricing Calculator

Correct Answer: D

AWS Pricing Calculator

Estimate the cost for your architecture solution.

Configure a cost estimate that fits your unique business or personal needs with AWS products and services.

327) How can Amazon EC2 Auto Scaling groups contribute to a web application's high availability?

- A. They automatically add more instances across multiple AWS Regions based on global demand of the application.
-  B. They automatically add or replace instances across multiple Availability Zones when the application needs it.
- C. They enable the application's static content to reside closer to end users.
- D. They are able to distribute incoming requests across a tier of web server instances.

Correct Answer: B

When the unhealthy Availability Zone returns to a healthy state, Auto Scaling automatically redistributes the application instances evenly across all of the designated Availability Zones.

328) A business has a stateless application workload that can withstand brief periods of outage. The program conducts calculations in massively parallel fashion.

Which Amazon EC2 pricing model should the business choose to save costs?

- A. On-Demand Instances
- B. Spot Instances
- C. Reserved Instances
- D. Dedicated Instances

Correct Answer: B

A Spot Instance is an instance that uses **spare EC2 capacity that is available for less than the On-Demand price**. Because Spot Instances enable you to request unused EC2 instances at steep discounts, you can lower your Amazon EC2 costs significantly. The hourly price for a Spot Instance is called a Spot price.

329) Which of the following is a method for enhancing AWS security? (Select two.)

- A. Using AWS Artifact
- B. Granting the broadest permissions to all IAM roles
- C. Running application code with AWS Cloud9
-  D. Enabling multi-factor authentication (MFA) with Amazon Cognito
-  E. Using AWS Trusted Advisor security checks

Correct Answer: DE

MFA adds extra security because it requires users to provide unique authentication from an AWS supported MFA mechanism in addition to their regular sign-in credentials when they access AWS websites or services: **Virtual MFA devices**. A software app that runs on a phone or other device and emulates a physical device.

330) Which AWS service or tool helps to centrally manage billing and allow controlled access to resources across AWS accounts?

-  A. AWS Identity and Access Management (IAM)
- B. AWS Organizations
- C. Cost Explorer
- D. AWS Budgets

Correct Answer: B

By default, organizations support **consolidated billing** features. Consolidated billing separates paying accounts and linked accounts. You can use AWS Organizations to set up a single payment method for all the AWS accounts in your organization through consolidated billing.

331) Under the AWS shared responsibility paradigm, the following parties are responsible for the security and patching of the guest operating system:

- A. AWS Support
-  B. the customer
- C. AWS Systems Manager
- D. AWS Config

Correct Answer: B

For example, a service such as Amazon Elastic Compute Cloud (Amazon EC2) is categorized as Infrastructure as a Service (IaaS) and, as such, requires the customer to perform all of the necessary security configuration and management tasks. Customers that deploy an Amazon EC2 instance are responsible for management of the guest operating system (including updates and security patches), any application software or utilities installed by the customer on the instances, and the configuration of the AWS-provided firewall (called a security group) on each instance.

332) What is an example of AWS Cloud's high availability?

- A. Consulting AWS technical support at any time day or night
-  B. Ensuring an application remains accessible, even if a resource fails
- C. Making any AWS service available for use by paying on demand
- D. Deploying in any part of the world using AWS Regions

Correct Answer: B

333) Which of the following are Amazon Virtual Private Cloud (Amazon VPC) resources?

-  A. Objects; access control lists (ACLs)
- B. Subnets; internet gateways
- C. Access policies; buckets
- D. Groups; roles

Correct Answer: B

A subnet is a range of IP addresses in your VPC. A subnet must reside in a single Availability Zone. After you add subnets, you can deploy AWS resources in your VPC.

A gateway connects your VPC to another network. For example, use an internet gateway to connect your VPC to the internet. Use a VPC endpoint to connect to AWS services privately, without the use of an internet gateway or NAT device.

334) Which aspects of AWS make it cost viable for workloads with fluctuating customer demand? (Select two.)

- A. High availability
- B. Shared security model
-  C. Elasticity
-  D. Pay-as-you-go pricing
- E. Reliability

Correct Answer: CD

335) A business wants to migrate 40TB of data from on-premises storage systems to Amazon S3. The transfer must occur as rapidly as feasible while maintaining a low cost. The firm has internet access at a speed of 100 megabits per second.

Which AWS service will satisfy these criteria?

-  A. AWS Snowball
- B. AWS Direct Connect
- C. AWS Storage Gateway
- D. Amazon S3 Transfer Acceleration

Correct Answer: A

Snowball is better for 40TB transfer. It will make transfer while keeping the costs at minimum. With 100 mbps, it will be quick.

336) Which of the following allows customers to programmatically access the power of AWS services?

-  A. AWS Command Line Interface (AWS CLI)
- B. AWS Trusted Advisor
- C. AWS CodeDeploy
- D. AWS Management Console

Correct Answer: A

The AWS Command Line Interface (AWS CLI) is a unified tool to manage your AWS services. With just one tool to download and configure, you can control multiple AWS services from the command line and automate them through scripts.

337) A business wishes to migrate some of its resources to the AWS Cloud. To comply with regulatory standards, data must stay on-premises and local. AWS and the company's resources must have a low latency.

Which Amazon Web Services (AWS) service or functionality may be leveraged to fulfill these requirements?

- A. AWS Local Zones
- B. Availability Zones
- C. AWS Outposts
- D. AWS Wavelength Zones

Correct Answer: C

With AWS Outposts, you can run some AWS services locally and connect to a broad range of services available in the local AWS Region. Run applications and workloads on premises using familiar AWS services, tools, and APIs. Outposts supports workloads and devices requiring low latency access to on-premises systems, local data processing, data residency, and application migration with local system interdependencies.

338) A business has specified the Amazon Web Services (AWS) resources it requires for a new application. The business must determine the expenses associated with operating the application on AWS.

What actions should the business take to comply with this stipulation?

- A. Take advantage of AWS on-demand pricing.
- B. Use the AWS Pricing Calculator to generate an approximate dollar amount.
- C. Use Amazon QuickSight to analyze current on-premises spending.
- D. Use Amazon AppStream 2.0 for real-time pricing analytics.

Correct Answer: B

Estimate your AWS billing

- If you plan to migrate significant infrastructure to AWS, use the [AWS Sales & Business Development contact form](#), and then choose **I need to speak to someone in sales**.
- To estimate a bill, use the [AWS Pricing Calculator](#). Choose **Create estimate**, and then choose your planned resources by service. The AWS Pricing Calculator provides an estimated cost per month. For more information, see [What is AWS Pricing Calculator?](#)
- To forecast your costs, use the [AWS Cost Explorer](#). Use [cost allocation tags](#) to divide your resources into groups, and then estimate the costs for each group.

339) For its steady-state database, a corporation runs a self-managed Oracle database directly on Amazon EC2. The corporation wishes to cut its computing expenses.

Which option should the business choose in order to optimize savings over a three-year period?

- A. EC2 Dedicated Instances
- B. EC2 Spot Instances
-  C. EC2 Reserved Instances
- D. EC2 On-Demand Instances

Correct Answer: C

Amazon EC2 Reserved Instances (RI) provide a significant discount (up to 72%) compared to On-Demand pricing and provide a capacity reservation when used in a specific Availability Zone.

340) A corporation is developing an application that must distribute photos and videos internationally with the least amount of delay possible.

Which strategy may the business take to do this in the most cost-effective manner?

-  A. Deliver the content through Amazon CloudFront.
- B. Store the content on Amazon S3 and enable S3 cross-region replication.
- C. Implement a VPN across multiple AWS Regions.
- D. Deliver the content through AWS PrivateLink.

Correct Answer: A

Amazon CloudFront is a content delivery network (CDN) operated by **Amazon Web Services**. Content delivery networks provide a globally-distributed network of proxy servers that cache content, such as web videos or other bulky media, more locally to consumers, thus improving access speed for downloading the content

341) Which AWS service or functionality gives advice to customers about frequently asked billing questions?

- ☐ A. AWS Marketplace
- ☒ B. AWS Knowledge Center
- ☐ C. Amazon Pinpoint
- ☐ D. Amazon Connect

Correct Answer: B

All **AWS** account owners have access to account and **billing** support free of charge.

342) Which AWS services enable an on-premises infrastructure to be extended to the AWS Cloud? (Select two.)

- ☐ A. Amazon EBS
- ☒ B. AWS Direct Connect
- ☐ C. Amazon CloudFront
- ☒ D. AWS Storage Gateway
- ☐ E. Amazon Connect

Correct Answer: BD

AWS Direct Connect is a **cloud service solution** that makes it easy to establish a dedicated network connection from your premises to AWS.

AWS Storage Gateway is a set of hybrid cloud storage services that provide on-premises access to virtually unlimited cloud storage.

343) Which AWS service enables the generation of warnings based on a monthly bill estimate?

- A. AWS Config
-  B. Amazon CloudWatch
- C. AWS X-Ray
- D. AWS CloudTrail

Correct Answer: B

You can monitor your estimated AWS charges by using Amazon CloudWatch. When you enable the monitoring of estimated charges for your AWS account, the estimated charges are calculated and sent several times daily to CloudWatch as metric data.

Billing metric data is stored in the US East (N. Virginia) Region and represents worldwide charges. This data includes the estimated charges for every service in AWS that you use, in addition to the estimated overall total of your AWS charges.

344) A company needs to identify the last time that a specific user accessed the AWS Management Console. Which AWS service will provide this information?

- A. Amazon Cognito
-  B. AWS CloudTrail
- C. Amazon Inspector
- D. Amazon GuardDuty

Correct Answer: B

AWS CloudTrail enables auditing, security monitoring, and operational troubleshooting by tracking user activity and API usage. CloudTrail logs, continuously monitors, and retains account activity related to actions across your AWS infrastructure, giving you control over storage, analysis, and remediation actions.

345) Which AWS service may be utilized to deploy the application?

- A. AWS AppSync
- B. AWS Batch
-  C. AWS CodePipeline
- D. AWS DataSync

Correct Answer: C

AWS CodePipeline is a continuous delivery service you can use to model, visualize, and automate the steps required to release your software. You can quickly model and configure the different stages of a software release process. CodePipeline automates the steps required to release your software changes continuously.

346) Which AWS service or functionality is by default highly available?

- A. Amazon EC2
- B. Amazon Aurora
- C. NAT instances
-  D. Amazon RDS

Correct Answer: D

Amazon Relational Database Service (Amazon RDS) supports two easy-to-use options for ensuring High Availability of your relational database. For your MySQL, MariaDB, PostgreSQL, Oracle, and SQL Server database (DB) instances, you can use **Amazon RDS Multi-AZ** deployments. When you provision a Multi-AZ DB instance, Amazon RDS automatically creates a primary DB instance and synchronously replicates the data to a standby instance in a different Availability Zone (AZ).

347) A business requires data migration from its on-premises environment to the AWS Cloud. The business demands extremely elastic and efficient connection.

Which AWS service satisfies these criteria?

- A. Amazon S3 Glacier
-  B. AWS Storage Gateway
- C. AWS Backup
- D. Amazon Elastic File System (Amazon EFS)

Correct Answer: B

AWS Storage Gateway is a hybrid cloud storage service that gives you on-premises access to virtually unlimited cloud storage. You can use Storage Gateway to simplify storage management and reduce costs for key hybrid cloud storage use cases. These include moving backups to the cloud, using on-premises file shares backed by cloud storage, and providing low-latency access to data in AWS for on-premises applications.

348) A company wants to perform sentiment analysis on customer service email messages that it receives. The company wants to identify whether the customer service engagement was positive or negative. Which AWS service should the company use to perform this analysis?

- A. Amazon Textract
- B. Amazon Translate
-  C. Amazon Comprehend
- D. Amazon Rekognition

Correct Answer: C

Amazon Comprehend uses natural language processing (NLP) to extract insights about the content of documents. It develops insights by recognizing the entities, key phrases, language, sentiments, and other common elements in a document.

349) What is the total amount of storage offered by Amazon S3?

- A. 100MB
- B. 5 GB
- C. 5 TB
-  D. Unlimited

Correct Answer: D

The total volume of data and number of objects you can store are unlimited. Individual Amazon S3 objects can range in size from a minimum of 0 bytes to a maximum of 5 TB.

350) Which AWS service or functionality is exclusively accessible to AWS Enterprise Support customers?

- A. AWS Trusted Advisor
- B. AWS Support case
-  C. AWS Concierge Support team
- D. Amazon Connect

Correct Answer: C

AWS Enterprise Support - SUPPORT CONCIERGE Dedicated team of enterprise account specialists to help with billing and account subjects

351) Which resource is the MOST EFFECTIVE for keeping current on AWS security announcements?

- A. AWS Personal Health Dashboard
- B. AWS Secrets Manager
- ➔ C. AWS Security Bulletins
- D. Amazon Inspector

Correct Answer: C

AWS makes public notifications in the form of [Security Bulletins](#), which are posted in the AWS Security website. Individuals, companies, and security teams typically post their advisories on their own websites and in other forums and when relevant, we will include links to those third-party resources in AWS Security Bulletins.

352) Which AWS services aid in application performance optimization by lowering latency associated with global content access? (Select two.)

- ➔ A. Amazon CloudFront
- B. AWS VPN
- C. AWS Direct Connect
- ➔ D. AWS Global Accelerator
- E. Amazon S3 Glacier

Correct Answer: AD

Amazon CloudFront is a content delivery network (CDN) service built for high performance, security, and developer convenience.

AWS Global Accelerator is a networking service that helps you improve the availability, performance, and security of your public applications. Global Accelerator provides two global static public IPs that act as a fixed entry point to your application endpoints, such as Application Load Balancers, Network Load Balancers, Amazon Elastic Compute Cloud (EC2) instances, and elastic IPs.

354) Which techniques are available for deriving AWS charges by department? (Select two.)

- ☐ A. Enable multi-factor authentication for the AWS account root user.
- ☒ B. Create separate accounts for each department.
- ☐ C. Use Reserved Instances whenever possible.
- ☒ D. Use tags to associate each instance with a particular department.
- ☐ E. Pay bills using purchase orders.

Correct Answer: *BD*

355) Which AWS services can be utilized to automate infrastructure? (Select two.)

- ☒ A. AWS CloudFormation
- ☐ B. Amazon CloudFront
- ☐ C. AWS Batch
- ☒ D. AWS OpsWorks
- ☐ E. Amazon QuickSight

Correct Answer: *AD*

356) The solutions architect of a business want to deploy a few Amazon EC2 instances in an already-existing AWS account. Before the firm can accept the request, it wants a cost estimate.

Which AWS tool should be used to generate the cost estimate?

-  A. AWS Pricing Calculator
- B. Cost Explorer
- C. AWS Cost and Usage Report
- D. AWS Budgets

Correct Answer: A

AWS Pricing Calculator is a web-based planning tool that you can use to create estimates for your AWS use cases. You can use it to model your solutions before building them, explore the AWS service price points, and review the calculations behind your estimates. You can use it to help you plan how you spend, find cost saving opportunities, and make informed decisions when using Amazon Web Services.

357) A company is migrating to Amazon S3. The company needs to transfer 60 TB of data from an on-premises data center to AWS within 10 days.

Which AWS service should the company use to accomplish this migration?

- A. Amazon S3 Glacier
- B. AWS Database Migration Service (AWS DMS)
-  C. AWS Snowball
- D. AWS Direct Connect

Correct Answer: C

358) What type of database is Amazon DynamoDB?

- A. In-memory
- B. Relational
-  C. Key-value
- D. Graph

Correct Answer: C

Amazon DynamoDB is a fully managed, serverless, key-value NoSQL database designed to run high-performance applications at any scale

359) A retail company has recently migrated its website to AWS. The company wants to ensure that it is protected from SQL injection attacks. The website uses an Application Load Balancer to distribute traffic to multiple Amazon EC2 instances. Which AWS service or feature can be used to create a custom rule that blocks SQL injection attacks?

- A. Security groups
-  B. AWS WAF
- C. Network ACLs
- D. AWS Shield

Correct Answer: B

360) Which AWS services enable users to link their AWS Cloud resources to their on-premises resources? (Select two.)

-  A. AWS VPN
- B. Amazon Connect
- C. Amazon Cognito
-  D. AWS Direct Connect
- E. AWS Managed Services

Correct Answer: AD

361) While an Amazon EC2 instance is only activated when it is required, it must stay active throughout the life of the procedure.

What is the most acceptable method of purchase?

- A. Dedicated Instances
- B. Spot Instances
-  C. On-Demand Instances
- D. Reserved Instances

Correct Answer: C

362) A business's operations are conducted on-site. The business want to anticipate the costs associated with hosting a major application on AWS.

Which AWS service or tool does the business have access to in order to collect this information?

-  A. AWS Pricing Calculator
- B. AWS Budgets
- C. AWS Trusted Advisor
- D. Cost Explorer

Correct Answer: A

363) How can a client strengthen the security of their Amazon Web Services (AWS) account logins? (Select two.)

- ☐ A. Configure AWS Certificate Manager
- ☒ B. Enable Multi-Factor Authentication (MFA)
- ☐ C. Use Amazon Cognito to manage access
- ☒ D. Configure a strong password policy
- ☐ E. Enable AWS Organizations

Correct Answer: *BD*

Your root account should always be protected by Multi-Factor Authentication (MFA). This additional layer of security helps protect against unauthorized logins to your account by requiring two factors: something you know (a password) and something you have (for example, an MFA device). AWS supports virtual and hardware MFA devices and U2F security keys.

364) A big firm often has a single Amazon Web Services (AWS) account.

What are the benefits of repurposing a single AWS account to create additional AWS accounts? (Select two.)

- ☒ A. It allows for administrative isolation between different workloads.
- ☐ B. Discounts can be applied on a quarterly basis by submitting cases in the AWS Management Console.
- ☐ C. Transitioning objects from Amazon S3 to Amazon S3 Glacier in separate AWS accounts will be less expensive.
- ☒ D. Having multiple accounts reduces the risks associated with malicious activity targeted at a single account.
- ☐ E. Amazon QuickSight offers access to a cost tool that provides application-specific recommendations for environments running in multiple accounts.

Correct Answer: *AD*

366) Which AWS services may be protected against typical online exploitations using AWS WAF? (Select two.)

- A. Amazon Route 53
-  B. Amazon CloudFront
- C. AWS Transfer Family
- D. AWS Site-to-Site VPN
-  E. Amazon API Gateway

Correct Answer: *BE*

367) A business uses an Amazon EC2 Auto Scaling policy in conjunction with an Application Load Balancer to restore sick apps running on Amazon EC2 instances automatically.

This activity pertains to which pillar of the AWS Well-Architected Framework?

- A. Security
- B. Performance efficiency
- C. Operational excellence
-  D. Reliability

Correct Answer: *D*

368) A business want to migrate current software licenses to AWS, however the licensing strategy involves licensing physical cores.

How can the organization do this in the AWS Cloud?

- A. Launch an Amazon EC2 instance with default tenancy.
-  B. Launch an Amazon EC2 instance on a Dedicated Host.
- C. Create an On-Demand Capacity Reservation.
- D. Purchase Dedicated Reserved Instances.

Correct Answer: *B*

369) Which service offers nearly infinite online storage for very durable objects?

- A. Amazon Redshift
- B. Amazon Elastic File System (Amazon EFS)
- C. Amazon Elastic Container Service (Amazon ECS)
-  D. Amazon S3

Correct Answer: D

370) Which of the following is an advantage that users experience when they move on-premises workloads to the AWS Cloud?

-  A. Elimination of expenses for running and maintaining data centers
- B. Price discounts that are identical to discounts from hardware providers
- C. Distribution of all operational controls to AWS
- D. Elimination of operational expenses

Correct Answer: A

371) Which AWS services are available for file storage? (Select two.)

-  A. Amazon S3
- B. AWS Lambda
-  C. Amazon Elastic Block Store (Amazon EBS)
- D. Amazon SageMaker
- E. AWS Storage Gateway

Correct Answer: AC

373) Which AWS advantage allows customers to construct cloud infrastructure that spans various geographic areas and is linked through a low-latency, high-performance network?

- A. Economies of scale
- B. Security
- C. Elasticity
- ▶ D. Global reach

Correct Answer: D

375) Which AWS IAM feature is used to link numerous users with a set of permissions?

- A. Multi-factor authentication
- ▶ B. Groups
- C. Password policies
- D. Access keys

Correct Answer: B

An IAM group is a collection of IAM users. You can use groups to specify permissions for a collection of users, which can make those permissions easier to manage for those users. For example, you could have a group called Admins and give that group the types of permissions that administrators typically need.

376) A social media firm has been tasked with the responsibility of launching a new function. Users will be able to share photographs that will be seen by other users worldwide with little delay.

Which AWS service or feature should the business use in order to achieve this demand the most cost-effectively?

A. Amazon DynamoDB global tables

B. AWS Direct Connect

 C. Amazon CloudFront

D. AWS Outposts

Correct Answer: C

Securely deliver content with low latency and high transfer speeds.

377) Which AWS service enables application decoupling?

A. AWS Config

 B. Amazon Simple Queue Service (Amazon SQS)

C. AWS Batch

D. Amazon Simple Email Service (Amazon SES)

Correct Answer: B

Amazon SQS refers to a **message queuing service that is used for decoupling and scaling microservices, distributed systems, and serverless applications**. SQS removes the complexity and overhead linked with managing and operating message-oriented middleware and empowers developers for focusing on differentiating work.

378) Amazon EC2 web servers connect to a traditional application operating in a corporate data center.

What phrase would be appropriate to use to describe this model?

- A. Cloud-native
- B. Partner network
-  C. Hybrid architecture
- D. Infrastructure as a service

Correct Answer: C

AWS hybrid cloud services deliver a consistent AWS experience wherever you need it – from the cloud, to on premises, and at the edge. Select from the broadest set of compute, networking, storage, security, identity, data integration, management, monitoring, and operations services to build hybrid architectures that meet your specific requirements and use cases.

379) Which design principle is included in the operational excellence pillar of the AWS Well-Architected Framework?

- A. Create annotated documentation.
-  B. Anticipate failure.
- C. Ensure performance efficiency.
- D. Optimize costs.

Correct Answer: B

380) Which of the following is the customer responsible for updating and patching, according to the AWS shared responsibility model?

-  A. Amazon FSx for Windows File Server
- B. Amazon WorkSpaces virtual Windows desktop
- C. AWS Directory Service for Microsoft Active Directory
- D. Amazon RDS for Microsoft SQL Server

Correct Answer: B

381) A previously utilized Amazon EC2 instance for development is no longer available and does not display in the AWS Management Console.

Which AWS service should be utilized to discover what activity resulted in the inaccessibility of this EC2 instance?

- A. Amazon CloudWatch Logs
- B. AWS Security Hub
- C. Amazon Inspector
-  D. AWS CloudTrail

Correct Answer: D

Visibility into your AWS account activity is a key aspect of security and operational best practices. You can use CloudTrail to view, search, download, archive, analyze, and respond to account activity across your AWS infrastructure. You can identify who or what took which action, what resources were acted upon, when the event occurred, and other details to help you analyze and respond to activity in your AWS account. Optionally, you can enable AWS CloudTrail Insights on a trail to help you identify and respond to unusual activity.

382) A Cloud Practitioner is required to retain data for seven years in order to comply with regulatory standards.

Which AWS service meets this need for the LEAST amount of money?

- A. Amazon S3
- B. AWS Snowball
- C. Amazon Redshift
-  D. Amazon S3 Glacier

Correct Answer: D

S3 Glacier Deep Archive is Amazon S3's lowest-cost storage class and supports long-term retention and digital preservation for data that may be accessed once or twice in a year. It is designed for customers "particularly those in highly-regulated industries, such as the Financial Services, Healthcare, and Public Sectors" that retain data sets for 7-10 years or longer to meet regulatory compliance requirements. S3 Glacier Deep Archive can also be used for backup and disaster recovery use cases, and is a cost-effective and easy-to-manage alternative to magnetic tape systems, whether they are on-premises libraries or off-premises services.

383) Why is an AWS Well-Architected review a critical part of the cloud design process?

- A. A Well-Architected review is mandatory before a workload can run on AWS.
-  B. A Well-Architected review helps identify design gaps and helps evaluate design decisions and related documents.
- C. A Well-Architected review is an audit mechanism that is a part of requirements for service level agreements.
- D. A Well-Architected review eliminates the need for ongoing auditing and compliance tests.

Correct Answer: B

384) A company implements an Amazon EC2 Auto Scaling policy along with an Application Load Balancer to automatically recover unhealthy applications that run on Amazon EC2 instances.

Which pillar of the AWS Well-Architected Framework does this action cover?

- A. Security
- B. Performance efficiency
- C. Operational excellence
-  D. Reliability

Correct Answer: D

385) What does AWS's obligation under the shared responsibility paradigm entail?

- A. Updating the network ACLs to block traffic to vulnerable ports.
- B. Patching operating systems running on Amazon EC2 instances.
-  C. Updating the firmware on the underlying EC2 hosts.
- D. Updating the security group rules to block traffic to the vulnerable ports.

Correct Answer: C

386) AWS CloudFormation is intended to assist the user in the following ways:

-  A. model and provision resources.
- B. update application code.
- C. set up data lakes.
- D. create reports for billing.

Correct Answer: A

AWS CloudFormation provides a common language for you to model and provision AWS and third party application resources in your cloud environment. AWS CloudFormation allows you to use programming languages or a simple text file to model and provision, in an automated and secure manner, all the resources needed for your applications across all regions and accounts. This gives you a single source of truth for your AWS and third party resources.

387) Which AWS Cloud benefit is shown by an architecture's ability to withstand failures with minimal downtime?

- A. Agility
- B. Elasticity
- C. Scalability
-  D. High availability

Correct Answer: D

388) Under the AWS shared responsibility model, which task is the customer's responsibility when managing AWS Lambda functions?

- ☒ A. Creating versions of Lambda functions
- ☐ B. Maintaining server and operating systems
- ☐ C. Scaling Lambda resources according to demand
- ☐ D. Updating the Lambda runtime environment

Correct Answer: A

389) A company needs to generate reports that can break down cloud costs by product, by company-defined tags, and by hour, day, and month.
Which AWS tool should the company use to meet these requirements?

- ☐ A. Reserved Instance utilization and coverage reports
- ☐ B. Savings Plans utilization reports
- ☐ C. AWS Budgets reports
- ☒ D. AWS Cost and Usage Reports

Correct Answer: D

390) A business wishes to grant access to Amazon RDS to one of its workers. Additionally, the corporation wishes to confine contact to the AWS CLI and AWS software development kits (SDKs).
Which combination of measures should the business do in order to satisfy these needs while adhering to the concept of least privilege? (Select two.)

- ☐ A. Create an IAM user and provide AWS Management Console access only.
- ☒ B. Create an IAM user and provide programmatic access only.
- ☐ C. Create an IAM role and provide AWS Management Console access only.
- ☐ D. Create an IAM policy with administrator access and attach it to the IAM user.
- ☒ E. Create an IAM policy with Amazon RDS access and attach it to the IAM user.

Correct Answer: BE

391) Which of the following guidelines constitutes a well-architected design philosophy for cloud application development?

- A. Keep static data closer to compute resources.
- B. Provision resources for peak capacity.
-  C. Design for automated recovery from failure.
- D. Use tightly coupled components.

Correct Answer: C

Automatically recover from failure: By monitoring a workload for key performance indicators (KPIs), you can trigger automation when a threshold is breached. These KPIs should be a measure of business value, not of the technical aspects of the operation of the service. This allows for automatic notification and tracking of failures, and for automated recovery processes that work around or repair the failure. With more sophisticated automation, it's possible to anticipate and remediate failures before they occur.

392) A company has a serverless application that includes an Amazon API Gateway API, an AWS Lambda function, and an Amazon DynamoDB database.

Which AWS service can the company use to trace user requests as they move through the application's components?

- A. AWS CloudTrail
- B. Amazon CloudWatch
- C. Amazon Inspector
-  D. AWS X-Ray

Correct Answer: D

AWS X-Ray - It provides a complete view of requests as they travel through your application and filters visual data across payloads, functions, traces, services, APIs, and more with no-code and low-code motions.

393) A company needs to set up a petabyte-scale data warehouse in the AWS Cloud. Which AWS service will meet this requirement?

- A. Amazon DynamoDB
- B. Amazon RDS
-  C. Amazon Redshift
- D. Amazon ElastiCache

Correct Answer: C

Amazon Redshift is a fast, fully managed, petabyte-scale data warehouse service that makes it simple and cost-effective to efficiently analyze all your data using your existing business intelligence tools.

394) What are the immediate advantages of AWS Cloud computing? (Select two.)

- A. Increased IT staff.
- ▶ B. Capital expenses are replaced with variable expenses.
- C. User control of infrastructure.
- ▶ D. Increased agility.
- E. AWS holds responsibility for security in the cloud.

Correct Answer: BD

395) The following are the economical advantages of using AWS: (Choose two.)

- ☒ A. reduced Total Cost of Ownership (TCO).
- ☐ B. increased capital expenditure (capex).
- ☒ C. reduced operational expenditure (opex).
- ☐ D. deferred payment plans for startups.
- ☐ E. business credit lines for startups.

Correct Answer: AC

396) Which AWS service is always provided at no charge?

- ☐ A. Amazon S3
- ☒ B. AWS Identity and Access Management (IAM)
- ☐ C. Elastic Load Balancers
- ☐ D. AWS WAF

Correct Answer: B

"You can interact with IAM through the web-based IAM console, the AWS Command Line Interface, or the AWS API or SDKs. IAM is offered at no additional charge. "

397) Which acts exemplify excellent practices for AWS IAM use? (Select two.)

- ☒ A. Configure a strong password policy
- ☐ B. Share the security credentials among users of AWS accounts who are in the same Region
- ☐ C. Use access keys to log in to the AWS Management Console
- ☒ D. Rotate access keys on a regular basis
- ☐ E. Avoid using IAM roles to delegate permissions

Correct Answer: AD

398) A business want to establish templates that it may reuse when deploying numerous AWS resources.

Which Amazon Web Services (AWS) offering or functionality can the business employ to achieve this requirement?

- A. AWS Marketplace
- B. Amazon Machine Image (AMI)
-  C. AWS CloudFormation
- D. AWS OpsWorks

Correct Answer: C

After you have your stacks and resources set up, you can reuse your templates to replicate your infrastructure in multiple environments.

[AWS CloudFormation](#) simplifies provisioning and management on AWS. You can create templates for the service or application architectures you want and have AWS CloudFormation use those templates for quick and reliable provisioning of the services or applications (called “stacks”). You can also easily update or replicate the stacks as needed

399) How can a business use AWS to lower its Total Cost of Ownership (TCO)?

-  A. By minimizing large capital expenditures
- B. By having no responsibility for third-party license costs
- C. By having no operational expenditures
- D. By having AWS manage applications

Correct Answer: A

AWS helps you reduce Total Cost of Ownership (TCO) by reducing the need to invest in large capital expenditures and providing a pay-as-you-go model that empowers you to invest in the capacity you need and use it only when the business requires it.

400) A business's website is hosted on Amazon EC2 instances. The firm must guarantee that the website is accessible to a worldwide audience and has a low latency for visitors.

Which Amazon Web Services (AWS) offering should the business employ to achieve these requirements?

- A. Amazon Route 53
-  B. Amazon CloudFront
- C. Elastic Load Balancing
- D. AWS Lambda

Correct Answer: B

CloudFront — Users can secure access to their websites and enable fast access worldwide.